

Chapter 5

Quasi-Newton Methods

5.1 Quasi-Newton Methods

We have seen that Newton's method $x_{k+1} = x_k - G_k^{-1}g_k$ is successful because it uses the Hessian which offers the useful curvature information. However, for various practical problems, the computing efforts of the Hessian matrices are very expensive, or the evaluation of the Hessian is difficult, even the Hessian is not available analytically. These lead to a class of methods that only uses the function values and the gradients of the objective function and that is closely related to Newton's method. Quasi-Newton method is such a class of methods which need not compute the Hessian, but generates a series of Hessian approximations, and at the same time maintains a fast rate of convergence.

Recall that in Chapter 3 the n -dimensional Newton's method $x_{k+1} = x_k - G_k^{-1}g_k$ comes from the one-dimensional Newton's method. Can we get any inspiration to the n -dimensional quasi-Newton method from the one-dimensional method? The answer is positive.

In Chapter 2, for quadratic interpolation with two points (2.4.6), we use interpolation condition (2.4.4) and obtain

$$\alpha_{k+1} = \alpha_k - \frac{\alpha_k - \alpha_{k-1}}{\phi'_k - \phi'_{k-1}} \phi'_k. \quad (5.1.1)$$

If we set

$$b_k = \frac{\phi'_k - \phi'_{k-1}}{\alpha_k - \alpha_{k-1}}, \quad (5.1.2)$$

then (5.1.1) can be written as

$$\alpha_{k+1} = \alpha_k - b_k^{-1} \phi'_k \quad (5.1.3)$$

which is also called the secant method. Comparing with Newton's form $\alpha_{k+1} = \alpha_k - [\phi''_k]^{-1} \phi'_k$ indicates that here b_k is used to approach ϕ''_k without computing ϕ''_k . Also, the convergence rate of the secant method is $\frac{1+\sqrt{5}}{2} \approx 1.618$ (see Theorem 2.4.1) which is fast. Now we apply this idea to the n -dimensional quasi-Newton method.

5.1.1 Quasi-Newton Equation

Instead of computing the Hessian G_k , we would like to construct Hessian approximation, for example, B_k in the quasi-Newton method. We hope that the sequence $\{B_k\}$ possesses positive definiteness, has the direction $d_k = -B_k^{-1}g_k$ down, and behaves like Newton's method. In addition, it is also required that its computation is convenient. What conditions does such a sequence $\{B_k\}$ satisfy? How to form $\{B_k\}$? In this subsection, we first reply the first question, and in the subsequent subsections we shall discuss the formations of B_k .

Let $f : R^n \rightarrow R$ be twice continuously differentiable on an open set $D \subset R^n$. Let the quadratic approximation of f at x_{k+1} be

$$f(x) \approx f(x_{k+1}) + g_{k+1}^T(x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^T G_{k+1}(x - x_{k+1}), \quad (5.1.4)$$

where $g_{k+1} \triangleq \nabla f(x_{k+1})$ and $G_{k+1} \triangleq \nabla^2 f(x_{k+1})$. Finding the derivative yields

$$g(x) \approx g_{k+1} + G_{k+1}(x - x_{k+1}). \quad (5.1.5)$$

Setting $x = x_k$, $s_k = x_{k+1} - x_k$ and $y_k = g_{k+1} - g_k$, we get

$$G_{k+1}^{-1}y_k \approx s_k. \quad (5.1.6)$$

Clearly, it is true that (5.1.6) holds exactly with equality for quadratic function f with the Hessian G , i.e.,

$$s_k = G^{-1}y_k, \text{ or } y_k = Gs_k. \quad (5.1.7)$$

Now we ask the produced inverse Hessian approximations H_{k+1} in the quasi-Newton method to satisfy this relation, i.e.,

$$H_{k+1}y_k = s_k, \quad (5.1.8)$$

which is called the quasi-Newton equation or quasi-Newton condition, where

$$s_k = x_{k+1} - x_k, \quad y_k = g_{k+1} - g_k. \quad (5.1.9)$$

In fact, if we consider the model function at x_{k+1} ,

$$m_{k+1}(x) = f(x_{k+1}) + g_{k+1}^T(x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^T B_{k+1}(x - x_{k+1}) \quad (5.1.10)$$

which satisfies the interpolation conditions

$$m_{k+1}(x_{k+1}) = f(x_{k+1}), \quad \nabla m_{k+1}(x_{k+1}) = g_{k+1}, \quad (5.1.11)$$

where $B_{k+1} = H_{k+1}^{-1}$ is an approximation to the Hessian G_{k+1} . Instead of the interpolation condition $\nabla^2 m_{k+1}(x_{k+1}) = G_{k+1}$ in Newton's method, we ask the model (5.1.10) to satisfy

$$\nabla m_{k+1}(x_k) = g_k, \quad (5.1.12)$$

that is

$$g_k = g_{k+1} + B_{k+1}(x_k - x_{k+1}).$$

So we have

$$B_{k+1}(x_{k+1} - x_k) = g_{k+1} - g_k$$

or

$$B_{k+1}s_k = y_k \quad (5.1.13)$$

which is also the quasi-Newton equation expressed in Hessian approximation form.

Premultiplying (5.1.13) by s_k^T gives

$$s_k^T B_{k+1} s_k = s_k^T y_k.$$

It means that if

$$s_k^T y_k > 0, \quad (5.1.14)$$

the matrix B_{k+1} is positive definite. Usually, (5.1.14) is called the curvature condition.

The above discussion tells us that the key point of the quasi-Newton method is to produce H_{k+1} (or B_{k+1}) by use of some convenient methods such that the quasi-Newton equation (5.1.8) (or (5.1.13)) holds. In general, such an H_{k+1} will be produced by updating H_k into H_{k+1} , which is our topic in the subsequent subsections. Now we state a general quasi-Newton algorithm below.

Algorithm 5.1.1 (*A general quasi-Newton algorithm*)

Step 1. Given $x_0 \in R^n, H_0 \in R^{n \times n}, 0 \leq \epsilon < 1, k := 0$.

Step 2. If $\|g_k\| \leq \epsilon$, stop.

Step 3. Compute

$$d_k = -H_k g_k. \quad (5.1.15)$$

Step 4. Find a step size $\alpha_k > 0$ by line search, and set $x_{k+1} = x_k + \alpha_k d_k$.

Step 5. Update H_k into H_{k+1} such that the quasi-Newton equation (5.1.8) holds.

Step 6. $k := k + 1$ and go to Step 2. $\square$

In the above algorithm, it is common to start the algorithm with $H_0 = I$, an identity matrix or set H_0 to be a finite-difference approximation to the inverse Hessian G_0^{-1} . If $H_0 = I$, the first iteration is just a steepest descent iteration. Sometimes, quasi-Newton method takes the form of Hessian approximation B_k . In this case, the Step 3 and Step 5 in Algorithm 5.1.1 have the following forms respectively.

Step 3*. Solve

$$B_k d = -g_k \text{ for } d_k. \quad (5.1.16)$$

Step 5*. Update B_k into B_{k+1} so that quasi-Newton equation (5.1.13) holds.

Next, we give some comparisons with Newton's method, which indicate that the quasi-Newton method is advantageous.

Comparison of quasi-Newton method vs Newton's method

quasi-Newton method	Newton's method
Only need the function values and gradients	Need the function values, gradients and Hessians
$\{H_k\}$ maintains positive definite for several updates	$\{G_k\}$ is not sure to be positive definite
Need $O(n^2)$ multiplications in each iteration	Need $O(n^3)$ multiplications in each iteration

As Newton's method is a steepest descent method under the norm $\|\cdot\|_{G_k}$, the quasi-Newton method is a steepest descent method under the norm $\|\cdot\|_{B_k}$, where B_k is the approximation of the Hessian G_k . In fact, d_k now is the solution of the minimization problem

$$\begin{aligned} \min \quad & g_k^T d \\ \text{s.t.} \quad & \|d\|_{B_k} \leq 1. \end{aligned} \quad (5.1.17)$$

From the inequality

$$(g_k^T d)^2 \leq (g_k^T B_k^{-1} g_k)(d^T B_k d),$$

it follows that when

$$d_k = -B_k^{-1} g_k = -H_k g_k,$$

$g_k^T d_k$ is the smallest.

By the way, since the metric matrices B_k are positive definite and always changed from iteration to iteration, the method is also called the variable metric method.

5.1.2 Symmetric Rank-One (SR1) Update

As we have seen, the key point of the quasi-Newton method is to generate H_{k+1} (or B_{k+1}) by means of the quasi-Newton equation. This subsection and the subsequent two subsections will discuss some typical and popular quasi-Newton updates. In this subsection we introduce a simple rank-one update that satisfies the quasi-Newton equation.

Let H_k be the inverse Hessian approximation of the k -th iteration. We try updating H_k into H_{k+1} , i.e.,

$$H_{k+1} = H_k + E_k, \quad (5.1.18)$$

where, usually, E_k is a matrix with lower rank. In the case of rank-one, we have

$$H_{k+1} = H_k + uv^T, \quad (5.1.19)$$

where $u, v \in R^n$. By quasi-Newton equation (5.1.8), we obtain

$$H_{k+1}y_k = (H_k + uv^T)y_k = s_k,$$

that is

$$(v^T y_k)u = s_k - H_k y_k. \quad (5.1.20)$$

This indicates that u must be in the direction of $s_k - H_k y_k$. Assume that $s_k - H_k y_k \neq 0$ (otherwise, H_k has satisfied the quasi-Newton equation) and that the vector v satisfies $v^T y_k \neq 0$, then it follows from (5.1.19) and (5.1.20) that

$$H_{k+1} = H_k + \frac{1}{v^T y_k} (s_k - H_k y_k) v^T. \quad (5.1.21)$$

Since the inverse Hessian approximation H_k is required to be symmetric, we can set simply $v = s_k - H_k y_k$ and get

$$H_{k+1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k} \quad (5.1.22)$$

which is called the symmetric rank-one update (SR1 update).

By the way, (5.1.21) is a general Broyden rank-one update in which, particularly, if $v = y_k$, (5.1.21) is called the Broyden rank-one update presented by Broyden (1965) for solving systems of nonlinear equations.

The distinct property of SR1 update is its natural quadratic termination, that is, for a quadratic function, it need not to do line search, but can be terminated within n steps, i.e., $H_n = G^{-1}$, where G is the Hessian of the quadratic function. This fact is proved by Theorem 5.1.2 below.

Theorem 5.1.2 (*Property Theorem of SR1 Update*) *Let $s_0, s_1, \dots, s_{n-1}$ be linearly independent. Then, for a quadratic function with a positive definite Hessian, SR1 method terminates at $n + 1$ steps, that is, $H_n = G^{-1}$.*

Proof. Let the Hessian G be positive definite. We can use

$$y_k = G s_k, k = 0, 1, \dots, n - 1, \quad (5.1.23)$$

that is shared by all proofs on quadratic termination.

First, by induction, we prove the hereditary property

$$H_i y_j = s_j, j = 0, 1, \dots, i - 1. \quad (5.1.24)$$

For $i = 1$, it is trivial from (5.1.22). Now suppose it is true for $i \geq 1$; we will prove it holds for $i + 1$.

From (5.1.22), we have

$$H_{i+1} y_j = H_i y_j + \frac{(s_i - H_i y_i)(s_i - H_i y_i)^T y_j}{(s_i - H_i y_i)^T y_i}. \quad (5.1.25)$$

When $j < i$, from the induction assumption and (5.1.23), we have

$$\begin{aligned}
 (s_i - H_i y_i)^T y_j &= s_i^T y_j - y_i^T H_i y_j \\
 &= s_i^T y_j - y_i^T s_j \\
 &= s_i^T G s_j - s_i^T G s_j \\
 &= 0.
 \end{aligned}$$

Then

$$H_{i+1} y_j = H_i y_j = s_j, \quad j < i.$$

When $j = i$, it is a direct consequence from (5.1.22) that

$$H_{i+1} y_i = s_i.$$

Therefore, (5.1.24) follows.

Furthermore, since

$$s_j = H_n y_j = H_n G s_j, \quad j = 0, 1, \dots, n-1$$

and s_j ($j = 0, 1, \dots, n-1$) are linearly independent, then $H_n G = I$, that is $H_n = G^{-1}$. $\square$

It is not difficult to find that SR1 update has the following characteristics.

1. SR1 update possesses natural quadratic termination.
2. SR1 update satisfies the hereditary property: $H_i y_j = s_j, j < i$.
3. SR1 update does not retain the positive definiteness of H_k . If and only if $(s_k - H_k y_k)^T y_k > 0$, SR1 update retains positive definiteness. However, this condition is difficult to guarantee. The remedy is that SR1 update can be used in the trust region framework since the trust region method does not require positive definiteness of the Hessian approximations (see Chapter 6).
4. Sometimes, the denominator $(s_k - H_k y_k)^T y_k$ is very small or zero, which results in serious numerical difficulty or even the algorithm is broken. This disadvantage restricts its applications. So, it is a topic deserving research how to modify SR1 update such that it possesses not only natural quadratic termination but also positive definiteness. A special

skipping strategy to prevent the SR1 update from breaking down is as follows. We use (5.1.22) only if

$$|(s_i - H_i y_i)^T y_i| \geq r \|s_i - H_i y_i\| \|y_i\|, \quad (5.1.26)$$

where $r \in (0, 1)$; otherwise we set $H_{i+1} = H_i$.

5. The SR1 update has a good behavior that it continues to generate good Hessian approximations, which is stated in the following theorem.

Theorem 5.1.3 *Let f be twice continuously differentiable, and its Hessian be bounded and Lipschitz continuous in a neighborhood of a point x^* . Let $\{x_k\}$ be a sequence of iterates with $x_k \rightarrow x^*$. Suppose that the skipping rule (5.1.26) holds for all k , and the steps s_k are uniformly linearly independent. Then the matrix sequence $\{B_k\}$ generated by SR1 update satisfies*

$$\lim_{i \rightarrow \infty} \|H_i - [\nabla^2 f(x^*)]^{-1}\| = 0. \quad (5.1.27)$$

5.1.3 DFP Update

DFP update is another typical update which is a rank-two update, i.e., H_{k+1} is formed by adding to H_k two symmetric matrices, each of rank one. Let us consider the symmetric rank-two update

$$H_{k+1} = H_k + a u u^T + b v v^T, \quad (5.1.28)$$

where $u, v \in R^n$, a and b are scalars to be determined. By the quasi-Newton equation (5.1.8),

$$H_k y_k + a u u^T y_k + b v v^T y_k = s_k. \quad (5.1.29)$$

Clearly, u and v are not uniquely determined, but their obvious choices are

$$u = s_k, \quad v = H_k y_k.$$

Then, from (5.1.29), we have

$$a = 1/u^T y_k = 1/s_k^T y_k, \quad b = -1/v^T y_k = -1/y_k^T H_k y_k.$$

Therefore

$$H_{k+1} = H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k}. \quad (5.1.30)$$

The formula (5.1.30) is the first quasi-Newton update proposed originally by Davidon [79] and developed later by Fletcher and Powell [137]. Hence it is called DFP update.

Now we state the quasi-Newton algorithm with DFP update (in brief, DFP method) as follows.

Algorithm 5.1.4 (*DFP method*)

Initial Step: Given $x_0 \in R^n$ an initial point, $H_0 \in R^{n \times n}$ a symmetric and positive definite matrix, $\epsilon > 0$ a termination scalar, $k := 0$.

k-th Step: For $k = 0, 1, \dots$,

1. If $\|g_k\| \leq \epsilon$, stop.
2. Compute $d_k = -H_k g_k$.
3. Compute the step size α_k .
4. Set $s_k = \alpha_k d_k$, $x_{k+1} = x_k + s_k$, $y_k = g_{k+1} - g_k$, and

$$H_{k+1} = H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k}.$$

5. $k := k + 1$, go to Step 1. $\square$

DFP method has the following important properties:

1. For a quadratic function (under exact line search)
 - (1) DFP update has quadratic termination, i.e., $H_n = G^{-1}$.
 - (2) DFP update has hereditary property, i.e., $H_i y_j = s_j$, $j < i$.
 - (3) DFP method generates conjugate directions; when $H_0 = I$, the method generates conjugate gradients.
2. For a general function
 - (1) DFP update maintains positive definiteness.
 - (2) Each iteration requires $3n^2 + O(n)$ multiplications.
 - (3) DFP method is superlinearly convergent.
 - (4) For a strictly convex function, under exact line search, DFP method is globally convergent.

The convergence properties of DFP method will be established in §5.3 and §5.4. In the remainder of this subsection we shall discuss the other two important properties: positive definiteness of the update and quadratic termination of the method.

The fact that quasi-Newton update retains positive definiteness is of importance in efficiency, numerical stability and global convergence. If the Hessian $G(x^*)$ is positive definite, the stationary point x^* is a strong minimizer. Hence, we hope Hessian approximation $\{B_k\}$ (or inverse Hessian approximation $\{H_k\}$) is positive definite. In addition, if $\{B_k\}$ (or $\{H_k\}$) is positive definite, the local quadratic model of f has a unique local minimizer, and the direction d_k from (5.1.15) or (5.1.16) is a descent direction. Usually, the update retaining positive definiteness means that if H_k (or B_k) is positive definite, then H_{k+1} (or B_{k+1}) is also positive definite. Such an update is also called positive definite update. Next, we discuss the positive definiteness of DFP update.

Theorem 5.1.5 (*Positive Definiteness of DFP Update*)

DFP update (5.1.30) retains positive definiteness if and only if $s_k^T y_k > 0$.

Proof. For the proof, we give two methods.

Proof (I) Sufficiency. We will prove

$$z^T H_k z > 0, \quad \forall z \neq 0 \quad (5.1.31)$$

by induction.

Obviously, H_0 is symmetric and positive definite. We now suppose that (5.1.31) holds for some $k \geq 0$ and set $H_k = LL^T$ as the Cholesky factorization of H_k . Let

$$a = L^T z, \quad b = L^T y_k. \quad (5.1.32)$$

Then by DFP update (5.1.30) we have

$$\begin{aligned} z^T H_{k+1} z &= z^T \left(H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} \right) z + z^T \frac{s_k s_k^T}{s_k^T y_k} z \\ &= \left[a^T a - \frac{(a^T b)^2}{b^T b} \right] + \frac{(z^T s_k)^2}{s_k^T y_k}. \end{aligned} \quad (5.1.33)$$

It is obvious from Cauchy-Schwartz inequality that

$$a^T a - \frac{(a^T b)^2}{b^T b} \geq 0. \quad (5.1.34)$$

In addition, the second term in (5.1.33) is also nonnegative because of $s_k^T y_k > 0$. Therefore we obtain that

$$z^T H_{k+1} z \geq 0.$$

Below, we must prove that at least one term in (5.1.33) is strictly larger than zero. Since $z \neq 0$, the equality holds in (5.1.34) if and only if a is parallel to b , equivalently, if and only if z is parallel to y_k . If z is parallel to y_k , we have $z = \beta y_k$, where $\beta \neq 0$, and

$$\frac{(z^T s_k)^2}{s_k^T y_k} = \beta^2 s_k^T y_k > 0,$$

which indicates that if z is parallel to y_k , i.e., if the first term in (5.1.33) equals zero, the second term must be strictly larger than zero. Thus, for any $z \neq 0$, we always have $z^T H_{k+1} z > 0$. The sufficiency follows.

In analogy, the necessity can be shown. $\square$

Proof (II). Let $H_k = LL^T$, $\bar{y} = L^T y_k$, $\bar{s} = L^{-1} s_k$. Then DFP update (5.1.30) can be written as

$$H_{k+1} = L W L^T, \quad (5.1.35)$$

where

$$W = I - \frac{\bar{y} \bar{y}^T}{\bar{y}^T \bar{y}} + \frac{\bar{s} \bar{s}^T}{\bar{s}^T \bar{y}}. \quad (5.1.36)$$

By the determinant relation (1.2.70) of update,

$$\det(W) = \frac{\bar{s}^T \bar{y}}{\bar{y}^T \bar{y}} = \frac{s_k^T y_k}{y_k^T H_k y_k},$$

which, together with (5.1.35), gives

$$\det(H_{k+1}) = \det(H_k) \frac{s_k^T y_k}{y_k^T H_k y_k}. \quad (5.1.37)$$

This implies that if H_k is positive definite, then $\det(H_{k+1}) > 0$ if and only if $s_k^T y_k > 0$.

Let

$$H_{k+1} = H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} = \bar{H} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k},$$

where $\bar{H} = H_k + \frac{s_k s_k^T}{s_k^T y_k}$. Since H_k is positive definite, we know by use of Theorem 1.2.17 that $s_k^T y_k > 0$ implies all eigenvalues of $\bar{H}$ are positive, i.e., $\bar{H}$ is positive definite. Using Theorem 1.2.17 again indicates that, at most, the smallest eigenvalue of H_{k+1} is not positive. Hence, $\det(H_{k+1})$ and the smallest eigenvalue of H_{k+1} have the same sign, which shows that H_{k+1} is positive definite if and only if $\det(H_{k+1}) > 0$. Therefore we have

$$s_k^T y_k > 0 \Leftrightarrow \det(H_{k+1}) > 0 \Leftrightarrow H_{k+1} \text{ is positive definite.} \quad \square$$

This theorem gives a sufficient and necessary condition of positive definite DFP update. By different definitions of positive definiteness and different algebraic tricks, we can establish this theorem. The interested readers may try different methods to give the proofs. The curvature condition $s_k^T y_k > 0$ for preserving positive definiteness is moderate, practical, and can be satisfied. For a quadratic positive definite function, obviously,

$$s_k^T y_k = s_k^T G s_k > 0.$$

For a strong convex function, the average Hessian

$$\bar{G}_k = \int_0^1 \nabla^2 f(x_k + \tau s_k) d\tau \quad (5.1.38)$$

is positive definite. So, from Taylor's formula

$$y_k = \nabla f(x_k + s_k) - \nabla f(x_k) = \int_0^1 \nabla^2 f(x_k + \tau s_k) s_k d\tau = \bar{G}_k s_k,$$

we have that

$$y_k^T s_k = s_k^T \bar{G}_k s_k > 0.$$

For a general function, we have

$$s_k^T y_k = g_{k+1}^T s_k - g_k^T s_k.$$

Note that $g_k^T s_k < 0$ is due to s_k being a descent direction. Using exact line search with $g_{k+1}^T s_k = 0$, we have $s_k^T y_k > 0$. When we use inexact line search, for example, if the rule (2.5.7) is satisfied, the condition $s_k^T y_k > 0$ can also be satisfied. In general, as long as we increase the precision of line search, we can make $g_{k+1}^T s_k$ small enough in magnitude to the desired degree.

From this theorem and the above discussion, it is obvious that, for Algorithm 5.1.4 with exact or inexact line search, the condition $s_k^T y_k > 0$ holds and therefore each update matrix H_k in DFP algorithm is positive definite. So, we have the following corollary.

Corollary 5.1.6 *Each matrix H_k generated by DFP Algorithm 5.1.4 is positive definite, and the directions $d_k = -H_k g_k$ are descent directions.*

Finally, we give a theorem on quadratic termination of DFP method. This theorem shows that, for a quadratic function with positive definite Hessian G , the directions generated from DFP method are conjugate, and the method terminates at n steps, that is $H_n = G^{-1}$.

Theorem 5.1.7 (*Quadratic Termination Theorem of DFP Method*)

Let $f(x)$ be a quadratic function with positive definite Hessian G . Then, if exact line search is used, the sequence $\{s_j\}$ generated from DFP method satisfies hereditary property, conjugate property and quadratic termination, that is, for $i = 0, 1, \dots, m$, where $m \leq n - 1$,

1. $H_{i+1}y_j = s_j$, $j = 0, 1, \dots, i$; (hereditary property)
2. $s_i^T G s_j = 0$, $j = 0, 1, \dots, i - 1$; (conjugate direction property)
3. The method terminates at $m + 1 \leq n$ steps. If $m = n - 1$, then $H_n = G^{-1}$.

Proof. We prove part (1) and (2) by induction. Clearly, when $i = 0$, it is trivial. Now suppose that part (1) and (2) hold for some i . We show that they also hold for $i + 1$. Since $g_{i+1} \neq 0$, by exact line search, the fact that $y_k = g_{k+1} - g_k = G(x_{k+1} - x_k) = G s_k$, ($1 \leq k \leq i$) and the induction hypothesis, we have, for $j \leq i$,

$$\begin{aligned}
 g_{i+1}^T s_j &= g_{j+1}^T s_j + \sum_{k=j+1}^i (g_{k+1} - g_k)^T s_j \\
 &= g_{j+1}^T s_j + \sum_{k=j+1}^i y_k^T s_j \\
 &= 0 + \sum_{k=j+1}^i s_k^T G s_j \\
 &= 0.
 \end{aligned} \tag{5.1.39}$$

Hence, by use of $s_{i+1} = -\alpha_{i+1} H_{i+1} g_{i+1}$, induction hypothesis in part (1) and (5.1.39), it follows that

$$s_{i+1}^T G s_j = -\alpha_{i+1} g_{i+1}^T H_{i+1} y_j$$

$$\begin{aligned}
&= -\alpha_{i+1} g_{i+1}^T s_j \\
&= 0,
\end{aligned} \tag{5.1.40}$$

which proves part (2) holds for $i + 1$.

Next, we prove that part (1) holds for $i + 1$, i.e.,

$$H_{i+2}y_j = s_j, \quad j = 0, 1, \dots, i + 1. \tag{5.1.41}$$

When $j = i + 1$, part (1) is immediate from DFP update (5.1.30), that is

$$H_{i+2}y_{i+1} = s_{i+1}. \tag{5.1.42}$$

When $j \leq i$, it follows from (5.1.40) and the induction hypothesis in part (1) that

$$\begin{aligned}
s_{i+1}^T y_j &= s_{i+1}^T G s_j = 0, \\
y_{i+1}^T H_{i+1} y_j &= y_{i+1}^T s_j = s_{i+1}^T G s_j = 0.
\end{aligned}$$

Then

$$\begin{aligned}
H_{i+2}y_j &= H_{i+1}y_j + \frac{s_{i+1}s_{i+1}^T y_j}{s_{i+1}^T y_{i+1}} - \frac{H_{i+1}y_{i+1}y_{i+1}^T H_{i+1}y_j}{y_{i+1}^T H_{i+1}y_{i+1}} \\
&= H_{i+1}y_j \\
&= s_j.
\end{aligned} \tag{5.1.43}$$

This, together with (5.1.42), shows (5.1.41). Therefore part (1) follows.

Finally, since s_i ($i = 0, 1, \dots, m$) are conjugate, the method is a conjugate direction method. Based on Theorem 4.1.3 of the conjugate direction method, the method terminates after m ($\leq n$) steps. When $m = n - 1$, since s_i ($i = 0, 1, \dots, n - 1$) are linearly independent, then part (1) means

$$H_n G s_j = H_n y_j = s_j, \quad j = 0, 1, \dots, n - 1$$

which implies $H_n = G^{-1}$. $\square$

From this theorem we see that DFP method is a conjugate direction method. If the initial approximation $H_0 = I$, the method becomes a conjugate gradient method. By the hereditary property, we have $H_{i+1}G s_j = s_j, j = 0, 1, \dots, i$, which also indicates that these s_j are eigenvectors of matrix $H_{i+1}G$ ($j = 0, 1, \dots, i$) corresponding to the eigenvalue 1.

DFP method is a seminal quasi-Newton method and has been widely used in many computer codes. It has played an important role in theoretical

analysis and numerical computing. However, further studies indicate that DFP method is numerically unstable, and sometimes produces numerically singular Hessian approximations. The other famous quasi-Newton update — BFGS update introduced in the next subsection will overcome these drawbacks and perform better than DFP update.

5.1.4 BFGS Update and PSB Update

In §5.1.1 we have seen that

$$H_{k+1}y_k = s_k \text{ and } B_{k+1}s_k = y_k \quad (5.1.44)$$

are the quasi-Newton equations with respect to inverse Hessian approximation and Hessian approximation respectively. Note that any approximation in (5.1.44) can be obtained from the other by means of exchanging $H_{k+1} \leftrightarrow B_{k+1}$ and $s_k \leftrightarrow y_k$. In analogy to the derivation of DFP update (5.1.30) about H_k , we can get

$$B_{k+1}^{(BFGS)} = B_k + \frac{y_k y_k^T}{y_k^T s_k} - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k}, \quad (5.1.45)$$

which is called BFGS update discovered independently by Broyden [27], Fletcher [125], Goldfarb [153] and Shanno [304]. In fact, if one makes directly simple exchanges $H_k \leftrightarrow B_k$ and $s_k \leftrightarrow y_k$, BFGS update (5.1.45) is just obtained from DFP update (5.1.30). Thus, BFGS update is also said to be a complement DFP update. Since $B_k s_k = -\alpha_k g_k$ and $B_k d_k = -g_k$, (5.1.45) can also be written as

$$B_{k+1}^{(BFGS)} = B_k + \frac{g_k g_k^T}{g_k^T d_k} + \frac{y_k y_k^T}{\alpha_k y_k^T d_k}. \quad (5.1.46)$$

By using twice the Sherman-Morrison formula (1.2.67), (5.1.45) will become as follows:

$$\begin{aligned} H_{k+1}^{(BFGS)} &= H_k + \left(1 + \frac{y_k^T H_k y_k}{s_k^T y_k} \right) \frac{s_k s_k^T}{s_k^T y_k} \\ &\quad - \frac{s_k y_k^T H_k + H_k y_k s_k^T}{s_k^T y_k} \\ &= H_k + \frac{(s_k - H_k y_k) s_k^T + s_k (s_k - H_k y_k)^T}{s_k^T y_k} \end{aligned} \quad (5.1.47)$$

$$-\frac{(s_k - H_k y_k)^T y_k}{(s_k^T y_k)^2} s_k s_k^T \quad (5.1.48)$$

$$= \left(I - \frac{s_k y_k^T}{s_k^T y_k} \right) H_k \left(I - \frac{y_k s_k^T}{s_k^T y_k} \right) + \frac{s_k s_k^T}{s_k^T y_k}. \quad (5.1.49)$$

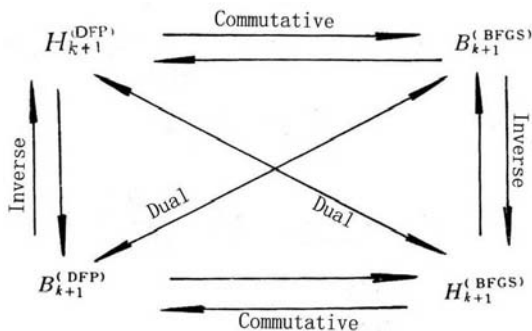
(5.1.47)–(5.1.49) are the three forms of BFGS update about H_k . Furthermore, by making exchanges $H_k \leftrightarrow B_k$ and $s_k \leftrightarrow y_k$ in (5.1.47)–(5.1.49), we can get three corresponding forms of DFP update about B_k :

$$\begin{aligned} B_{k+1}^{(DFP)} &= B_k + \left(1 + \frac{s_k^T B_k s_k}{y_k^T s_k} \right) \frac{y_k y_k^T}{y_k^T s_k} \\ &\quad - \frac{y_k s_k^T B_k + B_k s_k y_k^T}{y_k^T s_k} \end{aligned} \quad (5.1.50)$$

$$\begin{aligned} &= B_k + \frac{(y_k - B_k s_k) y_k^T + y_k (y_k - B_k s_k)^T}{y_k^T s_k} \\ &\quad - \frac{(y_k - B_k s_k)^T s_k}{(y_k^T s_k)^2} y_k y_k^T \end{aligned} \quad (5.1.51)$$

$$= \left(I - \frac{y_k s_k^T}{y_k^T s_k} \right) B_k \left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) + \frac{y_k y_k^T}{y_k^T s_k}. \quad (5.1.52)$$

The above discussions describe a method for finding its dual update from a given update. Given a quasi-Newton update H_{k+1} about H -form, by exchanging $H_k \leftrightarrow B_k$ and $s_k \leftrightarrow y_k$, we can get its dual update $B_{k+1}^{(D)}$ about B -form. Then, applying the Sherman-Morrison formula to $B_{k+1}^{(D)}$, we will produce the dual update $H_{k+1}^{(D)}$ of H_{k+1} about the H -form. Similarly, if we employ the same operations to the dual update $H_{k+1}^{(D)}$, the original update H_{k+1} will be restored. Notice that, for an H -form, the dual update of H_{k+1} is $H_{k+1}^{(D)}$. In addition, the dual operation maintains the quasi-Newton equation. The following figure represents the dual relation.

Figure 5.1.1 Duality of $H_{k+1}^{(DFP)}$ and $H_{k+1}^{(BFGS)}$

For SR1 update

$$H_{k+1}^{(SR1)} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k}, \quad (5.1.53)$$

exchanging $H_k \leftrightarrow B_k$ and $s_k \leftrightarrow y_k$ gives

$$B_{k+1}^{(D)} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}. \quad (5.1.54)$$

Then applying the Sherman-Morrison formula to (5.1.54), we see that the resultant $H_{k+1}^{(D)}$ is still the $H_{k+1}^{(SR1)}$, i.e., $H_{k+1}^{(D)} = H_{k+1}^{(SR1)}$. Thus SR1 update is self-dual. As we pointed out in §5.1.2, SR1 update does not retain the positive definiteness of the update. A self-dual update retaining the positive definiteness is called Hoshino update which will be given in (5.2.6) of §5.2.

The BFGS update is presently considered to be the best one of all quasi-Newton updates, which has all good properties of DFP update. In addition, when inexact line search (2.5.3) and (2.5.7) are used, BFGS method is globally convergent. Note that it is still an open problem whether DFP update has this property. The numerical performance of BFGS update is superior to that of DFP update. In particular, BFGS update can often work well in conjunction with some line searches with lower accuracy.

The next topic in this subsection is PSB update which is formally known as the Powell-symmetric-Broyden update due to Powell [260].

Let $B \in R^{n \times n}$ be a symmetric matrix. Consider the general Broyden rank-one update

$$C_1 = B + \frac{(y - Bs)c^T}{c^T s},$$

where $c \in R^n, c^T s \neq 0$. In general, C_1 is not symmetric. So, we consider a symmetrization:

$$C_2 = (C_1 + C_1^T)/2.$$

Now C_2 is symmetric but, in general, does not obey the quasi-Newton equation. Then we might continue the above process and generate the sequence $\{C_k\}$:

$$\begin{aligned} C_{2k+1} &= C_{2k} + \frac{(y - C_{2k}s)c^T}{c^T s}, \\ C_{2k+2} &= (C_{2k+1} + C_{2k+1}^T)/2, \quad k = 0, 1, \dots \end{aligned} \quad (5.1.55)$$

where $C_0 = B$. Here each C_{2k+1} is the closest matrix in $Q(y, s)$ to C_{2k} , and each C_{2k+2} is the closest symmetric matrix to C_{2k+1} , where $Q(y, s) = \{C \in R^{n \times n} \mid Cs = y\}$ is a matrix set satisfying the quasi-Newton equation. The Figure 5.1.2 illustrates the symmetrization process, where S denotes the set of symmetric matrices.

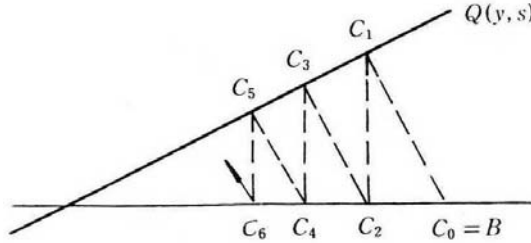


Figure 5.1.2 Production of the sequence C_k

Below, we show the limit of matrix sequence $\{C_k\}$ is

$$\bar{B} = B + \frac{(y - Bs)c^T + c(y - Bs)^T}{c^T s} - \frac{(y - Bs)^T s}{(c^T s)^2} cc^T \quad (5.1.56)$$

which satisfies symmetricity and the quasi-Newton equation.

Theorem 5.1.8 *Let $B \in R^{n \times n}$ be symmetric, $c, s, y \in R^n$ and $c^T s \neq 0$. Let the sequence $\{C_k\}$ be defined by (5.1.55), and $C_0 = B$. Then the sequence $\{C_k\}$ converges to $\bar{B}$ in (5.1.56).*

Proof. We only need to prove that the sequence $\{C_{2k}\}$ converges. Let $G_k = C_{2k}$. From (5.1.55), we have

$$G_{k+1} = G_k + \frac{1}{2} \frac{w_k c^T + c w_k^T}{c^T s}, \quad (5.1.57)$$

where $w_k = y - G_k s$. Note that

$$\begin{aligned} w_{k+1} &= y - G_{k+1} s \\ &= y - G_k s - \frac{1}{2} \frac{w_k c^T s + c w_k^T s}{c^T s} \\ &= \frac{1}{2} \left(I - \frac{c s^T}{c^T s} \right) w_k, \end{aligned}$$

that is

$$w_{k+1} = P w_k, \text{ where } P = \frac{1}{2} \left[I - \frac{c s^T}{c^T s} \right]. \quad (5.1.58)$$

Then it follows from Sherman-Morrison formula (1.2.67) that

$$\begin{aligned} \sum_{k=0}^{\infty} w_k &= \sum_{k=0}^{\infty} P^k (y - G_0 s) = \sum_{k=0}^{\infty} P^k (y - B s) \\ &= (I - P)^{-1} (y - B s) = 2 \left[I - \frac{1}{2} \frac{c s^T}{c^T s} \right] (y - B s) \\ &= 2(y - B s) - \frac{c s^T}{c^T s} (y - B s). \end{aligned} \quad (5.1.59)$$

Since

$$\lim_{k \rightarrow \infty} G_k = B + \sum_{k=0}^{\infty} (G_{k+1} - G_k), \quad (5.1.60)$$

and by (5.1.57) and (5.1.59), we get that the sequence $\{G_k\}$ is convergent. Note that

$$\begin{aligned} \sum_{k=0}^{\infty} (G_{k+1} - G_k) &= \frac{1}{2} \sum_{k=0}^{\infty} \frac{w_k c^T + c w_k^T}{c^T s} \\ &= \frac{1}{c^T s} \left[(y - B s) c^T - \frac{1}{2} \frac{s^T (y - B s)}{c^T s} c c^T + c (y - B s)^T - \frac{1}{2} \frac{(y - B s)^T s}{c^T s} c c^T \right] \\ &= \frac{1}{c^T s} [(y - B s) c^T + c (y - B s)^T] - \frac{(y - B s)^T s}{(c^T s)^2} c c^T, \end{aligned} \quad (5.1.61)$$

hence the conclusion (5.1.56) follows by (5.1.60) and (5.1.61). $\square$

(5.1.56) gives a class of rank-two update which is derived by a symmetrization process. If we add the subscripts, it can be written as

$$\begin{aligned} B_{k+1} = B_k + & \frac{(y_k - B_k s_k) c_k^T + c_k (y_k - B_k s_k)^T}{c_k^T s_k} \\ & - \frac{(y_k - B_k s_k)^T s_k}{(c_k^T s_k)^2} c_k c_k^T, \end{aligned} \quad (5.1.62)$$

which is called the general PSB update. In particular,

If $c_k = y_k - B_k s_k$, (5.1.62) is SR1 update (5.1.54).

If $c_k = y_k$, (5.1.62) is DFP update (5.1.51).

If $c_k = \frac{1}{w_k+1} y_k + \frac{w_k}{w_k+1} B_k s_k$, where $w_k = (y_k^T s_k / s_k^T B_k s_k)^{\frac{1}{2}}$, (5.1.62) is BFGS update (5.1.46).

If $c_k = s_k$, (5.1.62) is PSB update:

$$\begin{aligned} B_{k+1} = B_k + & \frac{(y_k - B_k s_k) s_k^T + s_k (y_k - B_k s_k)^T}{s_k^T s_k} \\ & - \frac{(y_k - B_k s_k)^T s_k}{(s_k^T s_k)^2} s_k s_k^T. \end{aligned} \quad (5.1.63)$$

Its dual update in H -form is

$$\begin{aligned} H_{k+1} = H_k + & \frac{(s_k - H_k y_k) y_k^T + y_k (s_k - H_k y_k)^T}{y_k^T y_k} \\ & - \frac{(s_k - H_k y_k)^T y_k}{(y_k^T y_k)^2} y_k y_k^T \end{aligned} \quad (5.1.64)$$

which is called Greenstadt update (see Greenstadt [163]).

PSB update (5.1.63) is important in theoretical research and practical computing. However, the drawback that PSB update does not retain the positive definiteness of updates hurts its performance in computing. Fortunately, the drawback can be avoided if we employ the trust region framework with PSB update.

5.1.5 The Least Change Secant Update

Various quasi-Newton updates obey the least change property which refers to the H_{k+1} (or B_{k+1}) being the minimum change to H_k (or B_k) consistent with the quasi-Newton equation if the change $H_{k+1} - H_k$ (or $B_{k+1} - B_k$) is measured under some norm. This property is helpful to maintain some information of the last iteration. By the way, by use of the property, we also can derive quasi-Newton update.

Theorem 5.1.9 *Let $B \in R^{n \times n}$, $s, y \in R^n$ and $s \neq 0$. Then Broyden rank-one update*

$$\bar{B} = B + \frac{(y - Bs)s^T}{s^T s} \quad (5.1.65)$$

is a unique solution of the minimization problem

$$\min\{\|\hat{B} - B\|_F : \hat{B}s = y\}. \quad (5.1.66)$$

Proof. [proof I] Since $y = \hat{B}s$, then

$$\begin{aligned} \|\bar{B} - B\| &= \left\| \frac{(y - Bs)s^T}{s^T s} \right\|_F = \left\| (\hat{B} - B) \frac{ss^T}{s^T s} \right\|_F \\ &\leq \|\hat{B} - B\|_F. \end{aligned} \quad (5.1.67)$$

Also, since the Frobenius norm is strictly convex and the set of matrix $\hat{B}$ satisfying the quasi-Newton equation is convex, then the solution of (5.1.66) is unique.

[proof II] Define $C = \hat{B} - B$ and let c_i^T be the i -th row of C . Then (5.1.66) can be represented as

$$\begin{aligned} \min \quad & \sum_{i=1}^n \|c_i^T\|_2^2 \\ \text{s.t.} \quad & c_i^T s = (y - Bs)_i, \quad i = 1, \dots, n \end{aligned} \quad (5.1.68)$$

where $(y - Bs)_i$ denotes the i -th component of $y - Bs$. Obviously, (5.1.68) can be divided into n subproblems

$$\begin{aligned} \min \quad & \|c_i^T\|_2^2 \\ \text{s.t.} \quad & c_i^T s = (y - Bs)_i. \end{aligned} \quad (5.1.69)$$

Solving (5.1.69) is equivalent to finding the Moore-Penrose inverse s^+ of s . Therefore

$$c_i^T = (y - Bs)_i s^+ = \frac{(y - Bs)_i s^T}{s^T s}$$

which indicates that (5.1.65) is the unique solution of (5.1.66). $\square$

This theorem shows that Broyden's rank-one update

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k) s_k^T}{s_k^T s_k} \quad (5.1.70)$$

is the unique solution of the minimization problem

$$\min\{\|\hat{B} - B_k\|_F : \hat{B} s_k = y_k\}. \quad (5.1.71)$$

Similarly,

$$H_{k+1} = H_k + \frac{(s_k - H_k y_k) y_k^T}{y_k^T y_k} \quad (5.1.72)$$

is the unique solution of the minimization problem

$$\min\{\|\hat{H} - H_k\|_F : \hat{H} y_k = s_k\}. \quad (5.1.73)$$

Next, we discuss the least change property of general symmetric rank-two update.

Theorem 5.1.10 *Let $B \in R^{n \times n}$ be symmetric, $c, s, y \in R^n$, and $c^T s > 0$. Assume that $M \in R^{n \times n}$ is a symmetric and nonsingular matrix satisfying*

$$Mc = M^{-1}s. \quad (5.1.74)$$

Then the general PSB update

$$\bar{B} = B + \frac{(y - Bs)c^T + c(y - Bs)^T}{c^T s} - \frac{(y - Bs)^T s}{(c^T s)^2} cc^T \quad (5.1.75)$$

is the unique solution of the minimization problem

$$\min\{\|\hat{B} - B\|_{M,F} : \hat{B}s = y, \hat{B}^T = \hat{B}\}, \quad (5.1.76)$$

where $\|B\|_{M,F} = \|MBM\|_F$.

Proof. Let $\hat{B}$ be a symmetric matrix obeying $y = \hat{B}s$. Let also $Mc = M^{-1}s = z$, $E = M(\hat{B} - B)M$, $\bar{E} = M(\bar{B} - B)M$. Left- and right-multiplying (5.1.75) by M yields

$$\bar{E} = \frac{Ezz^T + zz^TE}{z^Tz} - \frac{z^TEz}{(z^Tz)^2}zz^T.$$

Clearly, $\|\bar{E}z\|_2 = \|Ez\|_2$, and if $v \perp z$, then $\|\bar{E}v\|_2 \leq \|Ev\|_2$. Therefore $\|\bar{E}\|_F \leq \|E\|_F$. Also, note that the weighted Frobenius norm $\|\cdot\|_{M,F}$ is strictly convex and the matrix set $\{\hat{B} \mid \hat{B}s = y, \hat{B}^T = \hat{B}\}$ is convex, thus the general PSB update (5.1.75) is the unique solution of the problem (5.1.76). $\square$

In particular, some different choices of c in (5.1.75) give different conclusions.

Choosing $c = s$ (in this case, $M = I$), we get PSB update (5.1.63). Hence, Theorem 5.1.10 implies that $\bar{B}^{PSB}$ is the unique solution to the problem

$$\min_{\hat{B} \in R^{n \times n}} \{\|\hat{B} - B\|_F \mid \hat{B}s = y, \hat{B}^T = \hat{B}\}. \quad (5.1.77)$$

Choosing $c = y$ (in this case, M satisfies $M^{-2}s = y$), we get DFP update (5.1.50). Hence Theorem 5.1.10 implies that $\bar{B}^{DFP}$ is the unique solution to the problem

$$\min\{\|\hat{B} - B\|_{M,F} \mid \hat{B}s = y, \hat{B}^T = \hat{B}\}. \quad (5.1.78)$$

Similarly, by the dual technique, $\bar{H}^{BFGS}$ in (5.1.47) is the unique solution to the problem

$$\min\{\|\hat{H} - H\|_{M^{-1},F} \mid \hat{H}y = s, \hat{H}^T = \hat{H}\}. \quad (5.1.79)$$

As an exercise, it is not difficult to discuss the least change property of dual general PSB update.

5.2 The Broyden Class

From the last section we have seen that both DFP and BFGS updates are symmetric and positive definite rank-two updates consisting of $H_k y_k$ and s_k . It is natural to discuss their weighted (or convex) combinations which have the same type, and consider their behaviors.

Consider the update class

$$H_{k+1}^\phi = (1 - \phi)H_{k+1}^{DFP} + \phi H_{k+1}^{BFGS}, \quad (5.2.1)$$

where ϕ is a parameter. (5.2.1) is called the Broyden class of update. If $\phi \in [0, 1]$, (5.2.1) is called the Broyden convex class of update. Obviously, Broyden class (5.2.1) satisfies quasi-Newton equation (5.1.8). We can also write (5.2.1) in the following forms:

$$H_{k+1}^\phi = H_{k+1}^{DFP} + \phi v_k v_k^T \quad (5.2.2)$$

$$= H_{k+1}^{BFGS} + (\phi - 1)v_k v_k^T \quad (5.2.3)$$

$$= H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \phi v_k v_k^T, \quad (5.2.4)$$

where

$$v_k = (y_k^T H_k y_k)^{\frac{1}{2}} \left[\frac{s_k}{s_k^T y_k} - \frac{H_k y_k}{y_k^T H_k y_k} \right]. \quad (5.2.5)$$

In particular, in (5.2.4),

set $\phi = 0$, we get DFP update (5.1.30);

set $\phi = 1$, we get BFGS update (5.1.47);

set $\phi = \frac{s_k^T y_k}{(s_k - H_k y_k)^T y_k}$, we get SR1 update (5.1.22);

set

$$\phi = \frac{1}{1 \mp (y_k^T H_k y_k / s_k^T y_k)}, \quad (5.2.6)$$

we get Hoshino update.

Broyden class (5.2.2)–(5.2.4) can be derived directly by the quasi-Newton equation. Consider a general rank-two update consisting of s_k and $H_k y_k$:

$$H_{k+1} = H_k + a s_k s_k^T + b (H_k y_k s_k^T + s_k y_k^T H_k) + c H_k y_k y_k^T H_k, \quad (5.2.7)$$

where a, b, c are scalars to be determined. Using the quasi-Newton equation yields

$$\begin{aligned} 1 &= a s_k^T y_k + b y_k^T H_k y_k, \\ 0 &= 1 + b s_k^T y_k + c y_k^T H_k y_k. \end{aligned} \quad (5.2.8)$$

Here are two equations with three unknowns and one free degree. Set

$$b = -\phi/s_k^T y_k, \quad (5.2.9)$$

where ϕ is a parameter. Solving (5.2.8) and substituting the result into (5.2.7), we have

$$H_{k+1}^\phi = H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \phi v_k v_k^T = H_{k+1}^{DFP} + \phi v_k v_k^T,$$

where v_k is defined by (5.2.5). The above expression is just (5.2.2) and (5.2.4). By a slight arrangement, Broyden class has the following matrix form:

$$H_{k+1}^\phi = H_k + [s_k, H_k y_k] \begin{bmatrix} \frac{1+\phi y_k^T H_k y_k / s_k^T y_k}{s_k^T y_k} & -\frac{\phi}{s_k^T y_k} \\ -\frac{\phi}{s_k^T y_k} & \frac{\phi-1}{y_k^T H_k y_k} \end{bmatrix} [s_k, H_k y_k]^T. \quad (5.2.10)$$

Correspondingly, it is easy to produce Broyden class in B -form:

$$B_{k+1}^\theta = \theta B_{k+1}^{DFP} + (1-\theta) B_{k+1}^{BFGS} \quad (5.2.11)$$

$$= B_{k+1}^{BFGS} + \theta w_k w_k^T \quad (5.2.12)$$

$$= B_{k+1}^{DFP} + (\theta-1) w_k w_k^T \quad (5.2.13)$$

$$= B_k + \frac{y_k y_k^T}{s_k^T y_k} - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \theta w_k w_k^T, \quad (5.2.14)$$

where

$$w_k = (s_k^T B_k s_k)^{1/2} \left[\frac{y_k}{s_k^T y_k} - \frac{B_k s_k}{s_k^T B_k s_k} \right]. \quad (5.2.15)$$

Note that the relation between θ and ϕ is

$$\theta = (\phi-1)/(\phi-1-\phi\mu), \quad (5.2.16)$$

where

$$\mu = \frac{y_k^T H_k y_k s_k^T B_k s_k}{(s_k^T y_k)^2}. \quad (5.2.17)$$

Since $v_k^T y_k = 0$ and $w_k^T s_k = 0$, then (5.2.1)-(5.2.2) and (5.2.11)-(5.2.14) satisfy respectively the quasi-Newton equation (5.1.8) and (5.1.13) for any parameter ϕ and θ . In analogous to Theorem 5.1.2 and Theorem 5.1.5, we can show the quadratic termination property and positive definite property of Broyden class.

Theorem 5.2.1 (*Quadratic Termination Theorem of Broyden Class*) Let $f(x)$ be a quadratic function with positive definite Hessian G . Then, when exact line search is used, the Broyden class of update has hereditary property and conjugate direction property, that is, for $i = 0, 1, \dots, m, (m \leq n - 1)$,

$$\text{Hereditary property:} \quad H_{i+1}y_j = s_j, \quad j = 0, 1, \dots, i. \quad (5.2.18)$$

$$\text{Conjugate direction:} \quad s_i^T G s_j = 0, \quad j = 0, 1, \dots, i - 1. \quad (5.2.19)$$

The method terminates at m steps. If $m = n - 1$, then $H_n = G^{-1}$.

Proof. It is similar to the proof of Theorem 5.1.5. $\square$

Theorem 5.2.2 (*Positive Definiteness of Broyden Class of Update*) Let $\phi \geq 0$. If and only if $s_k^T y_k > 0$, Broyden class of update (5.2.2) retains the positive definiteness.

Proof. From Theorem 5.1.2, if and only if $s_k^T y_k > 0$, DFP update retains positive definiteness. Since $\phi \geq 0$, it follows from (5.2.3) and Theorem 1.2.17 that the smallest eigenvalue of H_{k+1}^ϕ is not less than the smallest one of H_{k+1}^{DFP} . Hence H_{k+1}^ϕ is positive definite. $\square$

This theorem shows that not all members of Broyden class retain the positive definiteness. Clearly, when $\phi \geq 0$, H_{k+1}^ϕ maintains its positive definiteness; when $\phi < 0$, it is possible that the update becomes singular. The following Theorem 5.2.3 gives a value $\bar{\phi}$, and says that as long as $\phi > \bar{\phi}$, H_{k+1}^ϕ will maintain positive definiteness. Such a value $\bar{\phi}$ is called the degenerate value of Broyden class, which makes $H_{k+1}^{\bar{\phi}}$ singular.

Theorem 5.2.3 *The degenerate value of Broyden class of update is*

$$\bar{\phi} = \frac{1}{1 - \mu} = \frac{1}{1 - y_k^T H_k y_k s_k^T B_k s_k / (s_k^T y_k)^2}. \quad (5.2.20)$$

Proof. Let $d_k = -H_k g_k, s_k = \alpha_k d_k$. When we use exact line search, $g_{k+1}^T d_k = 0 = g_{k+1}^T s_k$. Notice also that $g_{k+1} = y_k + g_k, v_k^T g_k = 0$, and using (5.2.5), we have

$$\begin{aligned} d_{k+1}^\phi &= -H_{k+1}^\phi g_{k+1} \\ &= -\left(H_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \phi v_k v_k^T \right) g_{k+1} \end{aligned}$$

$$\begin{aligned}
&= -H_k g_k - H_k y_k + \frac{y_k^T H_k (g_k + y_k)}{y_k^T H_k y_k} H_k y_k - \phi v_k^T g_k v_k \\
&= -H_k g_k + \frac{y_k^T H_k g_k}{y_k^T H_k y_k} H_k y_k - \phi v_k^T g_k v_k \\
&= d_k - \frac{y_k^T d_k}{y_k^T H_k y_k} H_k y_k - \phi v_k^T g_k v_k \\
&= \frac{d_k^T y_k}{(y_k^T H_k y_k)^{1/2}} \left[(y_k^T H_k y_k)^{1/2} \left(\frac{d_k}{d_k^T y_k} - \frac{H_k y_k}{y_k^T H_k y_k} \right) \right] - \phi v_k^T g_k v_k \\
&= \left(\frac{d_k^T y_k}{(y_k^T H_k y_k)^{1/2}} - \phi v_k^T g_k \right) v_k. \tag{5.2.21}
\end{aligned}$$

This shows that when exact line search is used, (5.2.21) holds. When $g_{k+1} \neq 0$, if $d_{k+1}^\phi = -H_{k+1}^\phi g_{k+1} = 0$, then ϕ is called the degenerate value of H_{k+1}^ϕ . By using $d_{k+1}^\phi = 0$ and (5.2.5), we obtain

$$\begin{aligned}
\phi &= \frac{y_k^T d_k}{(y_k^T H_k y_k)^{1/2} v_k^T g_k} \\
&= \frac{y_k^T d_k}{-g_k^T H_k y_k + (s_k^T g_k)(y_k^T H_k y_k)/s_k^T y_k} \\
&= \frac{1}{1 - \frac{(s_k^T B_k s_k)(y_k^T H_k y_k)}{(s_k^T y_k)^2}} \\
&= \frac{1}{1 - \mu}. \quad \square
\end{aligned}$$

(5.2.21) indicates that the parameter ϕ of Broyden class does not change the search direction, but only the length. Hence, we could expect that: any method of Broyden class is, in some degree, independent from the parameter ϕ . Dixon [107] proves: under exact line search, all updates of Broyden class ($\phi_k > \bar{\phi}$) generate the identical points, although for non-quadratic functions.

Theorem 5.2.4 *Let $f : R^n \rightarrow R$ be continuously differentiable, the level set $L(x_0) = \{x \mid f(x) \leq f(x_0)\}$ be bounded, and $H_0 \in R^{n \times n}$ be symmetric and positive definite. Let $\{H_k^\phi\}$ be a sequence generated by Broyden class, where $\phi_k > \bar{\phi}$ and $\bar{\phi}$ is the degenerate value of Broyden class. Assume that H_{k+1}^{BFGS} is an update obtained by applying BFGS update to H_k^ϕ . Then, under exact*

line search, Broyden class of update has the property: for all $k \geq 0$, x_{k+1} and H_{k+1}^{BFGS} are independent from parameters $\phi_0, \phi_1, \dots, \phi_{k-1}$.

Proof. We show this result by induction. For $k = 0$, it is trivially true. Now suppose it is true for $k \geq 0$, i.e., x_{k+1} and H_{k+1}^{BFGS} are independent from $\phi_0, \phi_1, \dots, \phi_{k-1}$. We shall show it is also true for $k + 1$.

From (5.2.21), the direction d_{k+1} generated by Broyden class does not depend on ϕ_k . Since $d_{k+1} \propto -H_{k+1}^{BFGS} g_{k+1}$, by the induction hypothesis, the direction d_{k+1} does not also depend on $\phi_0, \phi_1, \dots, \phi_{k-1}$. Then, by exact line search, $x_{k+2} = x_{k+1} + \alpha_{k+1} d_{k+1}$ does not depend on $\phi_0, \phi_1, \dots, \phi_{k-1}, \phi_k$. Now, from the assumption,

$$H_{k+2}^{BFGS} = \left(I - \frac{s_{k+1} y_{k+1}^T}{s_{k+1}^T y_{k+1}} \right) H_{k+1}^\phi \left(I - \frac{y_{k+1} s_{k+1}^T}{s_{k+1}^T y_{k+1}} \right) + \frac{s_{k+1} s_{k+1}^T}{s_{k+1}^T y_{k+1}}. \quad (5.2.22)$$

Note

$$H_{k+1}^\phi = H_{k+1}^{BFGS} + (\phi_k - 1) v_k v_k^T. \quad (5.2.23)$$

Since

$$\left[I - \frac{s_{k+1} y_{k+1}^T}{s_{k+1}^T y_{k+1}} \right] s_{k+1} = 0,$$

it follows from (5.2.21) that

$$\left[I - \frac{s_{k+1} y_{k+1}^T}{s_{k+1}^T y_{k+1}} \right] v_k = 0. \quad (5.2.24)$$

Then, substituting (5.2.23) into (5.2.22) and using (5.2.24) yield that H_{k+2}^{BFGS} can be defined by use of H_{k+1}^{BFGS} , s_{k+1} , and y_{k+1} . So, by induction hypothesis, H_{k+2}^{BFGS} is independent from $\phi_0, \phi_1, \dots, \phi_k$. We complete the proof. $\square$

To conclude this section, we give a brief introduction to Huang class of updates. Huang [180] presented a wider class of updates than Broyden class. In Broyden class, the update matrix sequence $\{H_k\}$ satisfies symmetricity and quasi-Newton equation, i.e.,

$$H_k^T = H_k \text{ and } H_{k+1} y_k = s_k. \quad (5.2.25)$$

However, in Huang class, the symmetricity condition is removed, and the update matrix $\{H_k\}$ is required to obey

$$H_{k+1} y_k = \rho s_k, \quad (5.2.26)$$

which is said to be a generalized quasi-Newton equation or a generalized quasi-Newton condition, where ρ is a parameter.

Huang class of updates can be described as follows:

$$H_{k+1} = H_k + s_k u_k^T + H_k y_k v_k^T, \quad (5.2.27)$$

where u_k and v_k satisfy

$$u_k = a_{11}s_k + a_{12}H_k^T y_k, \quad (5.2.28)$$

$$v_k = a_{21}s_k + a_{22}H_k^T y_k, \quad (5.2.29)$$

$$u_k^T y_k = \rho, \quad (5.2.30)$$

$$v_k^T y_k = -1. \quad (5.2.31)$$

There are five parameters $a_{11}, a_{12}, a_{21}, a_{22}$ and ρ , in which three parameters are free. Hence, in fact, Huang class of update depends on three parameters. In particular, if requiring $\{H_k\}$ symmetric and setting $\rho = 1$, then Huang class is just Broyden class. This means that Broyden class is a subclass of Huang class.

The main properties of Huang class of update are as follows:

- For positive definite and quadratic functions, Huang class generates conjugate directions and has quadratic termination property. All methods of Huang class generate the identical points.
- For general functions, the sequence generated by Huang class only depends on the parameter ρ .

Based on our experience, the generalized quasi-Newton equation (5.2.26) is important to present a good quasi-Newton method. The parameter ρ will play a big role on the iterative sequence and the properties of algorithms.

5.3 Global Convergence of Quasi-Newton Methods

In this section we discuss the global convergence for quasi-Newton methods. The global properties of quasi-Newton methods were established by Powell [262] and Powell [265]. These results have been extended to restricted Broyden's class by Byrd, Nocedal and Yuan [47]. We will study the global convergence of quasi-Newton methods under exact line search and inexact line search respectively in §5.3.1 and §5.3.2.

In the discussion of this section, we need the following assumptions:

Assumption 5.3.1 (a) $f : R^n \rightarrow R$ is twice continuously differentiable on convex set D .

(b) $f(x)$ is uniformly convex, i.e., there exist positive constants m and M such that for all $x \in L(x) = \{x | f(x) \leq f(x_0)\}$, which is convex, we have

$$m\|u\|^2 \leq u^T \nabla^2 f(x) u \leq M\|u\|^2, \quad \forall u \in R^n. \quad (5.3.1)$$

The assumption (b) implies that $\nabla^2 f(x)$ is positive definite on $L(x)$, and that f has a unique minimizer x^* in $L(x)$.

5.3.1 Global Convergence under Exact Line Search

We begin the discussion in case of exact line search.

Let

$$\bar{G} = \int_0^1 \nabla^2 f(x_k + \tau s_k) d\tau, \quad (5.3.2)$$

then we have from Taylor's theorem that

$$y_k = \bar{G} s_k. \quad (5.3.3)$$

Immediately, we have

$$m \leq \frac{y_k^T s_k}{\|s_k\|^2} = \frac{s_k^T \bar{G} s_k}{\|s_k\|^2} \leq M \quad (5.3.4)$$

and

$$\frac{1}{M} \leq \frac{\|s_k\|^2}{y_k^T s_k} \leq \frac{1}{m}. \quad (5.3.5)$$

Since also

$$\frac{\|y_k\|^2}{s_k^T y_k} = \frac{s_k^T \bar{G}_k^2 s_k}{s_k^T \bar{G}_k s_k},$$

if we let $z_k = \bar{G}_k^{\frac{1}{2}} s_k$, then

$$\frac{\|y_k\|^2}{s_k^T y_k} = \frac{z_k^T \bar{G}_k z_k}{z_k^T z_k} \leq M. \quad (5.3.6)$$

In addition, we have

$$\|y_k\| \leq \|\bar{G}\| \|s_k\|, \quad \|s_k\| \leq \|\bar{G}_k^{-1}\| \|y_k\|$$

which give

$$\frac{\|y_k\|}{\|s_k\|} \leq M \quad (5.3.7)$$

and

$$\frac{\|s_k\|}{\|y_k\|} \leq \frac{1}{m}. \quad (5.3.8)$$

Therefore, from the above discussion, we have

Lemma 5.3.2 *Let $f : R^n \rightarrow R$ satisfy Assumption 5.3.1. Then*

$$\frac{\|s_k\|}{\|y_k\|}, \frac{\|y_k\|}{\|s_k\|}, \frac{s_k^T y_k}{\|s_k\|^2}, \frac{s_k^T y_k}{\|y_k\|^2}, \frac{\|y_k\|^2}{s_k^T y_k}$$

are bounded.

Lemma 5.3.3 *Under exact line search, $\sum \|s_k\|^2$ and $\sum \|y_k\|^2$ are convergent.*

Proof. Let $\psi(\tau) = f(x_{k+1} - \tau s_k)$. From (5.3.1), it follows that $\psi''(\tau) \geq m\|s_k\|^2$. Note that the exact line search gives $\psi'(0) = 0$. Then we have

$$\psi(\tau) \geq \psi(0) + \frac{1}{2}m\|s_k\|^2\tau^2.$$

Taking $\tau = 1$, we deduces

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2}m\|s_k\|^2.$$

By summing this expression we have

$$\sum_{k=0}^{\infty} \|s_k\|^2 \leq 2\{f(x_0) - f(x^*)\}/m,$$

which implies $\sum \|s_k\|^2$ is convergent, where $f(x^*)$ is the minimum of $f(x)$.

By Lemma 5.3.2, we also obtain that $\sum \|y_k\|^2$ is convergent. $\square$

Lemma 5.3.4 *For all vectors x , the inequality*

$$\|g(x)\|^2 \geq m[f(x) - f(x^*)] \quad (5.3.9)$$

holds, where $f(x^)$ is the minimum of $f(x)$.*

Proof. Since the function

$$\psi(\tau) = f(x + \tau(x^* - x)), \quad (0 \leq \tau \leq 1)$$

is a convex function, then

$$f(x + \tau(x^* - x)) \geq f(x) + \tau(x^* - x)^T g(x).$$

In particular, set $\tau = 1$, then we have

$$f(x) - f(x^*) \leq -(x^* - x)^T g(x) \leq \|g(x)\| \|x^* - x\|. \quad (5.3.10)$$

By (5.3.5) and Cauchy-Schwartz inequality, we deduce

$$\begin{aligned} \|x^* - x\|^2 &\leq (x^* - x)^T (g(x^*) - g(x)) / m \\ &\leq \|x^* - x\| \|g(x^*) - g(x)\| / m, \end{aligned} \quad (5.3.11)$$

which gives

$$\|x^* - x\| \leq \|g(x^*) - g(x)\| / m = \|g(x)\| / m. \quad (5.3.12)$$

Substituting (5.3.12) into (5.3.10) establishes (5.3.9). $\square$

Theorem 5.3.5 *Suppose that $f(x)$ satisfies Assumption 5.3.1. Then, under exact line search, the sequence $\{x_k\}$ generated by DFP method converges to the minimizer x^* of f .*

Proof. Consider DFP formula of inverse Hessian approximation

$$H_{k+1} = H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \frac{s_k s_k^T}{s_k^T y_k} \quad (5.3.13)$$

and DFP formula of Hessian approximation

$$B_{k+1} = \left(I - \frac{y_k s_k^T}{s_k^T y_k} \right) B_k \left(I - \frac{s_k y_k^T}{s_k^T y_k} \right) + \frac{y_k y_k^T}{s_k^T y_k}. \quad (5.3.14)$$

Obviously, $B_{k+1} H_{k+1} = I$. By computing the trace of (5.3.14), we have

$$\text{Tr}(B_{k+1}) = \text{Tr}(B_k) - 2 \frac{s_k^T B_k y_k}{s_k^T y_k} + \frac{(s_k^T B_k s_k)(y_k^T y_k)}{(s_k^T y_k)^2} + \frac{y_k^T y_k}{s_k^T y_k}. \quad (5.3.15)$$

The middle two terms can be written as

$$\begin{aligned}
 & -2 \frac{s_k^T B_k y_k}{s_k^T y_k} + \frac{(s_k^T B_k s_k)(y_k^T y_k)}{(s_k^T y_k)^2} \\
 = & \alpha_k \left[\frac{2g_k^T y_k}{s_k^T y_k} + \frac{(-g_k^T s_k)(y_k^T y_k)}{(s_k^T y_k)^2} \right] \\
 = & \alpha_k \frac{2g_k^T y_k + y_k^T y_k}{s_k^T y_k} \\
 = & \frac{\|g_{k+1}\|^2 - \|g_k\|^2}{g_k^T H_k g_k}. \tag{5.3.16}
 \end{aligned}$$

Since $g_{k+1}^T s_k = 0$, then

$$\begin{aligned}
 g_{k+1}^T H_{k+1} g_{k+1} &= g_{k+1}^T \left[H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} \right] g_{k+1} \\
 &= g_k^T \left[H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} \right] g_k \\
 &= g_k^T \left[H_k - \frac{H_k g_k g_k^T H_k}{y_k^T H_k y_k} \right] g_k \\
 &= \frac{(g_k^T H_k g_k)(g_{k+1}^T H_k g_{k+1})}{g_k^T H_k g_k + g_{k+1}^T H_k g_{k+1}}.
 \end{aligned}$$

By finding the inverse number of the above expression, we get

$$\frac{1}{g_{k+1}^T H_{k+1} g_{k+1}} = \frac{1}{g_{k+1}^T H_k g_{k+1}} + \frac{1}{g_k^T H_k g_k}. \tag{5.3.17}$$

Using (5.3.16) and (5.3.17), then (5.3.15) becomes

$$\begin{aligned}
 \text{Tr}(B_{k+1}) &= \text{Tr}(B_k) + \frac{\|g_{k+1}\|^2}{g_{k+1}^T H_{k+1} g_{k+1}} - \frac{\|g_k\|^2}{g_k^T H_k g_k} \\
 &\quad - \frac{\|g_{k+1}\|^2}{g_{k+1}^T H_k g_{k+1}} + \frac{\|y_k\|^2}{s_k^T y_k}. \tag{5.3.18}
 \end{aligned}$$

By recurrence, we obtain

$$\begin{aligned}
 \text{Tr}(B_{k+1}) &= \text{Tr}(B_0) + \frac{\|g_{k+1}\|^2}{g_{k+1}^T H_{k+1} g_{k+1}} - \frac{\|g_0\|^2}{g_0^T H_0 g_0} \\
 &\quad - \sum_{j=0}^k \frac{\|g_{j+1}\|^2}{g_{j+1}^T H_j g_{j+1}} + \sum_{j=0}^k \frac{\|y_j\|^2}{s_j^T y_j}. \tag{5.3.19}
 \end{aligned}$$

Therefore, by Lemma 5.3.2, there exists a positive number M which is independent of k , such that

$$\text{Tr}(B_{k+1}) \leq \frac{\|g_{k+1}\|^2}{g_{k+1}^T H_{k+1} g_{k+1}} - \sum_{j=0}^k \frac{\|g_{j+1}\|^2}{g_{j+1}^T H_j g_{j+1}} + Mk. \quad (5.3.20)$$

In the left part, we will prove that if the theorem does not hold, then the sum of the last two terms in (5.3.20) is negative.

Now consider the trace of H_{k+1} . From (5.3.13), we have

$$\text{Tr}(H_{k+1}) = \text{Tr}(H_0) - \sum_{j=0}^k \frac{\|H_j y_j\|^2}{y_j^T H_j y_j} + \sum_{j=0}^k \frac{\|s_j\|^2}{s_j^T y_j}. \quad (5.3.21)$$

Since H_{k+1} is positive definite, the right-hand side of (5.3.21) is positive. By Lemma 5.3.2, there exists $m > 0$ which is independent of k , such that

$$\sum_{j=0}^k \frac{\|H_j y_j\|^2}{y_j^T H_j y_j} < \frac{k}{m}. \quad (5.3.22)$$

Note that

$$(y_j^T H_j y_j)^2 \leq \|H_j y_j\|^2 \|y_j\|^2 \quad (5.3.23)$$

and

$$\begin{aligned} y_j^T H_j y_j &= g_{j+1}^T H_j g_{j+1} + g_j^T H_j g_j + 2g_{j+1}^T d_j \\ &> g_{j+1}^T H_j g_{j+1} \end{aligned} \quad (5.3.24)$$

by the positive definiteness of H_j and exact line search, then by using (5.3.24), (5.3.23) and (5.3.22) in turn, we obtain

$$\sum_{j=0}^k \frac{g_{j+1}^T H_j g_{j+1}}{\|y_j\|^2} \leq \sum_{j=0}^k \frac{y_j^T H_j y_j}{\|y_j\|^2} \leq \sum_{j=0}^k \frac{\|H_j y_j\|^2}{y_j^T H_j y_j} \leq \frac{k}{m}. \quad (5.3.25)$$

By using Cauchy-Schwartz inequality and (5.3.25)

$$\begin{aligned} \sum_{j=0}^k \frac{\|g_{j+1}\|^2}{g_{j+1}^T H_j g_{j+1}} &\geq \left(\sum_{j=0}^k \frac{\|g_{j+1}\|}{\|y_j\|} \right)^2 \bigg/ \sum_{j=0}^k \frac{g_{j+1}^T H_j g_{j+1}}{\|y_j\|^2} \\ &\geq \frac{m}{k} \left(\sum_{j=0}^k \frac{\|g_{j+1}\|}{\|y_j\|} \right)^2. \end{aligned} \quad (5.3.26)$$

Now suppose that the theorem is not true, that is, there exists $\delta > 0$ such that for all sufficiently large k ,

$$\|g_k\| \geq \delta. \quad (5.3.27)$$

Also, by (5.3.11) and Theorem 2.2.9, there exists a constant $\eta > 0$ such that

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2}\eta\|s_k\|^2,$$

which gives $\|s_k\| \rightarrow 0$ and further $\|y_k\| \rightarrow 0$. Then, by (5.3.26) and (5.3.27), we deduce, for k sufficiently large, that

$$\sum_{j=0}^k \frac{\|g_{j+1}\|^2}{g_{j+1}^T H_j g_{j+1}} > Mk. \quad (5.3.28)$$

The above inequality implies that the sum of the last two terms in (5.3.20) is negative.

By (5.3.28) and (5.3.20), we immediately obtain

$$\text{Tr}(B_{k+1}) < \frac{\|g_{k+1}\|^2}{g_{k+1}^T H_{k+1} g_{k+1}}. \quad (5.3.29)$$

Note that, for a symmetric and positive definite matrix, the inverse of trace is the lower bound of the least eigenvalue of inverse of the matrix. Then, it follows from (5.3.29) that

$$\frac{g_{k+1}^T H_{k+1} g_{k+1}}{\|g_{k+1}\|^2} < \mu, \quad (5.3.30)$$

where μ is the lower bound of the least eigenvalue of H_{k+1} . However, from Theorem 1.2.10 on the property of Rayleigh quotient, we have

$$\frac{g_{k+1}^T H_{k+1} g_{k+1}}{\|g_{k+1}\|^2} > \mu, \quad (5.3.31)$$

which contradicts (5.3.30). This contradiction proves that $\{x_k\}$ converges to x^* and that our theorem holds. $\square$

5.3.2 Global Convergence under Inexact Line Search

Now, we turn to study the global convergence of BFGS method under inexact line search.

Let us rewrite BFGS method as follows:

$$x_{k+1} = x_k + s_k = x_k + \alpha_k d_k = x_k - \alpha_k B_k^{-1} g_k, \quad (5.3.32)$$

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{s_k^T y_k}. \quad (5.3.33)$$

Theorem 5.3.6 *Let x_0 and B_0 be a starting point and a symmetric positive definite initial matrix, respectively. Suppose that $f(x)$ satisfies Assumption 5.3.1. Then, under Wolfe-Powell inexact line search (2.5.3) and (2.5.7), the sequence $\{x_k\}$ generated by BFGS method converges to the minimizer x^* of f .*

Proof. By computing the trace and determinant of BFGS formula (5.3.33), we obtain that

$$\text{Tr}(B_{k+1}) = \text{Tr}(B_k) - \frac{\|B_k s_k\|}{s_k^T B_k s_k} + \frac{\|y_k\|^2}{y_k^T s_k} \quad (5.3.34)$$

and

$$\det(B_{k+1}) = \det(B_k) \frac{y_k^T s_k}{s_k^T B_k s_k}. \quad (5.3.35)$$

Let us define

$$m_k = \frac{y_k^T s_k}{s_k^T s_k}, \quad M_k = \frac{y_k^T y_k}{y_k^T s_k}. \quad (5.3.36)$$

It follows from (5.3.4) and (5.3.6) that

$$m \leq m_k \leq M, \quad m \leq M_k \leq M. \quad (5.3.37)$$

Let us also define

$$\cos \theta_k = \frac{s_k^T B_k s_k}{\|s_k\| \|B_k s_k\|}, \quad q_k = \frac{s_k^T B_k s_k}{s_k^T s_k}. \quad (5.3.38)$$

We then obtain that

$$\frac{\|B_k s_k\|^2}{s_k^T B_k s_k} = \frac{\|B_k s_k\|^2 \|s_k\|^2}{(s_k^T B_k s_k)^2} \frac{s_k^T B_k s_k}{\|s_k\|^2} = \frac{q_k}{\cos^2 \theta_k}. \quad (5.3.39)$$

In addition, we have from (5.3.36) that

$$\det(B_{k+1}) = \det(B_k) \frac{y_k^T s_k}{s_k^T s_k} \frac{s_k^T s_k}{s_k^T B_k s_k} = \det(B_k) \frac{m_k}{q_k}. \quad (5.3.40)$$

Now we introduce the following function of a positive definite matrix B_k :

$$\psi(B_k) = \text{Tr}(B_k) - \ln(\det(B_k)), \quad (5.3.41)$$

where $\ln(\cdot)$ denotes the natural logarithm. It is not difficult to show that $\psi(B_k) > 0$. By using (5.3.34)–(5.3.41), we have that

$$\begin{aligned} \psi(B_{k+1}) &= \text{Tr}(B_k) + M_k - \frac{q_k}{\cos^2 \theta_k} - \ln(\det(B_k)) - \ln m_k + \ln q_k \\ &= \psi(B_k) + (M_k - \ln m_k - 1) \\ &\quad + \left[1 - \frac{q_k}{\cos^2 \theta_k} + \ln \frac{q_k}{\cos^2 \theta_k} \right] + \ln \cos^2 \theta_k. \end{aligned} \quad (5.3.42)$$

Note that the function $h(t) = 1 - t + \ln t \leq 0$ for all $t > 0$. Hence the term inside the square brackets is nonpositive, and thus by summing both sides of (5.3.42), we have

$$0 < \psi(B_{k+1}) \leq \psi(B_1) + ck + \sum_{j=1}^k \ln \cos^2 \theta_j, \quad (5.3.43)$$

where the constant $c = M - \ln m - 1$ is assumed to be positive without loss of generality.

From Theorem 2.5.5, we have

$$\lim_{k \rightarrow \infty} \|g_k\| \cos \theta_k = 0. \quad (5.3.44)$$

If θ_k is bounded away from 90° , there is a positive constant δ such that

$$\cos \theta_k > \delta > 0, \quad \text{for } k \text{ sufficiently large,}$$

and thus we have our result.

Now assume, by contradiction, that $\cos \theta_k \rightarrow 0$. Then there exists $k_1 > 0$ such that for all $j > k_1$, we have

$$\ln \cos^2 \theta_j < -2c,$$

where c is the constant defined above.

By using (5.3.43), we deduce, for all $k > k_1$, that

$$\begin{aligned} 0 &< \psi(B_1) + ck + \sum_{j=1}^{k_1} \ln \cos^2 \theta_j + \sum_{j=k_1+1}^k (-2c) \\ &= \psi(B_1) + \sum_{j=1}^{k_1} \ln \cos^2 \theta_j + 2ck_1 - ck \\ &< 0, \end{aligned}$$

which gives a contradiction. Therefore the assumption $\cos \theta_j \rightarrow 0$ is not true, and there exists a subsequence $\{j_k\}$ such that

$$\{\cos \theta_{j_k}\} \geq \delta > 0,$$

which means

$$\liminf \|\nabla f(x_k)\| = 0. \quad (5.3.45)$$

Since the problem is strong convex, then (5.3.45) implies $x_k \rightarrow x^*$. $\square$

5.4 Local Convergence of Quasi-Newton Methods

In this section, we discuss local convergence of quasi-Newton methods. The convergence analysis in this section mainly makes use of Broyden, Dennis, and Moré [29], Dennis and Moré [91], Dennis and Moré [92], Nocedal and Wright [233] and others. In §5.4.1 we first consider solving $F(x) = 0$. The necessary and sufficient condition of superlinear convergence for solving $F(x) = 0$ is given in Theorem 5.4.3, which is basic and the most important in convergence analysis for quasi-Newton methods. Theorem 5.4.4 is a corollary of Theorem 5.4.3, and Lemma 5.4.5 gives the geometry of superlinear convergence for quasi-Newton methods. Then, we generalize the above results to minimization problems. We give superlinear convergence results in the case of basic iteration, exact line search and inexact line search respectively in Theorem 5.4.6, Theorem 5.4.7 and Theorem 5.4.8. In §5.4.2, we give linear convergence of general quasi-Newton methods by means of the bounded deterioration principle. In §5.4.3 the linear and superlinear convergence of SR1 method is established. In §5.5.4, we discuss the linear convergence of DFP method. In §5.4.5 and 5.4.6. we give the superlinear convergence results of BFGS and DFP methods respectively by different techniques. Finally, in §5.4.7, the local convergence of Broyden's class methods is discussed.

5.4.1 Superlinear Convergence of General Quasi-Newton Methods

First, we consider

$$F(x) = 0, \quad (5.4.1)$$

where $F : R^n \rightarrow R^n$ is a mapping. In convergence analysis, we often need the following assumption.

Assumption 5.4.1 (a) $F : R^n \rightarrow R^n$ is continuously differentiable on an open convex set $D \subset R^n$.

(b) There is $x^* \in D$ with $F(x^*) = 0$ and $F'(x^*)$ nonsingular.

(c) F' is Lipschitzian at x^* , i.e., there is a constant γ such that

$$\|F'(x) - F'(x^*)\| \leq \gamma \|x - x^*\|, \quad x \in D.$$

Second, we consider the minimization problem

$$\min_{x \in R^n} f(x). \quad (5.4.2)$$

If in Assumption 5.4.1, we replace $F(x)$ and $F'(x)$ by $g(x)$ and $\nabla^2 f(x)$ respectively, we get the following assumption for optimization problem (5.4.2):

Assumption 5.4.2 (a) $f : R^n \rightarrow R$ is twice continuously differentiable on an open convex set $D \subset R^n$.

(b) There is a strong local minimizer $x^* \in D$ with $\nabla^2 f(x^*)$ symmetric and positive definite.

(c) There is a neighborhood $N(x^*, \varepsilon)$ of x^* such that

$$\|\nabla^2 f(\bar{x}) - \nabla^2 f(x)\| \leq \gamma \|\bar{x} - x\|, \quad \forall x, \bar{x} \in N(x^*, \varepsilon).$$

Superlinear Convergence: Nonlinear System

We begin our discussion on a basic necessary and sufficient condition of superlinear convergence for a nonlinear system.

Theorem 5.4.3 Let $F : R^n \rightarrow R^n$ satisfy (a) and (b) in Assumption 5.4.1. Let $\{B_k\}$ be a sequence of nonsingular matrices. Suppose, for $x_0 \in D$, that the iterates generated by

$$x_{k+1} = x_k - B_k^{-1} F(x_k) \quad (5.4.3)$$

remain in D . $x_k \neq x^*$ ($\forall k \geq 0$). Suppose also that $\{x_k\}$ converges to x^* . Then $\{x_k\}$ converges to x^* at a superlinear rate if and only if

$$\lim_{k \rightarrow +\infty} \frac{\| [B_k - F'(x^*)](x_{k+1} - x_k) \|}{\|x_{k+1} - x_k\|} = 0. \quad (5.4.4)$$

Proof. Our idea is to prove the following equivalence:

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{\| [B_k - F'(x^*)]s_k \|}{\|s_k\|} = 0 &\Leftrightarrow \lim_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|s_k\|} = 0 \\ &\Leftrightarrow \lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0, \end{aligned} \quad (5.4.5)$$

where $s_k = x_{k+1} - x_k$.

First, suppose (5.4.4) holds. By (5.4.3), we have

$$\begin{aligned} &[B_k - F'(x^*)](x_{k+1} - x_k) \\ &= -F(x_k) - F'(x^*)(x_{k+1} - x_k) \\ &= [F(x_{k+1}) - F(x_k) - F'(x^*)(x_{k+1} - x_k)] - F(x_{k+1}). \end{aligned} \quad (5.4.6)$$

By taking the norm, dividing by $\|s_k\|$, and using Theorem 1.2.24, we obtain

$$\begin{aligned} \frac{\|F(x_{k+1})\|}{\|s_k\|} &\leq \frac{\|(B_k - F'(x^*))s_k\|}{\|s_k\|} + \frac{\|F(x_{k+1}) - F(x_k) - F'(x^*)s_k\|}{\|s_k\|} \\ &\leq \frac{\|(B_k - F'(x^*))s_k\|}{\|s_k\|} + \frac{\gamma}{2}(\|x_k - x^*\| + \|x_{k+1} - x^*\|). \end{aligned} \quad (5.4.7)$$

Since $\lim_{k \rightarrow \infty} x_k = x^*$, it follows from (5.4.4) that

$$\lim_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|s_k\|} = 0. \quad (5.4.8)$$

Since also $\lim_{k \rightarrow \infty} \|s_k\| = 0$, we have

$$F(x^*) = \lim_{k \rightarrow \infty} F(x_k) = 0.$$

Noting that $F'(x^*)$ is nonsingular, it follows from Theorem 1.2.25 that there is a $\beta > 0$ and $k_0 \geq 0$ such that $\forall k \geq k_0$, we have

$$\|F(x_{k+1})\| = \|F(x_{k+1}) - F(x^*)\| \geq \beta \|x_{k+1} - x^*\|.$$

Thus

$$\frac{\|F(x_{k+1})\|}{\|x_{k+1} - x_k\|} \geq \frac{\beta\|x_{k+1} - x^*\|}{\|x_{k+1} - x^*\| + \|x_k - x^*\|} = \beta \frac{r_k}{1 + r_k}, \quad (5.4.9)$$

where

$$r_k = \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|}.$$

Combining (5.4.8) and (5.4.9) implies that

$$\frac{r_k}{1 + r_k} \rightarrow 0$$

which gives

$$\lim_{k \rightarrow \infty} r_k = 0, \quad (5.4.10)$$

i.e., the sequence $\{x_k\}$ is convergent to x^* superlinearly.

Conversely, assume that $\{x_k\}$ converges superlinearly to x^* and $F(x^*) = 0$. By Theorem 1.2.25, there exist $\bar{\beta} > 0$ and $k_0 \geq 0$, such that $\forall k \geq k_0$, we have

$$\|F(x_{k+1})\| \leq \bar{\beta}\|x_{k+1} - x^*\|.$$

Since $\{x_k\}$ is convergent superlinearly, we have

$$\begin{aligned} 0 &= \lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \geq \lim_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\bar{\beta}\|x_k - x^*\|} \\ &= \lim_{k \rightarrow \infty} \frac{1}{\bar{\beta}} \frac{\|F(x_{k+1})\|}{\|x_{k+1} - x_k\|} \frac{\|x_{k+1} - x_k\|}{\|x_k - x^*\|}. \end{aligned}$$

By use of Theorem 1.5.2 giving $\lim_{k \rightarrow \infty} \|x_{k+1} - x_k\|/\|x_k - x^*\| = 1$, we obtain

$$\lim_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|x_{k+1} - x_k\|} = 0,$$

which gives (5.4.4) by means of (5.4.6). $\square$

Theorem 5.4.3 indicates that if B_k converges to $F'(x^*)$ along the direction s_k , then quasi-Newton methods converge superlinearly. This theorem is very important in analysis of quasi-Newton methods. Equation (5.4.4) is called the Dennis-Moré characterization of superlinear convergence. The following theorem shows, for the iteration (5.4.11), that the method is convergent superlinearly if and only if the sequence of steplength factors $\{\alpha_k\}$ converges to 1. The proof of Theorem 5.4.4 is completed by use of Theorem 5.4.3.

Theorem 5.4.4 *Let $F : R^n \rightarrow R^n$ satisfy the assumptions of Theorem 5.4.3. Let $\{B_k\}$ be a sequence of nonsingular matrices. Suppose, for $x_0 \in D$, that the iteration*

$$x_{k+1} = x_k - \alpha_k B_k^{-1} F(x_k) \quad (5.4.11)$$

remains in D and $\{x_k\}$ converges to x^ . If (5.4.4) holds, then $\{x_k\}$ converges to x^* superlinearly and $F(x^*) = 0$ if and only if $\{\alpha_k\}$ converges to 1.*

Proof. Necessity. Suppose that $\{x_k\}$ converges to x^* superlinearly and $F(x^*) = 0$. By Theorem 5.4.3, we have

$$\lim_{k \rightarrow \infty} \frac{\|[\alpha_k^{-1} B_k - F'(x^*)](x_{k+1} - x_k)\|}{\|x_{k+1} - x_k\|} = 0. \quad (5.4.12)$$

So, (5.4.4) implies that

$$\lim_{k \rightarrow \infty} \|(\alpha_k^{-1} - 1)B_k(x_{k+1} - x_k)\|/\|x_{k+1} - x_k\| = 0.$$

Since $B_k(x_{k+1} - x_k) = -\alpha_k F(x_k)$, the above equality can be written as

$$\lim_{k \rightarrow \infty} \|(\alpha_k - 1)F(x_k)\|/\|x_{k+1} - x_k\| = 0. \quad (5.4.13)$$

Noting that $F'(x^*)$ is nonsingular, it follows from Theorem 1.2.25 that there exists $\beta > 0$ such that $\|F(x_k)\| \geq \beta\|x_k - x^*\|$. Then, from (5.4.13), we obtain

$$\lim_{k \rightarrow \infty} |\alpha_k - 1| \frac{\beta\|x_k - x^*\|}{\|x_{k+1} - x_k\|} = 0. \quad (5.4.14)$$

Since also $\{x_k\}$ is convergent superlinearly, i.e., $\lim_{k \rightarrow \infty} \|x_{k+1} - x_k\|/\|x_k - x^*\| = 1$, we obtain immediately from (5.4.14) that $\{\alpha_k\} \rightarrow 1$.

Sufficiency. Suppose that $\{\alpha_k\} \rightarrow 1$. It follows from (5.4.4) that (5.4.12) holds. Therefore, from Theorem 5.4.3, we obtain that $\{x_k\}$ converges to x^* superlinearly and $F(x^*) = 0$. $\square$

This theorem suggests that when a method is required to be superlinearly convergent, we should ask for $\alpha_k \rightarrow 1$ as $k \rightarrow \infty$.

Next, we interpret the geometry of superlinear convergence of quasi-Newton methods, which is an equivalent and geometric representation of (5.4.4).

Let $s_k = x_{k+1} - x_k$. Let also Newton's iteration be $s_k^N = -F'(x_k)^{-1}F(x_k)$. Since $F(x_k) = -B_k s_k$, then

$$s_k - s_k^N = s_k + F'(x_k)^{-1}F(x_k) = F'(x_k)^{-1}[F'(x_k) - B_k]s_k. \quad (5.4.15)$$

By use of Assumption 5.4.1, we have that $\|F'(x_k)^{-1}\|$ is bounded above for x_k sufficiently close to x^* . Thus,

$$F'(x_k)^{-1}[F'(x_k) - B_k]s_k = O(\|[F'(x_k) - B_k]s_k\|) = o(\|s_k\|),$$

where we have used (5.4.4). Therefore (5.4.15) is equivalent to

$$\lim_{k \rightarrow \infty} \frac{\|s_k - s_k^N\|}{\|s_k\|} = 0. \quad (5.4.16)$$

The above (5.4.16) indicates that when $\{x_k\}$ converges superlinearly, the relative error of s_k should tend to zero. It is not difficult to prove that (5.4.16) is equivalent to the fact that s_k tends to s_k^N in both direction and length. For this, we introduce the following lemma.

Lemma 5.4.5 *Let $u, v \in R^n, u, v \neq 0$, and $\alpha \in (0, 1)$. If $\|u - v\| \leq \alpha\|u\|$, then $\langle u, v \rangle$ is positive and*

$$\left| 1 - \frac{\|v\|}{\|u\|} \right| \leq \alpha, \quad 1 - \left(\frac{\langle u, v \rangle}{\|u\|\|v\|} \right)^2 \leq \alpha^2. \quad (5.4.17)$$

Conversely, if $\langle u, v \rangle$ is positive and (5.4.17) holds, then

$$\|u - v\| \leq 3\alpha\|u\|. \quad (5.4.18)$$

Proof. First, assume that $\|u - v\| \leq \alpha\|u\|$. Then

$$\left| \frac{\|u\| - \|v\|}{\|u\|} \right| \leq \frac{\|u - v\|}{\|u\|} \leq \alpha,$$

which implies that the first inequality in (5.4.17) holds.

Let $\omega = \langle u, v \rangle / (\|u\|\|v\|)$. Since

$$\|v\|^2 - 2\langle u, v \rangle + \frac{\langle u, v \rangle^2}{\|v\|^2} = \left[\|v\| - \frac{\langle u, v \rangle}{\|v\|} \right]^2 \geq 0,$$

then

$$\|v\|^2 - 2\langle u, v \rangle \geq -\frac{\langle u, v \rangle^2}{\|v\|^2}.$$

So,

$$\|u - v\|^2 = \|u\|^2 - 2\|u\|\|v\|\omega + \|v\|^2 \quad (5.4.19)$$

$$\begin{aligned} &\geq \|u\|^2 - \frac{\langle u, v \rangle^2}{\|v\|^2} \\ &= \|u\|^2(1 - \omega^2). \end{aligned} \quad (5.4.20)$$

Therefore,

$$1 - \omega^2 \leq \frac{\|u - v\|^2}{\|u\|^2} \leq \alpha^2$$

giving the second inequality of (5.4.17). In addition, if $\omega \leq 0$, it follows from (5.4.19) that $\|u - v\| \geq \|u\|$, and therefore $\alpha \geq 1$. Hence, if $\alpha < 1$, we have that $\langle u, v \rangle$ is positive.

Conversely, if $\langle u, v \rangle$ is positive and (5.4.17) holds, then by using (5.4.17) and some manipulations, we obtain

$$\begin{aligned} \|u - v\|^2 &= (\|u\| - \|v\|)^2 + 2(1 - \omega)\|u\|\|v\| \\ &\leq \alpha^2\|u\|^2[1 + 2(1 + \alpha)], \end{aligned}$$

which gives (5.4.18) since $\alpha < 1$. $\square$

If (5.4.16) holds, we have, for given $\varepsilon \in (0, 1)$, that

$$\|s_k - s_k^N\| \leq \varepsilon \|s_k\|$$

when $k \geq k_0$. So, by Lemma 5.4.5, it follows that if $\langle s_k, s_k^N \rangle > 0$ and $k \geq k_0$, we have

$$\left| 1 - \frac{\|s_k^N\|}{\|s_k\|} \right| \leq \varepsilon$$

and

$$1 - \left(\frac{\langle s_k, s_k^N \rangle}{\|s_k\|\|s_k^N\|} \right)^2 \leq \varepsilon^2.$$

They show that (5.4.16) is equivalent to

$$\lim_{k \rightarrow \infty} \frac{\|s_k^N\|}{\|s_k\|} = \lim_{k \rightarrow \infty} \left\langle \frac{s_k}{\|s_k\|}, \frac{s_k^N}{\|s_k^N\|} \right\rangle = 1. \quad (5.4.21)$$

Therefore we have a conclusion: the necessary and sufficient condition of superlinear convergence of quasi-Newton method is that s_k approaches s_k^N in both length and direction.

Superlinear Convergence: Minimization Problem

Next, we consider minimization problem (5.4.2) and discuss the superlinear convergence in the case of basic iteration, exact line search, and inexact line search.

Completely similar to Theorem 5.4.3, for minimization problem (5.4.2), we have

Theorem 5.4.6 *Let $f : R^n \rightarrow R$ satisfy the assumptions (a) and (b) in Assumption 5.4.2. Consider iteration sequence*

$$x_{k+1} = x_k - B_k^{-1}g_k, \quad (5.4.22)$$

where $\{B_k\}$ is a sequence of symmetric and positive definite matrices. Assume that $\{x_k\}$ converges to x^* . Then $\{x_k\}$ converges superlinearly to x^* if and only if

$$\lim_{k \rightarrow \infty} \frac{\|[B_k - \nabla^2 f(x^*)]s_k\|}{\|s_k\|} = 0. \quad (5.4.23)$$

Proof. The proof is the same as for Theorem 5.4.3. $\square$

The following Theorem 5.4.7 shows the superlinear convergence of quasi-Newton method in the case of exact line search.

Theorem 5.4.7 *Let $f : R^n \rightarrow R$ satisfy conditions (a) and (b) in Assumption 5.4.2. Suppose $\{B_k\}$ is a sequence of symmetric and positive definite matrices. Consider, for a given $x_0 \in D$, the iteration*

$$x_{k+1} = x_k - \alpha_k B_k^{-1}g_k, \quad (5.4.24)$$

where α_k is determined by exact line search. If the sequence $\{x_k\}$ provided by (5.4.24) remains in D and $x_k \neq x^*$ ($\forall k \geq 0$), and if $x_k \rightarrow x^*$, then when

$$\lim_{k \rightarrow \infty} \frac{\|[B_k - \nabla^2 f(x^*)]s_k\|}{\|s_k\|} = 0, \quad (5.4.25)$$

we have $\alpha_k \rightarrow 1$ and $g(x^*) = 0$, hence $\{x_k\}$ converges to x^* superlinearly.

Proof. It is enough to prove $\alpha_k \rightarrow 1$ when (5.4.25) holds. Other conclusions can be obtained direct from Theorem 5.4.4.

Since $\nabla^2 f(x^*)$ is positive definite, there exists $m > 0$ such that

$$s_k^T \nabla^2 f(x^*) s_k \geq m \|s_k\|^2.$$

Therefore we only need to prove

$$(\alpha_k - 1)s_k^T \nabla^2 f(x^*)s_k = o(\|s_k\|^2). \quad (5.4.26)$$

From (1.2.111), we have

$$\|g_{k+1} - g_k - \nabla^2 f(x^*)s_k\| \leq \max_{0 \leq t \leq 1} \|\nabla^2 f(x_k + ts_k) - \nabla^2 f(x^*)\| \|s_k\|.$$

Then from $x_k \rightarrow x^*$ and the continuity of $\nabla^2 f(x)$, we obtain

$$\|g_{k+1} - g_k - \nabla^2 f(x^*)s_k\| = o(\|s_k\|)$$

which implies

$$g_{k+1}^T s_k - g_k^T s_k - s_k^T \nabla^2 f(x^*)s_k = o(\|s_k\|^2). \quad (5.4.27)$$

Since α_k is a steplength from exact line search, $g_{k+1}^T s_k = 0$. Also, noting that $B_k s_k = \alpha_k B_k d_k = -\alpha_k g_k$, we may write (5.4.27) as

$$\begin{aligned} s_k^T \nabla^2 f(x^*)s_k &= -g_k^T s_k + o(\|s_k\|^2) \\ &= \frac{1}{\alpha_k} s_k^T B_k s_k + o(\|s_k\|^2). \end{aligned} \quad (5.4.28)$$

From (5.4.25), we have

$$s_k^T [B_k - \nabla^2 f(x^*)]s_k = o(\|s_k\|^2). \quad (5.4.29)$$

So, combining (5.4.28) and (5.4.29) gives

$$\begin{aligned} (\alpha_k - 1)s_k^T \nabla^2 f(x^*)s_k &= s_k^T [B_k - \nabla^2 f(x^*)]s_k + o(\|s_k\|^2) \\ &= o(\|s_k\|^2) \end{aligned}$$

which proves (5.4.26). $\square$

About inexact line search, we consider Wolfe-Powell rule (2.5.3) and (2.5.7). By use of $d_k = -B_k g_k$, we employ the following rule: if

$$f(x_k - B_k^{-1}g_k) \leq f(x_k) - \rho g_k^T B_k^{-1}g_k, \quad (5.4.30)$$

$$g(x_k - B_k^{-1}g_k)^T B_k^{-1}g_k \leq \sigma g_k^T B_k^{-1}g_k \quad (5.4.31)$$

hold, take $\alpha_k = 1$; otherwise, take $\alpha_k > 0$ such that

$$f(x_k - \alpha_k B_k^{-1}g_k) \leq f(x_k) - \rho \alpha_k g_k^T B_k^{-1}g_k, \quad (5.4.32)$$

$$g(x_k - \alpha_k B_k^{-1}g_k)^T B_k^{-1}g_k \leq \sigma g_k^T B_k^{-1}g_k, \quad (5.4.33)$$

where $g(\cdot) = \nabla f(\cdot)$.

Theorem 5.4.8 *Let $f : R^n \rightarrow R$ satisfy conditions (a) and (b) in Assumption 5.4.2. Suppose that $\{B_k\}$ is a sequence of symmetric and positive definite matrices. For given $x_0 \in D$, consider the iteration (5.4.24), where α_k is determined by Wolfe-Powell rule (5.4.30)–(5.4.33). If the sequence $\{x_k\}$ produced by (5.4.24) remains in D and $x_k \neq x^*$ ($\forall k \geq 0$) and if $x_k \rightarrow x^*$, then when (5.4.25) holds, $\alpha_k \rightarrow 1$ and hence $\{x_k\}$ converges to x^* superlinearly.*

Proof. Now we only need to prove that for sufficiently large k , (5.4.30)–(5.4.33) hold, and thus $\alpha_k = 1$. The remainder is obtained from Theorem 5.4.4.

Since $B_k s_k = -\alpha_k g_k$, it follows from (5.4.25) that

$$\begin{aligned} 0 &= \lim_{k \rightarrow \infty} \frac{\| [B_k - \nabla^2 f(x^*)] s_k \|}{\| s_k \|} \\ &= \lim_{k \rightarrow \infty} \frac{\| g_k - \nabla^2 f(x^*) B_k^{-1} g_k \|}{\| B_k^{-1} g_k \|}. \end{aligned}$$

Then

$$\begin{aligned} &g_k^T B_k^{-1} g_k - (B_k^{-1} g_k)^T \nabla^2 f(x^*) (B_k^{-1} g_k) \\ &= (g_k - \nabla^2 f(x^*) B_k^{-1} g_k)^T (B_k^{-1} g_k) \\ &= o(\| B_k^{-1} g_k \|^2), \end{aligned}$$

that is

$$g_k^T B_k^{-1} g_k = (B_k^{-1} g_k)^T \nabla^2 f(x^*) (B_k^{-1} g_k) + o(\| B_k^{-1} g_k \|^2). \quad (5.4.34)$$

Since $\nabla^2 f(x^*)$ is positive definite, there exists $\eta > 0$ such that for sufficiently large k ,

$$g_k^T B_k^{-1} g_k \geq \eta \| B_k^{-1} g_k \|^2. \quad (5.4.35)$$

Then, from Taylor's expansion (1.2.103) and (5.4.34), we have

$$\begin{aligned} f(x_k - B_k^{-1} g_k) - f(x_k) &= -g_k^T B_k^{-1} g_k + \frac{1}{2} g_k^T B_k^{-1} g_k + o(\| B_k^{-1} g_k \|^2) \\ &= -\frac{1}{2} g_k^T B_k^{-1} g_k + o(\| B_k^{-1} g_k \|^2) \\ &\leq -\rho g_k^T B_k^{-1} g_k. \end{aligned} \quad (5.4.36)$$

Also, by (1.2.111) and a proof similar to (5.4.27), we get

$$\begin{aligned} & g(x_k - B_k^{-1}g_k)^T B_k^{-1}g_k - g_k^T B_k^{-1}g_k + (B_k^{-1}g_k)^T \nabla^2 f(x^*)(B_k^{-1}g_k) \\ &= o(\|B_k^{-1}g_k\|^2), \end{aligned}$$

which, together with (5.4.34), gives

$$g(x_k - B_k^{-1}g_k)^T B_k^{-1}g_k = o(\|B_k^{-1}g_k\|^2) \leq \sigma g_k^T B_k^{-1}g_k. \quad (5.4.37)$$

It follows from (5.4.36) and (5.4.37) that (5.4.30)-(5.4.31) hold, and thus $\alpha_k = 1$ for k sufficiently large. $\square$

5.4.2 Linear Convergence of General Quasi-Newton Methods

In this subsection, our goal is to discuss the local and linear convergence results of general quasi-Newton methods. Let the iterative scheme of general quasi-Newton methods be

$$x_{k+1} = x_k - B_k^{-1}F(x_k), \quad (5.4.38)$$

$$B_{k+1} \in U(x_k, B_k), \quad (5.4.39)$$

where $U(x_k, B_k)$ denotes a nonempty set of updates, $(x_k, B_k) \in \text{dom}U$, $\text{dom}U$ denotes the domain of U .

Theorem 5.4.9 *Let $F : R^n \rightarrow R^n$ satisfy the assumptions (a), (b) and (c) in Assumption 5.4.1, U an update function, such that for all $(x_k, B_k) \in \text{dom}U$ and $B_{k+1} \in U(x_k, B_k)$, we have that*

$$\|B_{k+1} - F'(x^*)\| \leq \|B_k - F'(x^*)\| + \frac{\gamma}{2}(\|x_{k+1} - x^*\| + \|x_k - x^*\|), \quad (5.4.40)$$

where γ is some constant, or that

$$\|B_{k+1} - F'(x^*)\| \leq [1 + \alpha_1 \sigma(x_k, x_{k+1})]\|B_k - F'(x^*)\| + \alpha_2 \sigma(x_k, x_{k+1}), \quad (5.4.41)$$

where α_1 and α_2 are some constants, and

$$\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}. \quad (5.4.42)$$

Then, there exist constants ε and δ , such that, for $\|x_0 - x^*\| < \varepsilon$ and $\|B_0 - F'(x^*)\| < \delta$, the iteration (5.4.38)-(5.4.39) is well-defined, and $\{x_k\}$ converges to x^* linearly.

Proof. First, we prove the conclusion for the given condition (5.4.40). Assume $\|F'(x^*)^{-1}\| \leq \beta$ and choose ε and δ such that

$$6\beta\delta < 1, \quad (5.4.43)$$

$$3\gamma\varepsilon \leq 2\delta. \quad (5.4.44)$$

To prove the local and linear convergence, we prove, by induction, that

$$\|B_k - F'(x^*)\| \leq (2 - 2^{-k})\delta, \quad (5.4.45)$$

$$\|x_{k+1} - x^*\| \leq \frac{1}{2}\|x_k - x^*\|. \quad (5.4.46)$$

For $k = 0$, (5.4.45) is obvious. Since the proof of (5.4.46) for $k = 0$ is the same as that in the following general case, we omit it here.

Now, suppose that (5.4.45) and (5.4.46) hold for $k = 0, 1, \dots, i-1$. For $k = i$, by assumption of induction and (5.4.40), we have

$$\begin{aligned} \|B_i - F'(x^*)\| &\leq \|B_{i-1} - F'(x^*)\| + \frac{\gamma}{2}(\|x_i - x^*\| + \|x_{i-1} - x^*\|) \\ &\leq (2 - 2^{-(i-1)})\delta + \frac{3}{4}\gamma\|x_{i-1} - x^*\|. \end{aligned} \quad (5.4.47)$$

From (5.4.46) and $\|x_0 - x^*\| < \varepsilon$, we have

$$\|x_{i-1} - x^*\| \leq 2^{-(i-1)}\|x_0 - x^*\| \leq 2^{-(i-1)}\varepsilon. \quad (5.4.48)$$

Substituting (5.4.48) into (5.4.47) and using (5.4.44) yield

$$\begin{aligned} \|B_i - F'(x^*)\| &\leq (2 - 2^{-(i-1)})\delta + \frac{3}{4}\gamma \cdot 2^{-(i-1)}\varepsilon \\ &\leq (2 - 2^{-(i-1)} + 2^{-i})\delta = (2 - 2^{-i})\delta, \end{aligned} \quad (5.4.49)$$

which proves (5.4.45).

To prove (5.4.46), we first show that B_i is invertible. In fact, since $\|F'(x^*)^{-1}\| \leq \beta$, it follows from (5.4.45) and (5.4.43) that

$$\begin{aligned} &\|F'(x^*)^{-1}[B_i - F'(x^*)]\| \\ &\leq \|F'(x^*)^{-1}\|\|B_i - F'(x^*)\| \\ &\leq \beta(2 - 2^{-i})\delta \leq 2\beta\delta \leq \frac{1}{3}. \end{aligned}$$

Then, by Von-Neumann Theorem 1.2.5, we know that B_i is invertible, and

$$\begin{aligned}\|B_i^{-1}\| &\leq \frac{\|F'(x^*)^{-1}\|}{1 - \|F'(x^*)^{-1}(B_i - F'(x^*))\|} \\ &\leq \frac{\beta}{1 - \frac{1}{3}} = \frac{3\beta}{2}.\end{aligned}\tag{5.4.50}$$

Thus, x_{i+1} is well-defined. Also,

$$\begin{aligned}B_i(x_{i+1} - x^*) &= B_i(x_i - x^*) - F(x_i) + F(x^*) \\ &= [-F(x_i) + F(x^*) + F'(x^*)(x_i - x^*)] \\ &\quad + [B_i - F'(x^*)(x_i - x^*)],\end{aligned}\tag{5.4.51}$$

which gives

$$\begin{aligned}\|x_{i+1} - x^*\| &\leq \|B_i^{-1}\|[\| -F(x_i) + F(x^*) + F'(x^*)(x_i - x^*) \| \\ &\quad + \|B_i - F'(x^*)\| \|x_i - x^*\|].\end{aligned}\tag{5.4.52}$$

By use of Theorem 1.2.22,

$$\| -F(x_i) + F(x^*) + F'(x^*)(x_i - x^*) \| \leq \frac{\gamma}{2} \|x_i - x^*\|^2.\tag{5.4.53}$$

So, (5.4.52), (5.4.50), (5.4.53) and (5.4.49) give

$$\|x_{i+1} - x^*\| \leq \frac{3}{2}\beta \left[\frac{\gamma}{2} \|x_i - x^*\| + (2 - 2^{-i})\delta \right] \|x_i - x^*\|.\tag{5.4.54}$$

Also, using (5.4.48) and (5.4.44), we have

$$\frac{\gamma}{2} \|x_i - x^*\| \leq 2^{-(i+1)}\gamma\varepsilon \leq \frac{2^{-i}}{3}\delta.$$

Substituting the above inequality into (5.4.54), we obtain

$$\begin{aligned}\|x_{i+1} - x^*\| &\leq \frac{3}{2}\beta \left[\frac{1}{3}2^{-i} + 2 - 2^{-i} \right] \delta \|x_i - x^*\| \\ &\leq 3\beta\delta \|x_i - x^*\| \\ &\leq \frac{1}{2} \|x_i - x^*\|.\end{aligned}$$

Therefore, the desired result (5.4.46) is proved.

Similarly, for the given condition (5.4.41), we also can prove our conclusion.

In fact, let $\|F'(x^*)\| \leq \beta$ and $r \in (0, 1)$. Choose $\varepsilon(r) = \varepsilon$ and $\delta(r) = \delta$, such that

$$(2\alpha_1\delta + \alpha_2)\frac{\varepsilon}{1-r} \leq \delta, \quad (5.4.55)$$

$$\beta(1+r)(\gamma\varepsilon + 2\delta) \leq r. \quad (5.4.56)$$

To prove the local and linear convergence, we still prove

$$\|B_k - F'(x^*)\| \leq 2\delta, \quad (5.4.57)$$

$$\|x_{k+1} - x^*\| \leq r\|x_k - x^*\| \quad (5.4.58)$$

by induction.

Obviously, for $k = 0$, the conclusion holds. Suppose that the conclusion holds for $k = 0, 1, \dots, i-1$. By (5.4.41), we have

$$\|B_{k+1} - F'(x^*)\| - \|B_k - F'(x^*)\| \leq 2\alpha_1\delta\varepsilon r^k + \alpha_2\varepsilon r^k.$$

Summing for $k = 0$ to $i-1$ yields

$$\|B_i - F'(x^*)\| \leq \|B_0 - F'(x^*)\| + (2\alpha_1\delta + \alpha_2)\frac{\varepsilon}{1-r}.$$

So, using (5.4.55) and $\|B_0 - F'(x^*)\| \leq \delta$, we obtain

$$\|B_i - F'(x^*)\| \leq 2\delta, \quad (5.4.59)$$

which proves (5.4.57).

To prove (5.4.58), first note that $\|B_i^{-1}\| \leq (1+r)\beta$ from (5.4.59) and Theorem 1.2.5. Then, by Theorem 1.2.24, we have

$$\begin{aligned} \|x_{i+1} - x^*\| &\leq \|B_i^{-1}\|[\|F(x_i) - F(x^*) - F'(x^*)(x_i - x^*)\| \\ &\quad + \|B_i - F'(x^*)\|\|x_i - x^*\|] \\ &\leq \beta(1+r)(\gamma\varepsilon + 2\delta)\|x_i - x^*\|. \end{aligned}$$

By using (5.4.56) we immediately obtain

$$\|x_{i+1} - x^*\| \leq r\|x_i - x^*\|.$$

So, (5.4.58) is proved. We complete the proof by induction. $\square$

Similarly, we have the following local and linear convergence theorem for update of the inverse Hessian approximation.

Theorem 5.4.10 *Let $F : R^n \rightarrow R^n$ satisfy the conditions (a), (b), and (c) in Assumption 5.4.1. Let U be an update function, such that for all $(x_k, H_k) \in \text{dom}U$ and $H_{k+1} \in U(x_k, H_k)$, we have that*

$$\|H_{k+1} - F'(x^*)^{-1}\| \leq \|H_k - F'(x^*)^{-1}\| + \frac{\gamma}{2}(\|x_{k+1} - x^*\| + \|x_k - x^*\|), \quad (5.4.60)$$

where γ is some constant, or that

$$\|H_{k+1} - F'(x^*)^{-1}\| \leq [1 + \alpha_1 \sigma(x_k, x_{k+1})] \|H_k - F'(x^*)^{-1}\| + \alpha_2 \sigma(x_k, x_{k+1}), \quad (5.4.61)$$

where α_1 and α_2 are some constants, and

$$\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}.$$

Then, there exist constants ε and δ , such that, for $\|x_0 - x^*\| < \varepsilon$ and $\|H_0 - F'(x^*)^{-1}\| < \delta$, the iteration

$$x_{k+1} = x_k - H_k F(x_k), \quad H_{k+1} \in U(x_k, H_k) \quad (5.4.62)$$

is well-defined and $\{x_k\}$ converges to x^* linearly.

As a consequence of the above two theorems, we give the following corollaries on superlinear convergence for general iterations.

Corollary 5.4.11 *Suppose that the assumptions of Theorem 5.4.9 hold. If some subsequence of $\{\|B_k - F'(x^*)\|\}$ converges to zero, then $\{x_k\}$ converges to x^* superlinearly.*

Proof. We hope to prove

$$\lim_{k \rightarrow +\infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0.$$

Let $r \in (0, 1)$. It follows from Theorem 5.4.9 that there exist $\varepsilon(r)$ and $\delta(r)$ such that $\|B_0 - F'(x^*)\| < \delta(r)$ and $\|x_0 - x^*\| < \varepsilon(r)$ imply that $\|x_{k+1} - x^*\| \leq r\|x_k - x^*\|, \forall k \geq 0$. From the assumption, we can choose $m > 0$ such that $\|B_m - F'(x^*)\| < \delta(r)$ and $\|x_m - x^*\| < \varepsilon(r)$. Hence $\|x_{k+1} - x^*\| \leq r\|x_k - x^*\|, \forall k \geq m$. Since $r \in (0, 1)$ is arbitrary, the conclusion is shown. $\square$

Similarly, we have

Corollary 5.4.12 *Suppose that the conditions of Theorem 5.4.10 hold. If some subsequence of $\{\|H_k - F'(x^*)^{-1}\|\}$ converges to zero, then $\{x_k\}$ converges to x^* superlinearly.*

5.4.3 Local Convergence of Broyden's Rank-One Update

In this section, we prove the linear convergence and superlinear convergence of Broyden's rank-one update

$$x_{k+1} = x_k - B_k^{-1}F(x_k), \quad (5.4.63)$$

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k)s_k^T}{s_k^T s_k}. \quad (5.4.64)$$

Theorem 5.4.13 *Let $F : R^n \rightarrow R^n$ satisfy the conditions (a), (b), and (c) in Assumption 5.4.1. Assume that there exist positive constants ε and δ such that $\|x_0 - x^*\| < \varepsilon$ and $\|B_0 - F'(x^*)\| < \delta$. Then the sequence $\{x_k\}$ generated by Broyden's rank-one update (5.4.63)–(5.4.64) is well-defined and convergent to x^* superlinearly.*

Proof. It is enough to prove, under the conditions of the theorem, that (5.4.40) and (5.4.4) are satisfied respectively.

First, we prove that B_{k+1} generated by Broyden's rank-one update satisfies (5.4.40).

By (5.4.63)–(5.4.64), we have

$$\begin{aligned} B_{k+1} - F'(x^*) &= B_k - F'(x^*) + \frac{(y_k - B_k s_k)s_k^T}{s_k^T s_k} \\ &= B_k - F'(x^*) + \frac{(F'(x^*)s_k - B_k s_k)s_k^T}{s_k^T s_k} + \frac{(y_k - F'(x^*)s_k)s_k^T}{s_k^T s_k} \\ &= (B_k - F'(x^*)) \left[I - \frac{s_k s_k^T}{s_k^T s_k} \right] + \frac{(y_k - F'(x^*)s_k)s_k^T}{s_k^T s_k}. \end{aligned} \quad (5.4.65)$$

Taking norms gives

$$\|B_{k+1} - F'(x^*)\| \leq \|B_k - F'(x^*)\| \left\| I - \frac{s_k s_k^T}{s_k^T s_k} \right\| + \frac{\|y_k - F'(x^*)s_k\|}{\|s_k\|}. \quad (5.4.66)$$

Note that

$$\left\| I - \frac{s_k s_k^T}{s_k^T s_k} \right\| = 1 \quad (5.4.67)$$

and

$$\begin{aligned} \|y_k - F'(x^*)s_k\| &= \|F(x_{k+1}) - F(x_k) - F'(x^*)s_k\| \\ &\leq \frac{\gamma}{2}(\|x_{k+1} - x^*\| + \|x_k - x^*\|)\|s_k\| \end{aligned} \quad (5.4.68)$$

by Theorem 1.2.24, we obtain immediately that

$$\|B_{k+1} - F'(x^*)\| \leq \|B_k - F'(x^*)\| + \frac{\gamma}{2}(\|x_{k+1} - x^*\| + \|x_k - x^*\|),$$

which is (5.4.40). The linear convergence is proved.

Next, we prove the superlinear convergence of Broyden's rank-one update by use of Theorem 5.4.3, that is, we want to prove that (5.4.4) holds.

Let $E_k = B_k - F'(x^*)$. From (5.4.65),

$$\|E_{k+1}\|_F \leq \left\| E_k \left(I - \frac{s_k s_k^T}{s_k^T s_k} \right) \right\|_F + \frac{\|(y_k - F'(x^*)s_k)s_k^T\|_F}{s_k^T s_k}. \quad (5.4.69)$$

Since

$$\begin{aligned} \left\| E_k \frac{s_k s_k^T}{s_k^T s_k} \right\|_F^2 &= \text{tr} \left(\frac{(E_k s_k)^T (E_k s_k)}{(s_k^T s_k)^2} s_k s_k^T \right) \\ &= \frac{\|E_k s_k\|^2}{\|s_k\|^4} \|s_k\|^2 = \frac{\|E_k s_k\|^2}{\|s_k\|^2}, \end{aligned}$$

we get

$$\begin{aligned} \|E_k\|_F^2 &= \left\| E_k \frac{s_k s_k^T}{s_k^T s_k} \right\|_F^2 + \left\| E_k \left(I - \frac{s_k s_k^T}{s_k^T s_k} \right) \right\|_F^2 \\ &= \frac{\|E_k s_k\|^2}{\|s_k\|^2} + \left\| E_k \left(I - \frac{s_k s_k^T}{s_k^T s_k} \right) \right\|_F^2. \end{aligned}$$

Hence

$$\left\| E_k \left(I - \frac{s_k s_k^T}{s_k^T s_k} \right) \right\|_F = \left(\|E_k\|_F^2 - \frac{\|E_k s_k\|^2}{\|s_k\|^2} \right)^{\frac{1}{2}}. \quad (5.4.70)$$

Since $(\alpha^2 - \beta^2)^{\frac{1}{2}} \leq \alpha - \beta^2/(2\alpha)$ for any $\alpha \geq |\beta| \geq 0$, (5.4.70) implies that

$$\left\| E_k \left(I - \frac{s_k s_k^T}{s_k^T s_k} \right) \right\|_F \leq \|E_k\|_F - \frac{1}{2\|E_k\|_F} \left(\frac{\|E_k s_k\|}{\|s_k\|} \right)^2. \quad (5.4.71)$$

Also, by means of Theorem 1.2.24,

$$\|y_k - F'(x^*)s_k\|_F \leq \frac{\gamma}{2}(\|x_{k+1} - x^*\| + \|x_k - x^*\|)\|s_k\|. \quad (5.4.72)$$

So, by using (5.4.71), (5.4.72), and (5.4.46), we can write (5.4.69) as

$$\|E_{k+1}\|_F \leq \|E_k\|_F - \frac{\|E_k s_k\|^2}{2\|E_k\|_F \|s_k\|^2} + \frac{3}{4}\gamma\|x_k - x^*\|,$$

which is

$$\frac{\|E_k s_k\|^2}{\|s_k\|^2} \leq 2\|E_k\|_F \left[\|E_k\|_F - \|E_{k+1}\|_F + \frac{3}{4}\gamma\|x_k - x^*\| \right]. \quad (5.4.73)$$

Recalling (5.4.45) and (5.4.46), we have that

$$\|E_k\|_F \leq 2\delta, \forall k \geq 0$$

and

$$\sum_{k=0}^{\infty} \|x_k - x^*\| \leq 2\varepsilon.$$

Thus, (5.4.73) can be written as

$$\frac{\|E_k s_k\|^2}{\|s_k\|^2} \leq 4\delta \left[\|E_k\|_F - \|E_{k+1}\|_F + \frac{3}{4}\gamma\|x_k - x^*\| \right]. \quad (5.4.74)$$

By summing both sides, we obtain

$$\begin{aligned} \sum_{k=0}^i \frac{\|E_k s_k\|^2}{\|s_k\|^2} &\leq 4\delta \left[\|E_0\|_F - \|E_{i+1}\|_F + \frac{3}{4}\gamma \sum_{k=0}^i \|x_k - x^*\| \right] \\ &\leq 4\delta \left[\|E_0\|_F + \frac{3}{2}\gamma\varepsilon \right] \\ &\leq 4\delta \left[\delta + \frac{3}{2}\gamma\varepsilon \right], \end{aligned} \quad (5.4.75)$$

which holds for any $i \geq 0$. Therefore

$$\sum_{k=0}^{\infty} \frac{\|E_k s_k\|^2}{\|s_k\|^2}$$

is finite and further

$$\lim_{k \rightarrow \infty} \frac{\|E_k s_k\|}{\|s_k\|} = 0, \quad (5.4.76)$$

which is (5.4.4). Then we have proved the superlinear convergence of Broyden's rank-one update by use of Theorem 5.4.3. $\square$

Similarly, for the following form of Broyden's rank-one update in inverse Hessian approximation:

$$x_{k+1} = x_k - H_k F(x_k), \quad (5.4.77)$$

$$H_{k+1} = H_k + \frac{(s_k - H_k y_k) y_k^T}{y_k^T y_k}, \quad (5.4.78)$$

we have the following theorem.

Theorem 5.4.14 *Let $F : R^n \rightarrow R^n$ satisfy the conditions (a), (b), and (c) in Assumption 5.4.1. Assume that there exist ε and δ such that $\|x_0 - x^*\| < \varepsilon$ and $\|H_0 - F'(x^*)^{-1}\| < \delta$. Then the sequence $\{x_k\}$ generated by Broyden's rank-one update (5.4.77)–(5.4.78) is well-defined and convergent to x^* superlinearly.*

5.4.4 Local and Linear Convergence of DFP Method

In this subsection and subsequent subsections, we discuss the local convergence of rank-two methods, which includes the linear and superlinear convergence, and local convergence under line search. Note that we introduce two different techniques to prove the superlinear convergence of BFGS and DFP methods respectively.

The DFP iteration we consider is

$$x_{k+1} = x_k - B_k^{-1} \nabla f(x_k), \quad (5.4.79)$$

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k) y_k^T + y_k (y_k - B_k s_k)^T}{y_k^T y_k} - \frac{(y_k - B_k s_k)^T s_k}{(y_k^T s_k)^2} y_k y_k^T. \quad (5.4.80)$$

To study the local convergence of DFP method, it is required to estimate $\|B_{k+1} - \nabla^2 f(x^*)\|$. As shown in the following theorem, there is a matrix $P = I - \frac{s_k y_k^T}{s_k^T y_k}$ in $B_{k+1} - \nabla^2 f(x^*)$. Since

$$\|P\|_2 = \frac{\|s_k\| \|y_k\|}{s_k^T y_k}, \quad (5.4.81)$$

it is a secant of the angle between y_k and s_k . In general, y_k and s_k is not parallel, so $\|P\|_2$ may be quite big, and it is not suitable to estimate

$\|B_{k+1} - \nabla^2 f(x^*)\|$ by means of l_2 norm. However, near x^* , $f(x)$ closes a quadratic function, and hence $A^{-\frac{1}{2}}y_k$ and $A^{\frac{1}{2}}s_k$ are approximately parallel, where $A = \nabla^2 f(x^*)$. It motivates us to use some weighted norm to estimate $\|B_{k+1} - \nabla^2 f(x^*)\|$. Then we define

$$\|E\|_{DFP} = \|E\|_{A^{-\frac{1}{2}}, F} = \|A^{-\frac{1}{2}}EA^{-\frac{1}{2}}\|_F. \quad (5.4.82)$$

Below, we first develop the linear convergence of DFP method.

Theorem 5.4.15 *Let $f : R^n \rightarrow R$ satisfy Assumption 5.4.2. Also let*

$$\mu\gamma\sigma(x_k, x_{k+1}) \leq \frac{1}{3} \quad (5.4.83)$$

in a neighborhood of x^ , where $\mu = \|\nabla^2 f(x^*)^{-1}\|$, $\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}$. Then, there exist $\varepsilon > 0$ and $\delta > 0$ such that for $\|x_0 - x^*\| < \varepsilon$ and $\|B_0 - \nabla^2 f(x^*)\|_{DFP} < \delta$, the iteration (5.4.79)–(5.4.80) of DFP method is well-defined, and the produced sequence $\{x_k\}$ converges to x^* linearly.*

Proof. Based on Theorem 5.4.9, to prove the linear convergence of DFP method, it is enough to prove

$$\begin{aligned} \|B_{k+1} - \nabla^2 f(x^*)\|_{DFP} &< [1 + \alpha_1\sigma(x_k, x_{k+1})]\|B_k - \nabla^2 f(x^*)\|_{DFP} \\ &\quad + \alpha_2\sigma(x_k, x_{k+1}), \end{aligned} \quad (5.4.84)$$

where α_1 and α_2 are positive constants independent of x_k and x_{k+1} , $\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}$.

Let $A = \nabla^2 f(x^*)$. From (5.4.79)–(5.4.80), it follows that

$$B_{k+1} - A = P^T(B_k - A)P + \frac{(y_k - As_k)y_k^T + y_k(y_k - As_k)^T P}{y_k^T s_k}, \quad (5.4.85)$$

where

$$P = I - \frac{s_k y_k^T}{s_k^T y_k}. \quad (5.4.86)$$

Note that $\|P\|_2 = \|s_k\|/\|y_k\|$, hence

$$\begin{aligned} \|P^T(B_k - A)P\|_{DFP} &\leq \|A^{\frac{1}{2}}PA^{-\frac{1}{2}}\|_2^2 \|B_k - A\|_{DFP} \\ &\leq \frac{1}{\omega_2} \|B_k - A\|_{DFP}, \end{aligned} \quad (5.4.87)$$

$$\left\| \frac{y_k(y_k - As_k)^T P}{y_k^T s_k} \right\|_{DFP} \leq \frac{1}{\omega^2} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|}, \quad (5.4.88)$$

$$\left\| \frac{(y_k - As_k)y_k^T}{y_k^T s_k} \right\|_{DFP} \leq \frac{1}{\omega} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|}, \quad (5.4.89)$$

where

$$\omega = \frac{y_k^T s_k}{\|A^{-\frac{1}{2}} y_k\| \|A^{\frac{1}{2}} s_k\|} = \frac{\langle A^{-\frac{1}{2}} y_k, A^{\frac{1}{2}} s_k \rangle}{\|A^{-\frac{1}{2}} y_k\| \|A^{\frac{1}{2}} s_k\|}. \quad (5.4.90)$$

Now, we estimate $\|B_{k+1} - A\|_{DFP}$ by using (5.4.87), (5.4.88) and (5.4.89), and have

$$\begin{aligned} \|B_{k+1} - A\|_{DFP} &\leq \frac{1}{\omega^2} \|B_k - A\|_{DFP} \\ &\quad + \frac{2}{\omega^2} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|}. \end{aligned} \quad (5.4.91)$$

Note from Theorem 1.2.24 that

$$\begin{aligned} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|} &\leq \frac{\|A^{-\frac{1}{2}}\| \|y_k - As_k\|}{\|s_k\| \|A^{-\frac{1}{2}}\|} \\ &= \mu \frac{\|y_k - As_k\|}{\|s_k\|} \\ &\leq \mu \gamma \sigma(x_k, x_{k+1}) \leq \frac{1}{3}. \end{aligned} \quad (5.4.92)$$

Also, by Lemma 5.4.5, we have

$$1 - \omega^2 \leq \left[\mu \frac{\|y_k - As_k\|}{\|s_k\|} \right]^2 \leq [\mu \gamma \sigma(x_k, x_{k+1})]^2.$$

Then, if x_k and x_{k+1} are in the neighborhood of x^* , then

$$1 - \omega^2 \leq [\mu \gamma \sigma(x_k, x_{k+1})]^2 < \frac{1}{2},$$

which is

$$\omega^2 > \frac{1}{2} > \mu \gamma \sigma(x_k, x_{k+1}).$$

Hence

$$\begin{aligned} \frac{1}{\omega^2} &= 1 + \frac{1 - \omega^2}{\omega^2} < 1 + \frac{[\mu\gamma\sigma(x_k, x_{k+1})]^2}{\mu\gamma\sigma(x_k, x_{k+1})} \\ &= 1 + \mu\gamma\sigma(x_k, x_{k+1}). \end{aligned}$$

So, the two terms in (5.4.91) satisfy respectively

$$\frac{1}{\omega^2} \|B_k - A\|_{DFP} < (1 + \mu\gamma\sigma(x_k, x_{k+1})) \|B_k - A\|_{DFP} \quad (5.4.93)$$

and

$$\begin{aligned} \frac{2}{\omega^2} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|} &< 2[1 + \mu\gamma\sigma(x_k, x_{k+1})] \mu\gamma\sigma(x_k, x_{k+1}) \\ &< 3\mu\gamma\sigma(x_k, x_{k+1}). \end{aligned} \quad (5.4.94)$$

Substituting (5.4.93) and (5.4.94) into (5.4.112) yields (5.4.113), where $\alpha_1 = \mu\gamma$, $\alpha_2 = 3\mu\gamma$. So, we complete the proof. $\square$

5.4.5 Superlinear Convergence of BFGS Method

In this subsection, we discuss the superlinear convergence of BFGS method. Let

$$\tilde{s}_k = G_*^{\frac{1}{2}} s_k, \quad \tilde{y}_k = G_*^{-\frac{1}{2}} y_k, \quad \tilde{B}_k = G_*^{-\frac{1}{2}} B_k G_*^{-\frac{1}{2}}, \quad (5.4.95)$$

where $G_* = G(x^*) = \nabla^2 f(x^*)$. Define

$$\cos \tilde{\theta}_k = \frac{\tilde{s}_k^T \tilde{B}_k \tilde{s}_k}{\|\tilde{s}_k\| \|\tilde{B}_k \tilde{s}_k\|}, \quad \tilde{q}_k = \frac{\tilde{s}_k^T \tilde{B}_k \tilde{s}_k}{\|\tilde{s}_k\|^2}, \quad (5.4.96)$$

and define

$$\tilde{M}_k = \frac{\|\tilde{y}_k\|^2}{\tilde{y}_k^T \tilde{s}_k}, \quad \tilde{m}_k = \frac{\tilde{y}_k^T \tilde{s}_k}{\tilde{s}_k^T \tilde{s}_k}. \quad (5.4.97)$$

By pre- and postmultiplying the BFGS update (5.1.45) by $G_*^{-\frac{1}{2}}$, we obtain

$$\tilde{B}_{k+1} = \tilde{B}_k - \frac{\tilde{B}_k \tilde{s}_k \tilde{s}_k^T \tilde{B}_k}{\tilde{s}_k^T \tilde{B}_k \tilde{s}_k} + \frac{\tilde{y}_k \tilde{y}_k^T}{\tilde{y}_k^T \tilde{s}_k}. \quad (5.4.98)$$

Since this expression has precisely the same form as the BFGS formula, it follows from the argument leading to (5.3.42) that

$$\begin{aligned}\psi(\tilde{B}_{k+1}) &= \psi(\tilde{B}_k) + (\tilde{M}_k - \ln \tilde{m}_k - 1) \\ &= \left[1 - \frac{\tilde{q}_k}{\cos^2 \tilde{\theta}_k} + \ln \frac{\tilde{q}_k}{\cos^2 \tilde{\theta}_k} \right] + \ln \cos^2 \tilde{\theta}_k.\end{aligned}\quad (5.4.99)$$

Noting that

$$y_k - G_* s_k = (\bar{G}_k - G_*) s_k,$$

where

$$\bar{G}_k = \int_0^1 \nabla^2 f(x_k + \tau s_k) d\tau,$$

we obtain

$$\tilde{y}_k - \tilde{s}_k = G_*^{-\frac{1}{2}} (\bar{G}_* - G_k) G_*^{-\frac{1}{2}} \tilde{s}_k.$$

Assuming that the Hessian matrix G is Lipschitz at x^* , then we have

$$\|\tilde{y}_k - \tilde{s}_k\| \leq \|G_*^{-\frac{1}{2}}\|^2 \|\tilde{s}_k\| \|\bar{G}_k - G_*\| \leq \|G_*^{-\frac{1}{2}}\|^2 \|\tilde{s}_k\| L \epsilon_k,$$

which gives

$$\frac{\|\tilde{y}_k - \tilde{s}_k\|}{\|\tilde{s}_k\|} \leq \bar{c} \epsilon_k \quad (5.4.100)$$

for some positive constant $\bar{c}$, where

$$\epsilon_k = \max\{\|x_{k+1} - x^*\|, \|x_k - x^*\|\}. \quad (5.4.101)$$

Now we are in a position to prove the superlinear convergence theorem.

Theorem 5.4.16 *Let f be twice continuously differentiable and the Hessian matrix G be Lipschitz continuous at x^* . Suppose that the sequence generated by the BFGS algorithm converges to a minimizer x^* and that the condition*

$$\sum_{k=1}^{\infty} \|x_k - x^*\| < \infty \quad (5.4.102)$$

holds. Then $\{x_k\}$ converges to x^ at a superlinear rate.*

Proof. By (5.4.100), we have

$$\|\tilde{y}_k\| - \|\tilde{s}_k\| \leq \bar{c}\epsilon_k \|\tilde{s}_k\|, \quad \|\tilde{s}_k\| - \|\tilde{y}_k\| \leq \bar{c}\epsilon_k \|\tilde{s}_k\|,$$

which give

$$(1 - \bar{c}\epsilon_k)\|\tilde{s}_k\| \leq \|\tilde{y}_k\| \leq (1 + \bar{c}\epsilon_k)\|\tilde{s}_k\|. \quad (5.4.103)$$

By squaring (5.4.100) and using (5.4.103), we obtain

$$(1 - \bar{c}\epsilon_k)^2 \|\tilde{s}_k\|^2 - 2\tilde{y}_k^T \tilde{s}_k + \|\tilde{s}_k\|^2 \leq \|\tilde{y}_k\|^2 - 2\tilde{y}_k^T \tilde{s}_k + \|\tilde{s}_k\|^2 \leq \bar{c}^2 \epsilon_k^2 \|\tilde{s}_k\|^2,$$

and therefore

$$2\tilde{y}_k^T \tilde{s}_k \geq (1 - 2\bar{c}\epsilon_k + \bar{c}^2 \epsilon_k^2 + 1 - \bar{c}^2 \epsilon_k^2) \|\tilde{s}_k\|^2 = 2(1 - \bar{c}\epsilon_k) \|\tilde{s}_k\|^2.$$

It follows from the definition of $\tilde{m}_k$ that

$$\tilde{m}_k = \frac{\tilde{y}_k^T \tilde{s}_k}{\|\tilde{s}_k\|^2} \geq 1 - \bar{c}\epsilon_k. \quad (5.4.104)$$

Combining (5.4.103) and (5.4.104) gives also that

$$\tilde{M}_k = \frac{\|\tilde{y}_k\|^2}{\tilde{y}_k^T \tilde{s}_k} \leq \frac{1 + \bar{c}\epsilon_k}{1 - \bar{c}\epsilon_k}. \quad (5.4.105)$$

Since $x_k \rightarrow x^*$, we have that $\epsilon_k \rightarrow 0$. Thus by (5.4.105) there exists a positive constant $c > \bar{c}$ such that the following inequalities hold for all sufficient large k :

$$\tilde{M}_k \leq 1 + \frac{2\bar{c}}{1 - \bar{c}\epsilon_k} \epsilon_k \leq 1 + c\epsilon_k. \quad (5.4.106)$$

Making use of the nonpositiveness of the function $h(t) = 1 - t + \ln t$ gives

$$\frac{-x}{1-x} - \ln(1-x) = h\left(\frac{1}{1-x}\right) \leq 0.$$

Now for k large enough we can assume that $\bar{c}\epsilon_k < \frac{1}{2}$, and by using the above inequality we have

$$\ln(1 - \bar{c}\epsilon_k) \geq \frac{-\bar{c}\epsilon_k}{1 - \bar{c}\epsilon_k} \geq -2\bar{c}\epsilon_k.$$

This relation and (5.4.104) imply that for sufficiently large k , we have

$$\ln \tilde{m}_k \geq \ln(1 - \bar{c}\epsilon_k) \geq -2\bar{c}\epsilon_k > -2c\epsilon_k. \quad (5.4.107)$$

We can now deduce from (5.4.99), (5.4.106), and (5.4.107) that

$$0 < \psi(\tilde{B}_{k+1}) \leq \psi(\tilde{B}_k) + 3c\epsilon_k + \ln \cos^2 \tilde{\theta}_k + \left[1 - \frac{\tilde{q}_k}{\cos^2 \tilde{\theta}_k} + \ln \frac{\tilde{q}_k}{\cos^2 \tilde{\theta}_k} \right]. \quad (5.4.108)$$

By summing this expression and making use of (5.4.102) we have that

$$\begin{aligned} & \sum_{j=0}^{\infty} \left(\ln \frac{1}{\cos^2 \tilde{\theta}_j} - \left[1 - \frac{\tilde{q}_j}{\cos^2 \tilde{\theta}_j} + \ln \frac{\tilde{q}_j}{\cos^2 \tilde{\theta}_j} \right] \right) \\ & \leq \psi(\tilde{B}_0) + 3c \sum_{j=0}^{\infty} \epsilon_j < +\infty. \end{aligned}$$

Since the term in the square brackets is nonpositive, and since $\ln(1/\cos^2 \tilde{\theta}_j) \geq 0$ for all j , we obtain

$$\lim_{j \rightarrow \infty} \ln \frac{1}{\cos^2 \tilde{\theta}_j} = 0, \quad \lim_{j \rightarrow \infty} \left(1 - \frac{\tilde{q}_j}{\cos^2 \tilde{\theta}_j} + \ln \frac{\tilde{q}_j}{\cos^2 \tilde{\theta}_j} \right) = 0,$$

which imply that

$$\lim_{j \rightarrow \infty} \cos \tilde{\theta}_j = 1, \quad \lim_{j \rightarrow \infty} \tilde{q}_j = 1. \quad (5.4.109)$$

By use of these limits we can obtain that

$$\begin{aligned} \frac{\|G_*^{-\frac{1}{2}}(B_k - G_*)s_k\|^2}{\|G_*^{\frac{1}{2}}s_k\|^2} &= \frac{\|(\tilde{B}_k - I)\tilde{s}_k\|^2}{\|\tilde{s}_k\|^2} \\ &= \frac{\|\tilde{B}_k\tilde{s}_k\|^2 - 2\tilde{s}_k^T \tilde{B}_k \tilde{s}_k + \tilde{s}_k^T \tilde{s}_k}{\tilde{s}_k^T \tilde{s}_k} \\ &= \frac{\tilde{q}_k^2}{\cos^2 \tilde{\theta}_k} - 2\tilde{q}_k + 1 \\ &\rightarrow 0. \end{aligned} \quad (5.4.110)$$

Then we conclude that

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - G_*)s_k\|}{\|s_k\|} = 0 \quad (5.4.111)$$

which shows that the rate of convergence is superlinear. $\square$

5.4.6 Superlinear Convergence of DFP Method

We first give the following three lemmas.

Lemma 5.4.17 *Let $M \in R^{n \times n}$ be a nonsingular symmetric matrix. If, for $\beta \in [0, 1/3]$, the inequality*

$$\|My_k - M^{-1}s_k\| \leq \beta \|M^{-1}s_k\| \quad (5.4.112)$$

holds, then for any nonzero matrix $E \in R^{n \times n}$, we have

$$(a) \quad (1 - \beta) \|M^{-1}s_k\|^2 \leq y_k^T s_k \leq (1 + \beta) \|M^{-1}s_k\|^2, \quad (5.4.113)$$

$$(b) \quad \left\| E \left[I - \frac{(M^{-1}s_k)(M^{-1}s_k)^T}{y_k^T s_k} \right] \right\|_F \leq \sqrt{1 - \alpha\theta^2} \|E\|_F, \quad (5.4.114)$$

$$(c) \quad \begin{aligned} & \left\| E \left[I - \frac{M^{-1}s_k(My_k)^T}{y_k^T s_k} \right] \right\|_F \\ & \leq \left[\sqrt{1 - \alpha\theta^2} + (1 - \beta)^{-1} \frac{\|My_k - M^{-1}s_k\|}{\|M^{-1}s_k\|} \right] \|E\|_F, \end{aligned} \quad (5.4.115)$$

where

$$\alpha = \frac{1 - 2\beta}{1 - \beta^2} \in \left[\frac{3}{8}, 1 \right], \quad \theta = \frac{\|EM^{-1}s_k\|}{\|E\|_F \|M^{-1}s_k\|} \in [0, 1]. \quad (5.4.116)$$

Proof. Note that

$$y_k^T s_k = (My_k)^T (M^{-1}s_k) = (My_k - M^{-1}s_k)^T M^{-1}s_k + \|M^{-1}s_k\|^2. \quad (5.4.117)$$

Also, it follows from Cauchy-Schwartz inequality and (5.4.112) that

$$|(My_k - M^{-1}s_k)^T M^{-1}s_k| \leq \beta \|M^{-1}s_k\|^2. \quad (5.4.118)$$

Then, combining (5.4.117) and (5.4.118) gives the first conclusion (a).

Now, we prove (b). By using the property (1.2.71) of Frobenius norm of a rank-one update, we have

$$\|E(I - uv^T)\|_F^2 = \|E\|_F^2 - 2v^T E^T E u + \|Eu\|^2 \|v\|^2.$$

In particular,

$$\begin{aligned} & \left\| E \left[I - \frac{(M^{-1}s_k)(M^{-1}s_k)^T}{y_k^T s_k} \right] \right\|_F^2 \\ &= \|E\|_F^2 + (-2y_k^T s_k + \|M^{-1}s_k\|^2) \frac{\|EM^{-1}s_k\|}{(y_k^T s_k)^2}. \end{aligned}$$

Using (a) and (5.4.116) yields

$$\begin{aligned} & \left\| E \left[I - \frac{(M^{-1}s_k)(M^{-1}s_k)^T}{y_k^T s_k} \right] \right\|_F^2 \\ &\leq \|E\|_F^2 - \left(\frac{1-2\beta}{1-\beta} \right) \frac{\|EM^{-1}s_k\|^2}{y_k^T s_k} \\ &\leq \|E\|_F^2 - \alpha \left(\frac{\|EM^{-1}s_k\|}{\|M^{-1}s_k\|} \right)^2 \\ &= \|E\|_F^2 (1 - \alpha\theta^2), \end{aligned}$$

which shows (b).

Finally, we prove (c) by means of (b). It is enough to prove that

$$\begin{aligned} & \left\| E \frac{M^{-1}s_k(M^{-1}s_k - My_k)^T}{y_k^T s_k} \right\|_F \\ &\leq (1-\beta)^{-1} \left(\frac{\|My_k - M^{-1}s_k\|}{\|M^{-1}s_k\|} \right) \|E\|_F. \end{aligned} \quad (5.4.119)$$

Since

$$\left\| \frac{M^{-1}s_k(M^{-1}s_k - My_k)^T}{y_k^T s_k} \right\|_F \leq \frac{\|M^{-1}s_k\| \|M^{-1}s_k - My_k\|}{y_k^T s_k},$$

then we obtain (5.4.119) by using (a). $\square$

Lemma 5.4.18 *Let $\{\phi_k\}$ and $\{\delta_k\}$ be sequences of nonnegative numbers satisfying*

$$\phi_{k+1} \leq (1 + \delta_k)\phi_k + \delta_k \quad (5.4.120)$$

and

$$\sum_{k=1}^{\infty} \delta_k < +\infty, \quad (5.4.121)$$

then $\{\phi_k\}$ converges.

Proof. We first prove that $\{\phi_k\}$ is bounded above. Let

$$\mu_k = \prod_{j=1}^{k-1} (1 + \delta_j).$$

Obviously, $\mu_k \geq 1$. Inequality (5.4.121) indicates that there exists a constant μ such that $\mu_k \leq \mu$. By using (5.4.120), we have

$$\frac{\phi_{k+1}}{\mu_{k+1}} \leq \frac{\phi_k}{\mu_k} + \frac{\delta_k}{\mu_{k+1}} \leq \frac{\phi_k}{\mu_k} + \delta_k.$$

Hence

$$\frac{\phi_{m+1}}{\mu_{m+1}} \leq \frac{\phi_1}{\mu_1} + \sum_{k=1}^m \delta_k.$$

From (5.4.121) and the boundedness of $\{\mu_k\}$, we obtain that $\{\phi_k\}$ is bounded.

Since $\{\phi_k\}$ is bounded, then there is at least a limit point. Suppose that there are two subsequences $\{\phi_{k_n}\}$ and $\{\phi_{k_m}\}$, which converge to ϕ' and ϕ'' respectively. We can show that $\phi' \leq \phi''$, and that $\phi'' \leq \phi'$ by symmetry. Thus $\phi' = \phi''$ and $\{\phi_k\}$ is convergent.

In fact, let ϕ be a bound of $\{\phi_k\}$. Let also, for example, $k_n \geq k_m$. From (5.4.120), we have

$$\phi_{k_n} - \phi_{k_m} \leq (1 + \phi) \sum_{j=k_m}^{k_n-1} \delta_j.$$

By the selection of k_n , we have

$$\phi' - \phi_{k_m} \leq (1 + \phi) \sum_{j=k_m}^{\infty} \delta_j.$$

By the selection of k_m , we have

$$\phi' - \phi'' \leq 0.$$

Therefore $\phi' \leq \phi''$. Similarly, by symmetry, we obtain $\phi'' \leq \phi'$. We complete the proof. $\square$

We have known that if $f : R^n \rightarrow R$ satisfies Assumption 5.4.2, then (5.4.84) holds. Let $\|B_k - A\|_{DFP} = \phi_k$ and $\max\{\alpha_1 \sigma(x_k, x_{k+1}), \alpha_2 \sigma(x_k, x_{k+1})\} = \delta_k$. Then (5.4.121) holds. Thus, it follows from Lemma 5.4.18 that the limit

$$\lim_{k \rightarrow +\infty} \|B_k - A\|_{DFP} \quad (5.4.122)$$

exists.

Lemma 5.4.19 *Under the assumptions of Theorem 5.4.15, there exist positive constants β_1, β_2 , and β_3 , such that $\forall x_k, x_{k+1} \in N(x^*, \varepsilon)$, we have*

$$\|B_{k+1} - \nabla^2 f(x^*)\|_{DFP} \leq \left[\sqrt{1 - \beta_1 \theta_k^2} + \beta_2 \sigma(x_k, x_{k+1}) \right] \|B_k - \nabla^2 f(x^*)\|_{DFP} + \beta_3 \sigma(x_k, x_{k+1}), \quad (5.4.123)$$

where

$$\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}, \quad (5.4.124)$$

$$\theta_k = \frac{\|\nabla^2 f(x^*)^{-\frac{1}{2}} [B_k - \nabla^2 f(x^*)^{\frac{1}{2}}] s_k\|}{\|B_k - \nabla^2 f(x^*)\|_{DFP} \|\nabla^2 f(x^*)^{\frac{1}{2}} s_k\|}. \quad (5.4.125)$$

Proof. Write $A = \nabla^2 f(x^*)$. From (5.4.85), we have

$$\begin{aligned} \|B_{k+1} - A\|_{DFP} &\leq \|P^T(B_k - A)P\|_{DFP} + \left\| \frac{(y_k - As_k)y_k^T}{y_k^T s_k} \right\|_{DFP} \\ &\quad + \left\| \frac{y_k(y_k - As_k)^T P}{y_k^T s_k} \right\|_{DFP}. \end{aligned} \quad (5.4.126)$$

Let

$$Q = I - \frac{A^{\frac{1}{2}} s_k y_k^T A^{-\frac{1}{2}}}{y_k^T s_k}, \quad E_k = A^{-\frac{1}{2}}(B_k - A)A^{-\frac{1}{2}}. \quad (5.4.127)$$

Then

$$\begin{aligned} \|P^T(B_k - A)P\|_{DFP} &= \|(A^{-\frac{1}{2}} P^T A^{\frac{1}{2}})(A^{-\frac{1}{2}}(B_k - A)A^{-\frac{1}{2}})(A^{\frac{1}{2}} P A^{-\frac{1}{2}})\|_F \\ &= \|Q^T E Q\|_F. \end{aligned}$$

Similar to the proof of Theorem 5.4.15, we know that there exist α_3 and $\alpha_4 > 0$ such that

$$\begin{aligned} \left\| \frac{(y_k - As_k)y_k^T}{y_k^T s_k} \right\|_{DFP} &\leq \frac{1}{\omega} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|} \leq \alpha_3 \sigma(x_k, x_{k+1}), \\ \left\| \frac{y_k(y_k - As_k)^T P}{y_k^T s_k} \right\|_{DFP} &\leq \frac{1}{\omega^2} \frac{\|A^{-\frac{1}{2}} y_k - A^{\frac{1}{2}} s_k\|}{\|A^{\frac{1}{2}} s_k\|} \leq \alpha_4 \sigma(x_k, x_{k+1}). \end{aligned}$$

If we let $\beta_3 = \alpha_3 + \alpha_4$, then (5.4.126) becomes

$$\|B_{k+1} - A\|_{DFP} \leq \|Q^T E Q\|_F + \beta_3 \sigma(x_k, x_{k+1}). \quad (5.4.128)$$

Since

$$\frac{\|A^{-\frac{1}{2}}y_k - A^{\frac{1}{2}}s_k\|}{\|A^{\frac{1}{2}}s_k\|} \leq \mu\gamma\sigma(x_k, x_{k+1}) \leq \frac{1}{3},$$

then, by use of Lemma 5.4.17, we obtain

$$\|Q^T EQ\|_F \leq \left[1 + (1 - \beta)^{-1} \frac{\|A^{-\frac{1}{2}}y_k - A^{\frac{1}{2}}s_k\|}{\|A^{\frac{1}{2}}s_k\|} \right] \|Q^T E\|_F.$$

Note that $\|Q^T E\|_F = \|E^T Q\|_F = \|EQ\|_F$, thus, by using Lemma 5.4.17 once more, we obtain

$$\|EQ\|_F \leq \left[\sqrt{1 - \alpha\theta_k^2} + (1 - \beta)^{-1} \frac{\|A^{-\frac{1}{2}}y_k - A^{\frac{1}{2}}s_k\|}{\|A^{\frac{1}{2}}s_k\|} \right] \|E\|_F,$$

where θ_k is defined by (5.4.125). Then

$$\begin{aligned} \|Q^T EQ\|_F &\leq \left[\sqrt{1 - \alpha\theta_k^2} + \frac{5}{2}(1 - \beta)^{-1} \frac{\|A^{-\frac{1}{2}}y_k - A^{\frac{1}{2}}s_k\|}{\|A^{\frac{1}{2}}s_k\|} \right] \|E\|_F \\ &\leq [\sqrt{1 - \beta_1\theta_k^2} + \beta_2\sigma(x_k, x_{k+1})] \|E\|_F, \end{aligned} \quad (5.4.129)$$

where $\beta_1 = \alpha$, $\beta_2 = \frac{5}{2}(1 - \beta)^{-1}\mu\gamma$. Substituting (5.4.129) into (5.4.128), we deduce the desired result (5.4.123). The proof is complete. $\square$

Using the above three lemmas, we can establish the following superlinear convergence theorem of DFP method.

Theorem 5.4.20 *Under the assumptions of Theorem 5.4.15, DFP method defined by (5.4.79)–(5.4.80) is convergent superlinearly.*

Proof. Since $(1 - \beta_1\theta_k^2)^{\frac{1}{2}} \leq 1 - (\beta_1/2)\theta_k^2$, then (5.4.123) can be written as

$$\begin{aligned} (\beta_1\theta_k^2/2)\|B_k - A\|_{DFP} &\leq \|B_k - A\|_{DFP} - \|B_{k+1} - A\|_{DFP} \\ &\quad + [\beta_2\|B_k - A\|_{DFP} + \beta_3]\sigma(x_k, x_{k+1}). \end{aligned}$$

Summing both sides yields

$$\begin{aligned} \frac{1}{2}\beta_1 \sum_{k=1}^{\infty} \theta_k^2 \|B_k - A\|_{DFP} &\leq \|B_1 - A\|_{DFP} + \beta_2 \sum_{k=1}^{\infty} \sigma(x_k, x_{k+1}) \|B_k - A\|_{DFP} \\ &\quad + \beta_3 \sum_{k=1}^{\infty} \sigma(x_k, x_{k+1}). \end{aligned}$$

Since, from Theorem 5.4.15, $\{x_k\}$ is linearly convergent, then $\sum_{k=1}^{\infty} \sigma(x_k, x_{k+1}) < \infty$. Also, since $\{\|B_k - A\|_{DFP}\}$ is bounded, then

$$\frac{\beta_1}{2} \sum_{k=1}^{\infty} \theta_k^2 \|B_k - A\|_{DFP} < \infty.$$

By (5.4.56), the limit $\lim_{k \rightarrow \infty} \|B_k - A\|_{DFP}$ exists. Hence, if some subsequence of $\{\|B_k - A\|_{DFP}\}$ converges to zero, the whole sequence converges to zero. Therefore

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - A)s_k\|}{\|s_k\|} = 0,$$

and the conclusion holds. Otherwise, if $\|B_k - A\|_{DFP} \geq \omega > 0, \forall k \geq k_0$, then $\theta_k \rightarrow 0$. Note that

$$\begin{aligned} \frac{\|(B_k - A)s_k\|}{\|s_k\|} &\leq \frac{\|A^{\frac{1}{2}}\| \|A^{-\frac{1}{2}}(B_k - A)s_k\|}{\|A^{\frac{1}{2}}\|^{-1} \|A^{\frac{1}{2}}s_k\|} \\ &= \|A\| \|B_k - A\|_{DFP} \frac{\|A^{-\frac{1}{2}}(B_k - A)s_k\|}{\|B_k - A\|_{DFP} \|A^{\frac{1}{2}}s_k\|} \\ &= \|A\| \|B_k - A\|_{DFP} \theta_k, \end{aligned}$$

then, by using $\theta_k \rightarrow 0$, we immediately obtain

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - A)s_k\|}{\|s_k\|} = 0.$$

Hence $\{x_k\}$ is convergent superlinearly. We complete the proof. $\square$

Similarly, we can state the superlinear convergence theorem for BFGS method.

Theorem 5.4.21 *Under the assumptions of Theorem 5.4.15, the sequence $\{x_k\}$ generated by BFGS method (5.4.79) and (5.1.45) is convergent to x^* superlinearly.*

It is not difficult to describe the above theorems in inverse Hessian approximations, which proofs are left to interested readers as an exercise.

We consider BFGS update in inverse Hessian approximation (5.1.48), now written again as

$$x_{k+1} = x_k - H_k g_k, \tag{5.4.130}$$

$$\begin{aligned}
 H_{k+1} = H_k &+ \frac{(s_k - H_k y_k) s_k^T + s_k (s_k - H_k y_k)^T}{s_k^T y_k} \\
 &- \frac{(s_k - H_k y_k)^T y_k}{(s_k^T y_k)^2} s_k s_k^T.
 \end{aligned} \tag{5.4.131}$$

We employ the weighted norm

$$\|E\|_{BFGS} = \|E\|_{A^{1/2}, F} = \|A^{1/2} E A^{1/2}\|_F, \tag{5.4.132}$$

where $A = \nabla^2 f(x^*)$.

Theorem 5.4.22 *Let $f : R^n \rightarrow R$ satisfy Assumption 5.4.2. Also let*

$$\mu \gamma \sigma(x_k, x_{k+1}) \leq \frac{1}{3} \tag{5.4.133}$$

in a neighborhood of x^ , where $\mu = \|\nabla^2 f(x^*)^{-1}\|$ and $\sigma(x_k, x_{k+1}) = \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\}$. Then, there exist $\varepsilon > 0$ and $\delta > 0$ such that for $\|x_0 - x^*\| < \varepsilon$ and $\|H_0 - \nabla^2 f(x^*)^{-1}\|_{BFGS} < \delta$, BFGS method (5.4.130)–(5.4.131) is well-defined, and the produced sequence $\{x_k\}$ converges to x^* linearly. Further, if $\sum_{k=0}^{\infty} \|x_k - x^*\| < +\infty$, then the sequence $\{x_k\}$ converges to x^* superlinearly.*

5.4.7 Local Convergence of Broyden's Class Methods

Finally, in this section, we discuss local convergence of Broyden's class methods.

Byrd, Nocedal and Yuan [47] proved the superlinear convergence of Broyden's class method. We state the theorem without proof.

Theorem 5.4.23 *Suppose that $f : R^n \rightarrow R$ is twice continuously differentiable on a convex set D and that $f(x)$ is uniformly convex, i.e., there exists $m > 0$ such that for any $x \in R^n$ and $u \in R^n$,*

$$u^T \nabla^2 f(x) u \geq m \|u\|^2.$$

Suppose also that there is a neighborhood $N(x^, \varepsilon)$ of x^* , such that*

$$\|\nabla^2 f(\bar{x}) - \nabla^2 f(x)\| \leq \gamma \|\bar{x} - x\|, \quad \forall x, \bar{x} \in N(x^*, \varepsilon).$$

Then, for any positive definite matrix B_0 , when line search satisfies Wolfe-Powell rule (5.4.30)–(5.4.33), the sequence $\{x_k\}$ generated by the restricted Broyden's class ($\theta \in (0, 1)$) converges to x^ superlinearly.*

For Broyden's class with exact line search, we have

Theorem 5.4.24 *Suppose that the assumptions of Theorem 5.4.23 hold. When the exact line search is employed, the sequence $\{x_k\}$ generated by Broyden's class method converges to x^* superlinearly.*

Byrd, Liu, and Nocedal [43] established the following superlinear characterization in which the superlinear characterization (5.4.25) is replaced by (5.4.135) and (5.4.136).

Theorem 5.4.25 *Let iterates generated by*

$$x_{k+1} = x_k - \alpha_k B_k^{-1} g_k$$

converge to x^ with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ positive definite. Then*

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0 \quad (5.4.134)$$

if and only if

$$\lim_{k \rightarrow \infty} \cos^2 \langle B_k^{-1} g_k, -\nabla^2 f(x^*)^{-1} g_k \rangle = 1 \quad (5.4.135)$$

and

$$\lim_{k \rightarrow \infty} \frac{s_k^T B_k s_k}{\alpha_k s_k^T y_k} = 1. \quad (5.4.136)$$

Proof. Suppose that (5.4.134) holds, then we have

$$\lim_{k \rightarrow \infty} \cos^2 \langle B_k^{-1} g_k, x_k - x^* \rangle = 1. \quad (5.4.137)$$

Note also that

$$\lim_{k \rightarrow \infty} \cos^2 \langle x_k - x^*, -\nabla^2 f(x^*)^{-1} g_k \rangle = 1. \quad (5.4.138)$$

Hence (5.4.135) holds.

By (5.4.134) and the positive definiteness of $\nabla^2 f(x^*)$, we have

$$\lim_{k \rightarrow \infty} \frac{\|g_k + y_k\|}{\|g_k\|} = 0,$$

which implies

$$\lim_{k \rightarrow \infty} \frac{s_k^T g_k + s_k^T y_k}{\|s_k\| \|g_k\|} = 0.$$

Therefore

$$\lim_{k \rightarrow \infty} \frac{-s_k^T g_k}{s_k^T y_k} = 1, \quad (5.4.139)$$

which means (5.4.136).

Conversely, assume that (5.4.135) and (5.4.136) hold. By (5.4.135) and (5.4.138) we deduce that (5.4.137) holds. Also, (5.4.136) means (5.4.139). Then, we obtain

$$\lim_{k \rightarrow \infty} \frac{s_k^T g_k + s_k^T \nabla^2 f(x^*) s_k}{s_k^T y_k} = 0,$$

which is

$$\lim_{k \rightarrow \infty} \frac{s_k^T \nabla^2 f(x^*) [s_k + \nabla^2 f(x^*)^{-1} g_k]}{s_k^T \nabla^2 f(x^*) s_k} = 0. \quad (5.4.140)$$

Then, (5.4.140) and (5.4.135) gives

$$\lim_{k \rightarrow \infty} \frac{\|s_k + \nabla^2 f(x^*)^{-1} g_k\|}{\|s_k\|} = 0, \quad (5.4.141)$$

which is equivalent to (5.4.134). We complete the proof. $\square$

5.5 Self-Scaling Variable Metric (SSVM) Methods

5.5.1 Motivation to SSVM Method

We have seen that DFP method is a typical rank-two quasi-Newton method. However, numerical experiments show that its implementation is not ideal. Why? Below, we would like to give some analysis.

First, we clarify that the single-step convergence Theorem 3.1.5 of the steepest descent method is also true for various Newton-like methods. Let

$$f(x) = \frac{1}{2} x^T G x - b^T x, \quad (5.5.1)$$

where G is an $n \times n$ symmetric and positive definite matrix. Let the Newton-like method be defined by

$$x_{k+1} = x_k - \alpha_k H_k g_k, \quad (5.5.2)$$

where

$$g_k = G x_k - b, \quad (5.5.3)$$

$$\alpha_k = g_k^T H_k g_k / g_k^T H_k G H_k g_k, \quad (5.5.4)$$

then we have the following theorem.

Theorem 5.5.1 *Let x^* be a minimizer of the quadratic function (5.5.1), and let Newton-like methods be defined by (5.5.2). Then, the single-step convergence rate satisfies the following bound:*

$$\frac{f(x_{k+1}) - f(x^*)}{f(x_k) - f(x^*)} \leq \frac{(\lambda_1 - \lambda_n)^2}{(\lambda_1 + \lambda_n)^2}, \quad (5.5.5)$$

$$E(x_{k+1}) \leq \frac{(\lambda_1 - \lambda_n)^2}{(\lambda_1 + \lambda_n)^2} E(x_k), \quad (5.5.6)$$

where $E(x_k) = \frac{1}{2}(x_k - x^*)^T G(x_k - x^*)$, λ_1 and λ_n are the largest and the smallest eigenvalues of matrix $H_k G$ respectively.

Proof. Since

$$x^* = x_k - G^{-1}g_k \quad (5.5.7)$$

and

$$f(x_k) - f(x^*) = \frac{1}{2}g_k^T G^{-1}g_k, \quad (5.5.8)$$

and since the exact line search factor α_k is represented by (5.5.4), we have

$$f(x_{k+1}) = f(x_k) - \frac{1}{2}\alpha_k^2 g_k^T H_k G H_k g_k$$

and

$$f(x_{k+1}) - f(x^*) = \frac{1}{2}g_k^T G^{-1}g_k - \frac{1}{2}\alpha_k^2 g_k^T H_k G H_k g_k.$$

Hence

$$\begin{aligned} & \frac{f(x_{k+1}) - f(x^*)}{f(x_k) - f(x^*)} \\ &= 1 - \frac{(g_k^T H_k g_k)^2}{(g_k^T G^{-1}g_k)(g_k^T H_k G H_k g_k)} \\ &= 1 - \frac{(z_k^T z_k)^2}{(z_k^T (H_k^{-\frac{1}{2}})^T G^{-1} H_k^{-\frac{1}{2}} z_k)(z_k^T H_k^{\frac{1}{2}} G (H_k^{\frac{1}{2}})^T z_k)}, \end{aligned} \quad (5.5.9)$$

where $z_k = H_k^{\frac{1}{2}}g_k$. Then the conclusion (5.5.5) is obtained by using Kantorovich Theorem 3.1.10.

Similarly, we have

$$\frac{E(x_k) - E(x_{k+1})}{E(x_k)} = \frac{(z_k^T z_k)^2}{(z_k^T T_k z_k)(z_k^T T_k^{-1} z_k)},$$

where $T_k = H_k^{\frac{1}{2}} G H_k^{\frac{1}{2}}$. By using Kantorovich Theorem 3.1.10 and noting that $H_k G$ and T_k are similar, we immediately obtain the conclusion (5.5.6). $\square$

From Theorem 5.5.1, we may see that if the condition number $\kappa(T_k)$ is very large, the single-step convergence rate will be very slow. In order to obtain a rapid rate in every iteration, we should make

$$\left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right)^2 \text{ or } \left[\frac{\kappa(T_k) - 1}{\kappa(T_k) + 1} \right]^2 \quad (5.5.10)$$

as small as possible, where $\kappa(T_k) = \lambda_1/\lambda_n$.

Second, let us observe carefully the DFP method. It is not difficult to see the fact that, usually, the eigenvalues of $H_0 G$ are greater than 1, and that DFP method and Broyden class method make one eigenvalue to being 1 in essence in each iteration. Hence, in the iterative procedure, a non-ideal eigen-ratio of $\{H_k G\}$ is produced. Also since $H_k G$ and T_k are similar, the eigen-ratio of $\{T_k\}$ is also non-ideal.

In fact, if we let

$$R_k = G^{\frac{1}{2}} H_k G^{\frac{1}{2}}, \quad r_k = G^{\frac{1}{2}} s_k, \quad (5.5.11)$$

then R_k is similar to $H_k G$, and further to T_k . By using $y_k = G^{\frac{1}{2}} s_k$, the DFP formula (5.1.30) is equivalent to

$$R_{k+1} = R_k - \frac{R_k r_k r_k^T R_k}{r_k^T R_k r_k} + \frac{r_k r_k^T}{r_k^T r_k}. \quad (5.5.12)$$

Let the eigenvalues of R_k satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n > 0$. Let

$$P = R_k - \frac{R_k r_k r_k^T R_k}{r_k^T R_k r_k} \quad (5.5.13)$$

with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$. Obviously, $P r_k = 0$. Then we have

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_n \geq \mu_n = 0. \quad (5.5.14)$$

From (5.5.12), it follows that

$$R_{k+1} = P + \frac{r_k r_k^T}{r_k^T r_k} \quad (5.5.15)$$

and

$$R_{k+1} r_k = r_k. \quad (5.5.16)$$

Since r_k is the eigenvector of P , and since P is symmetric, then all other eigenvectors of P are orthogonal to r_k . So, the unique different eigenvalue between R_{k+1} and P is the eigenvalue associated to r_k , which is 1. This shows that DFP method moves one eigenvalue of R_k to 1 in each iteration. Note that R_k is similar to $H_k G$, thus, it implies that if all eigenvalues of $H_0 G$ are greater than 1, then the eigen-ratio of $H_k G$ will worsen.

However, if $1 \in [\lambda_n, \lambda_1]$, then, it follows from the above discussion that the eigenvalues $\mu_1, \mu_2, \dots, \mu_{n-1}$ of R_{k+1} and 1 will be contained in $[\lambda_n, \lambda_1]$. Hence, in this case, the eigen-ratio of $H_k G$ will not worsen. This conclusion is true for updates of Broyden class with $0 \leq \phi \leq 1$.

Theorem 5.5.2 *Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be eigenvalues of $H_k G$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$. Suppose that $1 \in [\lambda_n, \lambda_1]$. Then, for any ϕ with $0 \leq \phi \leq 1$, the eigenvalues of $H_{k+1}^\phi G$ are contained in $[\lambda_n, \lambda_1]$, where H_{k+1}^ϕ is the Broyden class update defined by (5.2.4).*

Proof. The case $\phi = 0$ has been proved as before.

Now we consider the case $\phi = 1$ (BFGS update). The BFGS formula (5.1.45) can be written as

$$H_{k+1}^{-1} = H_k^{-1} + \frac{y_k y_k^T}{s_k^T y_k} - \frac{H_k^{-1} s_k s_k^T H_k^{-1}}{s_k^T H_k^{-1} s_k},$$

which is equivalent to

$$R_{k+1}^{-1} = R_k^{-1} - \frac{R_k^{-1} r_k r_k^T R_k^{-1}}{r_k^T R_k^{-1} r_k} + \frac{r_k r_k^T}{r_k^T r_k}. \quad (5.5.17)$$

Since the eigenvalues of R_k^{-1} satisfy

$$\frac{1}{\lambda_1} \leq \frac{1}{\lambda_2} \leq \dots \leq \frac{1}{\lambda_n},$$

then we have $1 \in [1/\lambda_1, 1/\lambda_n]$. Similar to the above discussion, we know that if the eigenvalues of R_{k+1}^{-1} satisfy $1/\mu_1 \leq 1/\mu_2 \leq \dots \leq 1/\mu_n$, then these eigenvalues are contained in $[1/\lambda_1, 1/\lambda_n]$. Hence, we have that $1/\lambda_1 \leq 1/\mu_1$ and $1/\lambda_n \geq 1/\mu_n$, i.e., $\mu_n \geq \lambda_n$ and $\mu_1 \leq \lambda_1$. This shows that all eigenvalues of R_{k+1} are contained in $[\lambda_n, \lambda_1]$. Therefore, the conclusion holds for $\phi = 1$.

Finally, we know that Broyden class updating formula (5.2.4) is equivalent to

$$R_{k+1}^\phi = R_k - \frac{R_k r_k r_k^T R_k}{r_k^T R_k r_k} + \frac{r_k r_k^T}{r_k^T r_k} + \phi u_k u_k^T, \quad (5.5.18)$$

where

$$u_k = G^{\frac{1}{2}} v_k = (r_k^T R_k r_k)^{\frac{1}{2}} \left[\frac{r_k}{r_k^T r_k} - \frac{R_k r_k}{r_k^T R_k r_k} \right]. \quad (5.5.19)$$

Clearly, the eigenvalues of R_{k+1}^ϕ are increasing monotonically as k increases. Since, for $\phi = 0$ and $\phi = 1$, the eigenvalues of R_{k+1}^ϕ are contained in $[\lambda_n, \lambda_1]$, then, for $0 \leq \phi \leq 1$, the eigenvalues of R_{k+1}^ϕ are also contained in $[\lambda_n, \lambda_1]$. Thus, from the fact that R_{k+1}^ϕ and $H_{k+1}^\phi G$ are similar, we obtain the conclusion. $\square$

The above theorem says that if we scale the matrix H_k such that the eigenvalues of $H_k G$ satisfy $1 \in [\lambda_n, \lambda_1]$, the eigenvalue structure of R_{k+1}^ϕ will be improved.

Obviously, for a quadratic function, it is enough to scale only the initial matrix H_0 . However, in general, it is useful to scale each H_k .

5.5.2 Self-Scaling Variable Metric (SSVM) Method

In this section we describe SSVM method due to Oren [237]. Multiplying H_k by γ_k and then replacing H_k by $\gamma_k H_k$ in (5.2.2) yield

$$H_{k+1}^{(\phi, \gamma_k)} = \left(H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \phi v_k v_k^T \right) \gamma_k + \frac{s_k s_k^T}{s_k^T y_k}, \quad (5.5.20)$$

where

$$v_k = (y_k^T H_k y_k)^{\frac{1}{2}} [s_k / s_k^T y_k - H_k y_k / y_k^T H_k y_k],$$

where ϕ is a parameter of Broyden class and γ_k a self-scaling parameter. The formula (5.5.20) is referred to as the self-scaling variable metric (SSVM) formula. When $\gamma_k = 1$, it is reduced to Broyden class update.

Algorithm 5.5.3 (*SSVM Algorithm*)

Step 0. Given an initial matrix H_0 and a starting point x_0 . Set $k = 0$.

Step 1. Set $d_k = -H_k g_k$.

Step 2. Find stepsize α_k , and set $x_{k+1} = x_k + \alpha_k d_k$, compute g_{k+1} and set $y_k = g_{k+1} - g_k$.

Step 3. Choose Broyden's class parameter $\phi \geq 0$ and self-scaling parameter $\gamma_k > 0$, and compute $H_{k+1}^{(\phi, \gamma_k)}$ by (5.5.20).

Step 4. $k := k + 1$, go to Step 1. $\square$

Similar to the discussion of DFP method in §5.1, we can prove that the SSVM method has the following properties. The proof is omitted.

Theorem 5.5.4 (*Properties of SSVM Method*)

1. If H_k is positive definite and $s_k^T y_k > 0$, then when $\phi \geq 0$ and $\gamma_k > 0$, the matrix $H_{k+1}^{(\phi, \gamma_k)}$ produced by (5.5.20) is positive definite.
2. If $f(x)$ is a quadratic function with Hessian G , the vectors $s_0, s_1, \dots, s_{n-1}$ produced by SSVM method are G -conjugate, i.e., satisfy

$$s_i^T G s_j = 0, \quad i \neq j; \quad i, j = 0, 1, \dots, n-1, \quad (5.5.21)$$

and for each k , $s_0, s_1, \dots, s_k$ are the eigenvalues of $H_{k+1}^{(\phi, \gamma_k)} G$, i.e., satisfy

$$H_{k+1}^{(\phi, \gamma_k)} G s_i = \bar{\gamma}_{i,k} s_i, \quad 0 < i < k, \quad (5.5.22)$$

where $\bar{\gamma}_{i,k} = \prod_{j=i+1}^k \gamma_j$, $\bar{\gamma}_{ii} = 1$.

This theorem shows that although the property $H_n^{(\phi, \gamma_{n-1})} = G^{-1}$ is not retained for quadratic functions by SSVM method, the property of conjugate directions is still retained. Therefore, for quadratic functions, the sequence generated from SSVM method converges to a minimizer in at most n steps.

5.5.3 Choices of the Scaling Factor

Now, the problem is how to choose a suitable scaling factor. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$ be eigenvalues of $H_k G$. Clearly, they are also the eigenvalues of R_k . We hope to choose a suitable scaling factor which is used to multiply H_k , such that 1 is contained among the new eigenvalues and thus the eigen-structure is improved. Therefore we get $\kappa(R_{k+1}^\phi) \leq \kappa(R_k)$. The following theorem is a consequence of Theorem 5.5.2.

Theorem 5.5.5 *Let $\phi \in [0, 1]$ and $\gamma_k > 0$. Let R_k and R_{k+1}^ϕ be defined respectively by (5.5.11) and (5.5.24). Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ and $\mu_1^\phi \geq \mu_2^\phi \geq \dots \geq \mu_n^\phi$ be eigenvalues of R_k and R_{k+1}^ϕ respectively. Then the following statements hold.*

1. *If $\gamma_k \lambda_n \geq 1$, then $\mu_n^\phi = 1$ and $1 \leq \gamma_k \lambda_{i+1} \leq \mu_i^\phi \leq \gamma_k \lambda_i, i = 1, 2, \dots, n-1$.*
2. *If $\gamma_k \lambda_1 \leq 1$, then $\mu_1^\phi = 1$ and $\gamma_k \lambda_i \leq \mu_i^\phi \leq \gamma_k \lambda_{i-1} \leq 1, i = 2, 3, \dots, n$.*
3. *If $\gamma_k \lambda_n \leq 1 \leq \gamma_k \lambda_1$ and i_0 is an index with $\gamma_k \lambda_{i_0+1} \leq 1 \leq \gamma_k \lambda_{i_0}$, then*

$$\begin{aligned} \gamma_k \lambda_1 \geq \mu_1^\phi \geq \gamma_k \lambda_2 &\geq \mu_2^\phi \geq \dots \geq \gamma_k \lambda_{i_0} \geq \mu_{i_0} \geq 1 \geq \mu_{i_0+1} \\ &\geq \gamma_k \lambda_{i_0+1} \geq \dots \geq \gamma_k \lambda_n, \end{aligned} \quad (5.5.23)$$

and there is at least one eigenvalue in $\mu_{i_0}^\phi$ and $\mu_{i_0+1}^\phi$ which equals 1.

Proof. This theorem is a direct consequence of Theorem 5.5.2. Since SSVM method is equivalent to

$$R_{k+1}^\phi = \left(R_k - \frac{R_k r_k r_k^T R_k}{r_k^T R_k r_k} + \phi u_k u_k^T \right) \gamma_k + \frac{r_k r_k^T}{r_k^T r_k}, \quad (5.5.24)$$

where r_k and u_k are defined respectively by (5.5.11) and (5.5.19), the above expression is just obtained by replacing R_k by $\gamma_k R_k$ in (5.5.18). Therefore, from Theorem 5.5.2, replacing $\lambda_1, \lambda_2, \dots, \lambda_n$ by use of $\gamma_k \lambda_1, \dots, \gamma_k \lambda_n$ gives our conclusion. $\square$

Corollary 5.5.6 *Let $\phi \in [0, 1]$ and $\gamma_k = 1$. Then*

$$|\mu_k^\phi - 1| \leq |\lambda_k - 1|. \quad (5.5.25)$$

Proof. From Theorem 5.5.5, for $\gamma_k = 1$, one of the following cases will hold:

(a) $\lambda_i \geq \mu_i^\phi \geq 1$;

(b) $\lambda_i \leq \mu_i^\phi \leq 1$.

Hence the conclusion (5.5.25) is obtained. $\square$

Obviously, if we choose γ_k such that

$$\lambda_n \leq \frac{1}{\gamma_k} \leq \lambda_1, \quad (5.5.26)$$

we have

$$\gamma_k \lambda_n \leq 1 \leq \gamma_k \lambda_1, \quad (5.5.27)$$

which says that 1 is included in the interval of scaled eigenvalues. In addition, we have

Corollary 5.5.7 *Let $\phi \in [0, 1]$ and $\gamma_k > 0$. Let $\kappa(\cdot)$ denote the condition number. If $\lambda_n \leq \frac{1}{\gamma_k} \leq \lambda_1$, then, for (5.5.24), we have*

$$\kappa(R_{k+1}^\phi) \leq \kappa(R_k). \quad (5.5.28)$$

Proof. From Theorem 5.5.5 (3), it follows that

$$\gamma_k \lambda_1 \geq \mu_1^\phi \geq 1 \geq \mu_n^\phi \geq \gamma_k \lambda_n, \quad (5.5.29)$$

which gives

$$\frac{\mu_1^\phi}{\mu_n^\phi} \leq \frac{\lambda_1}{\lambda_n}.$$

Thus, we complete the proof. $\square$

In the above discussion about the condition of γ_k , we always restrict the Broyden class parameter $\phi \in [0, 1]$. In fact, this restriction is sufficient and also necessary for the statement that if $\lambda_n \leq \frac{1}{\gamma_k} \leq \lambda_1$, then $\kappa(R_{k+1}^\phi) \leq \kappa(R_k)$ and $H_{k+1}^{(\phi, \gamma_k)}$ is positive definite.

Corollary 5.5.7 says that $\lambda_n \leq \frac{1}{\gamma_k} \leq \lambda_1$ is a suitable requirement to choose a scaling factor. Note that

$$\frac{r_k^T R_k r_k}{r_k^T r_k} = \frac{y_k^T H_k y_k}{s_k^T y_k}$$

and

$$\lambda_n \leq \frac{r_k^T R_k r_k}{r_k^T r_k} \leq \lambda_1,$$

it follows that

$$\gamma_k = \frac{s_k^T y_k}{y_k^T H_k y_k} \quad (5.5.30)$$

is a suitable scaling factor. Similarly, since

$$\frac{r_k^T R_k^{-1} r_k}{r_k^T r_k} = \frac{s_k^T H_k^{-1} s_k}{s_k^T y_k}$$

and

$$\frac{1}{\lambda_1} \leq \frac{r_k^T R_k^{-1} r_k}{r_k^T r_k} \leq \frac{1}{\lambda_n},$$

we have that

$$\gamma_k = \frac{s_k^T H_k^{-1} s_k}{s_k^T y_k} = -\frac{\alpha_k s_k^T g_k}{s_k^T y_k} = \frac{s_k^T g_k}{g_k^T H_k y_k} \quad (5.5.31)$$

is also a suitable scaling factor. Noting that when α_k is an optimal stepsize, we have that $s_k^T y_k = -s_k^T g_k$, and thus

$$\gamma_k = \alpha_k. \quad (5.5.32)$$

The above (5.5.32) shows an interesting fact, that we may choose directly an optimal stepsize as a scaling factor.

For any $\omega \in [0, 1]$,

$$\gamma_k = (1 - \omega) \frac{s_k^T y_k}{y_k^T H_k y_k} + \omega \frac{s_k^T H_k^{-1} s_k}{s_k^T y_k} \quad (5.5.33)$$

is a convex combination of (5.5.30) and (5.5.31). Hence (5.5.33) gives a convex class of suitable scaling factors. For this convex class, Oren [239] presented the following switch rule of parameters ϕ and ω .

If $\frac{s_k^T y_k}{y_k^T H_k y_k} > 1$, choose $\phi = 1$ and $\omega = 0$,

(i.e., $\phi = 1, \gamma_k = s_k^T y_k / y_k^T H_k y_k$).

If $\frac{s_k^T H_k^{-1} s_k}{s_k^T y_k} < 1$, choose $\phi = 0, \omega = 1$.

(i.e., $\phi = 0, \gamma_k = \frac{s_k^T H_k^{-1} s_k}{s_k^T y_k}$).

$$\text{If } \frac{s_k^T y_k}{y_k^T H_k y_k} \leq 1 \leq \frac{s_k^T H_k^{-1} s_k}{s_k^T y_k}, \text{ choose}$$

$$\omega = \phi = \frac{s_k^T y_k (y_k^T H_k y_k - s_k^T y_k)}{(s_k^T H_k^{-1} s_k)(y_k^T H_k y_k) - (s_k^T y_k)^2}, \text{ (i.e., } \gamma_k = 1\text{)}.$$

Another technique is an initial scaling method presented by Shanno and Phua [306]. At the beginning, set $H_0 = I$, and the stepsize α_0 is determined by some line search, such that the objective function descends sufficiently. Before computing H_1 , instead of H_0 , we use

$$\hat{H}_0 = \alpha_0 H_0, \quad (5.5.34)$$

and compute H_1 from $\hat{H}_0$, where α_0 is a stepsize or determined by

$$\alpha_0 := \gamma_0 = \frac{s_0^T y_0}{y_0^T H_0 y_0}. \quad (5.5.35)$$

The difference between the initial scaling and SSVM is that SSVM does process scaling in each iteration, but the initial scaling method does only at the beginning. Numerical experiments show that the initial scaling is simple and effective for a lot of problems in which the curvature changes smoothly.

By the way, a special self-scaling BFGS formula

$$B_{k+1} = \frac{s_k^T y_k}{s_k^T B_k s_k} \left(B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} \right) + \frac{y_k y_k^T}{s_k^T y_k} \quad (5.5.36)$$

is used widely in practice.

5.6 Sparse Quasi-Newton Methods

Schubert [303] first extended quasi-Newton update to an unsymmetric sparse matrix and proposed a sparse quasi-Newton method for solving nonlinear equations. Powell and Toint [276], Toint [341] derived sparse quasi-Newton update respectively, and Steihaug [321] presented a sparse quasi-Newton method with preconditioning and established the convergence.

The sparse quasi-Newton method requires generating sparse quasi-Newton updates which have the same (or similar) sparsity pattern as the true Hessian. It means that the current Hessian approximation B_k reflects the nonzero structure of the true Hessian, i.e.,

$$(B_k)_{ij} = 0 \text{ for } (i, j) \in I, \quad (5.6.1)$$

where

$$I \triangleq \{(i, j) \mid [\nabla^2 f(x)]_{ij} = 0\} \quad (5.6.2)$$

is a set of integer pairs. We also define

$$J \triangleq \{(i, j) \mid [\nabla^2 f(x)]_{ij} \neq 0\}. \quad (5.6.3)$$

It says that J , a set of integer pairs, is a complement of I . So, we demand that B_{k+1} satisfies the quasi-Newton condition

$$B_{k+1}s_k = y_k, \quad (5.6.4)$$

and keeps symmetry and sparsity. Neglecting the subscript, we would like to find $\bar{B}$, such that

$$\bar{B} = B + E, \quad (5.6.5)$$

where E satisfies

$$Es = y - Bs, \quad (5.6.6)$$

$$E = E^T, \quad (5.6.7)$$

$$E_{ij} = 0, \quad (i, j) \in I, \quad (5.6.8)$$

where E_{ij} are elements of the matrix E . If we determined E , we can get $\bar{B}$ from (5.6.5). However, (5.6.6)–(5.6.8) cannot determine completely the matrix E . So, to this end, we require that $\bar{B}$ is as close as possible to B in Frobenius norm. Therefore, we consider the following minimization problem:

$$\min \quad \frac{1}{2} \|E\|_F^2 \quad (5.6.9)$$

$$\text{s.t.} \quad Es = r, \quad (5.6.10)$$

$$E = E^T, \quad (5.6.11)$$

$$E_{ij} = 0, \quad (i, j) \in I, \quad (5.6.12)$$

where r is assumed to be

$$r = y - Bs. \quad (5.6.13)$$

In the left part of the section, we denote the j -th component of the vector s by s_j , and define the component of vector $s(i)$ as

$$s(i)_j = \begin{cases} s_j, & (i, j) \in J \\ 0, & (i, j) \in I. \end{cases} \quad (5.6.14)$$

Then the condition (5.6.10) can be written as

$$\sum_{j=1}^n E_{ij} s(i)_j = r_i, \quad i = 1, \dots, n. \quad (5.6.15)$$

In order to let E be symmetric, take

$$E = \frac{1}{2}(A + A^T). \quad (5.6.16)$$

Then the problem (5.6.9)-(5.6.12) becomes the following problem: finding a matrix A , such that

$$\min \quad \frac{1}{8} \|A + A^T\|_F \quad (5.6.17)$$

$$\text{s.t.} \quad \sum_{j=1}^n (A_{ij} + A_{ji}) s(i)_j = 2r_i, \quad i = 1, \dots, n, \quad (5.6.18)$$

where A_{ij} denote the elements of A .

Now, we discuss solving the problem (5.6.17)-(5.6.18). The Lagrangian function is

$$\begin{aligned} \Phi(A, \lambda) &= \frac{1}{8} \sum_{i=1}^n \sum_{j=1}^n (A_{ij}^2 + A_{ji}^2 + 2A_{ij}A_{ji}) \\ &\quad - \sum_{i=1}^n \lambda_i \left[\sum_{j=1}^n (A_{ij} + A_{ji}) s(i)_j - 2r_i \right]. \end{aligned} \quad (5.6.19)$$

Setting the derivative with respect to A_{ij} to be zero, we have

$$\begin{aligned} \frac{\partial \Phi(A, \lambda)}{\partial A_{ij}} &= \frac{1}{2} (A_{ij} + A_{ji}) - \lambda_i s(i)_j - \lambda_j s(j)_i = 0, \\ i, j &= 1, \dots, n. \end{aligned} \quad (5.6.20)$$

By using (5.6.16), the above expression is just

$$E_{ij} = \lambda_i s(i)_j + \lambda_j s(j)_i, \quad i, j = 1, \dots, n. \quad (5.6.21)$$

In place of (5.6.18), we employ (5.6.15). Substituting (5.6.21) into (5.6.15), we obtain

$$\sum_{j=1}^n [\lambda_i s(i)_j + \lambda_j s(j)_i] s(i)_j = r_i, \quad i = 1, \dots, n, \quad (5.6.22)$$

which is

$$\lambda_i \sum_{j=1}^n [s(i)_j]^2 + \sum_{j=1}^n \lambda_j s(j)_i s(i)_j = r_i, \quad i = 1, \dots, n. \quad (5.6.23)$$

Thus, we derive the update formula

$$\bar{B} = B + E, \quad (5.6.24)$$

which is, from (5.6.21), that

$$\bar{B} = B + \sum_{i=1}^n \lambda_i [e_i s(i)^T + s(i) e_i^T], \quad (5.6.25)$$

where e_i is the i -th unit vector and λ is a Lagrange multiplier vector satisfying

$$Q\lambda = r, \quad (5.6.26)$$

where

$$Q = \sum_{i=1}^n (s(i)^T s e_i + e_i^T s s(i)) e_i^T. \quad (5.6.27)$$

In fact, as long as we notice that

$$\begin{aligned} Q\lambda &= r = Es = \sum_{i=1}^n \lambda_i [e_i s(i)^T s + s(i) e_i^T s] \\ &= \sum_{i=1}^n [s(i)^T s e_i + e_i^T s s(i)] e_i^T \lambda, \end{aligned}$$

we can immediately obtain (5.6.27).

The matrix Q defined above satisfies symmetry, sparsity and positive definiteness. The properties of symmetry and sparsity can be seen direct from (5.6.27). As to the positive definiteness of Q , we give the following theorem.

Theorem 5.6.1 *If all vectors $s(i)$ ($i = 1, \dots, n$) are nonzero, then the matrix Q is positive definite, that is*

$$z^T Q z > 0, \quad \forall z \in R^n, z \neq 0. \quad (5.6.28)$$

Proof. Take $z \neq 0, z \in R^n$. Let z_i denote the components of vector z . From (5.6.27),

$$\begin{aligned}
 z^T Q z &= \sum_{i=1}^n \sum_{j=1}^n z_i^T Q_{ij} z_j \\
 &= \sum_{i=1}^n \sum_{j=1}^n z_i s(i)_j s(j)_i z_j + \sum_{i=1}^n \sum_{j=1}^n [s(i)_j]^2 z_i^2 \\
 &= \sum_{(i,j) \in J} [z_i s_i s_j z_j + z_i^2 s_i^2] \\
 &= \frac{1}{2} \sum_{(i,j) \in J} [s_i z_j + s_j z_i]^2 \\
 &= 2 \sum_{i=1}^n z_i^2 s_i^2 + \frac{1}{2} \sum_{\substack{(i,j) \in J \\ i \neq j}} (z_i s_j + z_j s_i)^2 \\
 &\geq 0.
 \end{aligned} \tag{5.6.29}$$

Suppose that $z^T Q z = 0$; since $z \neq 0$, there exists a component of z , for example, $z_k \neq 0$, such that by (5.6.29) we have

$$z_k s_k = 0, \tag{5.6.30}$$

$$z_k s_j + z_j s_k = 0, (k, j) \in J, j \neq k. \tag{5.6.31}$$

Thus, $s_k = 0$. Furthermore, $s_j = 0, j \neq k, (k, j) \in J$. This is equivalent to $s(k) = 0$, which contradicts the assumption. We complete the proof. $\square$

Since Q is positive definite, it follows from (5.6.21) and (5.6.26) that

$$E_{ij} = (Q^{-1}r)_i s(i)_j + (Q^{-1}r)_j s(j)_i, \tag{5.6.32}$$

which can be written as

$$E_{ij} = \begin{cases} 0, & (i, j) \in I, \\ \lambda_i s_j + \lambda_j s_i, & (i, j) \in J. \end{cases} \tag{5.6.33}$$

The above discussion gives the derivation of general sparse quasi-Newton update.

Now, we turn to the sparse PSB update.

Let $F : R^n \rightarrow R^n$. For solving sparse nonlinear equations $F(x) = 0$, Schubert [303] first suggested that Broyden's rank-one update

$$\bar{B} = B + \frac{(y - Bs)s^T}{s^T s} \tag{5.6.34}$$

can be written in the following form

$$\bar{B} = B + \sum_{i=1}^n e_i e_i^T \frac{(y - Bs)s^T}{s^T s}, \quad (5.6.35)$$

which is an update by row, where e_i is the i -th unit vector. By use of notation $s(i)$, one knows that

$$\bar{B} = B + \sum_{i=1}^n e_i e_i^T \frac{(y - Bs)s(i)^T}{s(i)^T s} \quad (5.6.36)$$

satisfies the quasi-Newton condition $\bar{B}s = y$, and has the sparsity pattern desired.

The general form of Schubert sparse update is

$$\bar{B} = B + \sum_{i=1}^n \alpha_i e_i z(i)^T, \quad (5.6.37)$$

where

$$\alpha_i = \frac{e_i^T (y - Bs)}{s(i)^T s}, \quad z(i)_j = \begin{cases} z_j, & (i, j) \in J, \\ 0, & (i, j) \in I. \end{cases} \quad (5.6.38)$$

Now we employ symmetrization to (5.6.37) and deduce that

$$\bar{B} = B + \sum_{i=1}^n \alpha_i (e_i z(i)^T + z(i) e_i^T). \quad (5.6.39)$$

Let us choose α_i , such that $\bar{B}$ satisfies the quasi-Newton condition. Obviously, $\bar{B}$ is symmetric and satisfies sparsity.

Similar to the discussion before, we can obtain that α satisfies

$$T\alpha = r, \quad (5.6.40)$$

where

$$T = \sum_{i=1}^n [z(i)^T s e_i + e_i^T s z(i)] e_i^T. \quad (5.6.41)$$

In particular, setting $z(i) = s(i)$, we immediately get (5.6.25)–(5.6.27), which is sparse PSB update.

Next, let us proceed to the sparse BFGS update.

For clarity, we repeat the BFGS update given in (5.1.45):

$$\bar{B} = B + \frac{yy^T}{s^T y} - \frac{Bss^T B}{s^T B s}, \quad (5.6.42)$$

where B is assumed to have some sparsity pattern. Since the $\bar{B}$ defined by the above formula has not such a structure, we modify it and make it have this kind of sparsity structure. Define

$$\hat{B} = \bar{B} + E. \quad (5.6.43)$$

We demand that $\hat{B}$ satisfies the following conditions:

- (i) $\hat{B}$ satisfies the quasi-Newton condition.
- (ii) $\hat{B}$ is symmetric.
- (iii) $\hat{B}$ is the closest to $\bar{B}$ in Frobenius norm.

So, we consider the following minimization problem:

$$\min \quad \|E\|_F = \frac{1}{2} \text{Tr}(E^T E) \quad (5.6.44)$$

$$\text{s.t.} \quad Es = 0, \quad (5.6.45)$$

$$E_{ij} = -\bar{B}_{ij}, \quad (i, j) \in I, \quad (5.6.46)$$

$$E = E^T. \quad (5.6.47)$$

To solve (5.6.44)–(5.6.47), we define the Lagrange function Φ as follows:

$$\begin{aligned} \Phi(E, \mu, \Lambda, \lambda) &= \frac{1}{2} \text{Tr}(E^T E) - \text{Tr}(Es\mu^T) - \text{Tr}(\Lambda(E - E^T)) \\ &\quad - \sum_{(i,j) \in I} \lambda_{ij} \text{Tr}(E + \bar{B}) e_j e_i^T \\ &= \frac{1}{2} \text{Tr}(E^T E) - \text{Tr}(Es\mu^T) - \text{Tr}(\Lambda(E - E^T)) \\ &\quad - \text{Tr}(\Lambda^T(E + \bar{B})), \end{aligned} \quad (5.6.48)$$

where μ is the multiplier vector, Λ and Δ are multiplier matrices, and λ_{ij} are the elements of the matrix Δ . When $(i, j) \in J$, $\lambda_{ij} = 0$.

Differentiating (5.6.48) and setting $\frac{\partial \Phi}{\partial E} = 0$, we have

$$\frac{\partial \Phi}{\partial E} = E - s\mu^T - \Lambda^T + \Lambda - \Delta = 0, \quad (5.6.49)$$

which gives

$$E = s\mu^T + \Delta - \Lambda + \Lambda^T \quad (5.6.50)$$

and

$$E^T = \mu s^T + \Delta^T - \Lambda^T + \Lambda. \quad (5.6.51)$$

By using (5.6.47), we get

$$E - E^T = s\mu^T + \Delta - \Delta^T + 2(\Lambda^T - \Lambda) = 0, \quad (5.6.52)$$

that is

$$\Lambda - \Lambda^T = \frac{1}{2}(s\mu^T - \mu s^T + \Delta - \Delta^T). \quad (5.6.53)$$

By use of (5.6.50) and (5.6.53), we have

$$E = \frac{1}{2}(s\mu^T + \mu s^T + \Delta + \Delta^T), \quad (5.6.54)$$

which gives, by (5.6.46), that

$$e_i^T E e_j = \frac{1}{2}(e_i^T \mu s^T e_j + e_i^T s \mu^T e_j + \lambda_{ij} + \lambda_{ji}) = -\bar{B}_{ij},$$

that is

$$\lambda_{ij} + \lambda_{ji} = -2\bar{B}_{ij} - e_i^T \mu s^T e_j - e_i^T s \mu^T e_j, \quad (i, j) \in I. \quad (5.6.55)$$

The above expression can be written in matrix form:

$$\Delta + \Delta^T = -2\bar{B}_{ij}^{(I)} - \sum_{i=1}^n e_i e_i^T (\mu \hat{s}(i)^T + s \hat{\mu}(i)^T), \quad (5.6.56)$$

where

$$\bar{B}_{ij}^{(I)} = \begin{cases} \bar{B}_{ij}, & (i, j) \in I, \\ 0, & (i, j) \in J, \end{cases} \quad (5.6.57)$$

$$\hat{s}(i)_j = \begin{cases} s_j, & (i, j) \in I, \\ 0, & (i, j) \in J, \end{cases} \quad \hat{\mu}(i)_j = \begin{cases} \mu_j, & (i, j) \in I, \\ 0, & (i, j) \in J. \end{cases} \quad (5.6.58)$$

By (5.6.54) and (5.6.56), we deduce that

$$\begin{aligned} E &= \frac{1}{2} \left[\sum_{i=1}^n e_i e_i^T (\mu s^T + s \mu^T) - 2\bar{B}^{(I)} - \sum_{i=1}^n e_i e_i^T (\mu \hat{s}(i)^T + s \hat{\mu}(i)^T) \right] \\ &= \frac{1}{2} \left[\sum_{i=1}^n e_i e_i^T (\mu s(i)^T + s \mu(i)^T) - 2\bar{B}^{(I)} \right], \end{aligned} \quad (5.6.59)$$

where

$$\mu(i)_j = \begin{cases} \mu_i, & (i, j) \in J, \\ 0, & (i, j) \in I. \end{cases} \quad (5.6.60)$$

Also, by (5.6.45), we have

$$Es = \frac{1}{2} \left[\sum_{i=1}^n e_i e_i^T (\mu s(i)^T + s \mu(i)^T) - 2\bar{B}^{(I)} \right] s = 0, \quad (5.6.61)$$

which is

$$\sum_{i=1}^n e_i e_i^T (\mu s(i)^T s + s \mu(i)^T s) = 2\bar{B}^{(I)} s. \quad (5.6.62)$$

Note that

$$\sum_{i=1}^n e_i e_i^T s \mu(i)^T = \sum_{i=1}^n \mu_i s(i) e_i^T,$$

and we can rewrite (5.6.62) as

$$\sum_{i=1}^n \mu_i (e_i s(i)^T + s(i) e_i^T) s = t, \quad (5.6.63)$$

where $t = 2\bar{B}^{(I)} s$.

Then, provided that we solve (5.6.63) for μ_i and substitute μ_i into (5.6.59), we can deduce that

$$E = \frac{1}{2} \sum_{i=1}^n \mu_i (e_i s(i)^T + s(i) e_i^T) - \bar{B}^{(I)}. \quad (5.6.64)$$

Thus,

$$\begin{aligned} \hat{B} &= \bar{B} + E \\ &= \bar{B} + \frac{1}{2} \sum_{i=1}^n \mu_i (e_i s(i)^T + s(i) e_i^T) - \bar{B}^{(I)} \\ &= \bar{B}^{(J)} + \frac{1}{2} \sum_{i=1}^n \mu_i (e_i s(i)^T + s(i) e_i^T) \end{aligned} \quad (5.6.65)$$

where

$$\bar{B}_{ij}^{(J)} = \begin{cases} \bar{B}_{ij}, & (i, j) \in J, \\ 0, & (i, j) \in I. \end{cases} \quad (5.6.66)$$

The formula (5.6.65) is said to be sparse BFGS update. Similarly, we can derive the sparse update for other quasi-Newton updates.

Note that the above formula (5.6.65) is obtained by minimization of problem (5.6.44)–(5.6.47) in Frobenius norm. Instead, we consider this minimization problem in the weighted Frobenius norm, i.e., consider the problem

$$\min \|E\|_{W,F} = \frac{1}{2} \text{Tr}(WE^TWE) \quad (5.6.67)$$

$$\text{s.t. } Es = 0, \quad (5.6.68)$$

$$E_{ij} = -\bar{B}_{ij}, \quad (i, j) \in I, \quad (5.6.69)$$

$$E = E^T. \quad (5.6.70)$$

Then, corresponding to (5.6.54), we have

$$E = \frac{1}{2}[z(s^T M) + (Ms)z^T + M(\Delta + \Delta^T)M], \quad (5.6.71)$$

where $M = W^{-1}$, $z = M\mu$.

Set $p = Ms$. We can obtain that if and only if $M(\Delta + \Delta^T)M$ and $\Delta + \Delta^T$ have the same sparsity pattern, the solution of (5.6.67)–(5.6.70) is

$$\hat{B} = \bar{B}^{(I)} + \sum_{i=1}^n z_i(e_i p(i)^T + p(i)e_i^T), \quad (5.6.72)$$

where

$$p(i)_j = \begin{cases} p_j, & (i, j) \in J, \\ 0, & (i, j) \in I, \end{cases} \quad (5.6.73)$$

z_i is the solution of the equations

$$\sum_{i=1}^n z_i(e_i p(i)^T + p(i)e_i^T)s = 2\bar{B}^{(I)}s. \quad (5.6.74)$$

Clearly, if W is a positive definite and diagonal matrix, $M(\Delta + \Delta^T)M$ and $\Delta + \Delta^T$ have the same sparsity structure.

Toint [341] considered sparse quasi-Newton update in the case that the weighted matrix is a non-diagonal matrix. For solving efficiently the sparse equations about μ_i , Steihaug [321] presented a preconditioned conjugate gradient method to solve the linear equations.

An alternative approach is to relax the quasi-Newton equation, making sure that it is approximately satisfied along the last few steps rather than

requiring it to hold strictly on the latest step. Define the $n \times m$ matrices S_k and Y_k by

$$S_k = [s_{k-m}, \dots, s_{k-1}], \quad Y_k = [y_{k-m}, \dots, y_{k-1}]. \quad (5.6.75)$$

We ask B_{k+1} to be a solution of

$$\min \quad \|\bar{B}S_k - Y_k\|_F^2 \quad (5.6.76)$$

$$\text{s.t.} \quad \bar{B} = \bar{B}^T, \quad (5.6.77)$$

$$\bar{B}_{ij} = 0, \quad (i, j) \in I. \quad (5.6.78)$$

In general, sparse quasi-Newton methods lost some advantages of dense quasi-Newton methods.

- (1) Because of the complexity of the sparse pattern, the modified matrix E is a rank- n matrix, rather than a rank-two matrix.
- (2) To compute the matrix E , we must solve a sparse linear equation about μ_i .
- (3) The positive definiteness of the update matrix $\{B_k\}$ cannot be guaranteed.
- (4) So far, the numerical performance is not ideal.

We think that it is still a challenging topic to solve large-scale optimization problems by studying sparse quasi-Newton methods.

5.7 Limited Memory BFGS Method

Limited memory quasi-Newton methods are useful for solving large-scale optimization problems. For large-scale problems, the methods save only a few n -dimensional vectors, instead of storing and computing fully dense $n \times n$ approximations of the Hessian. Since BFGS method is the most efficient method for solving unconstrained optimization problems, in this section we consider the limited memory BFGS method, known as L-BFGS, which is based on BFGS method.

As we know that the BFGS formula for inverse Hessian approximation H_k is

$$H_{k+1} = \left(I - \frac{s_k y_k^T}{s_k^T y_k} \right) H_k \left(I - \frac{y_k s_k^T}{s_k^T y_k} \right) + \frac{s_k s_k^T}{s_k^T y_k}. \quad (5.7.1)$$

Set

$$\rho_k = \frac{1}{s_k^T y_k}, \quad V_k = I - \rho_k y_k s_k^T, \quad (5.7.2)$$

then

$$H_{k+1} = V_k^T H_k V_k + \rho_k s_k s_k^T. \quad (5.7.3)$$

The above equation says that the matrix H_{k+1} is obtained by updating H_k using the pair $\{s_k, y_k\}$. In L-BFGS method we save implicitly a modified version of H_k by storing m pairs $\{s_i, y_i\} (i = k - m, k - m + 1, \dots, k - 1)$.

In the following, we describe the expression of the updating matrix H_k of the k -th iteration in L-BFGS method.

Choose some initial Hessian approximation $H_k^{(0)}$ for the k -th iteration. We apply the formula (5.7.3) m times repeatedly, i.e.,

$$H_k^{(j+1)} = V_{k-m+j}^T H_k^{(j)} V_{k-m+j} + \rho_{k-m+j} s_{k-m+j} s_{k-m+j}^T, \quad j = 0, 1, \dots, m-1, \quad (5.7.4)$$

and obtain

$$\begin{aligned} H_k &= (V_{k-1}^T \cdots V_{k-m}^T) H_k^{(0)} (V_{k-m} V_{k-m+1} \cdots V_{k-1}) \\ &\quad + \rho_{k-m} (V_{k-1}^T \cdots V_{k-m+1}^T) s_{k-m} s_{k-m}^T (V_{k-m+1} \cdots V_{k-1}) \\ &\quad + \rho_{k-m+1} (V_{k-1}^T \cdots V_{k-m+2}^T) s_{k-m+1} s_{k-m+1}^T (V_{k-m+2} \cdots V_{k-1}) \\ &\quad + \cdots \\ &\quad + \rho_{k-1} s_{k-1} s_{k-1}^T. \end{aligned} \quad (5.7.5)$$

It follows from the above expression that if we know pairs $\{s_i, y_i\} (i = k - m, k - m + 1, \dots, k - 1)$, we can compute H_k . In fact, we need not compute and save H_k explicitly, instead, we only save the pairs $\{s_i, y_i\}$ and compute $H_k g_k$, where g_k is the gradient of f at x_k . So, we have

$$\begin{aligned} H_k g_k &= (V_{k-1}^T \cdots V_{k-m}^T) H_k^{(0)} (V_{k-m} V_{k-m+1} \cdots V_{k-1}) g_k \\ &\quad + \rho_{k-m} (V_{k-1}^T \cdots V_{k-m+1}^T) s_{k-m} s_{k-m}^T (V_{k-m+1} \cdots V_{k-1}) g_k \\ &\quad + \rho_{k-m+1} (V_{k-1}^T \cdots V_{k-m+2}^T) s_{k-m+1} s_{k-m+1}^T (V_{k-m+2} \cdots V_{k-1}) g_k \\ &\quad + \cdots \\ &\quad + \rho_{k-1} s_{k-1} s_{k-1}^T g_k. \end{aligned} \quad (5.7.6)$$

Since

$$V_i g_k = (I - \rho_i y_i s_i^T) g_k, \quad i = k - 1, k - 2, \dots, k - m,$$

we have the following algorithm to compute $H_k g_k$.

Algorithm 5.7.1 (*L-BFGS two-loop recursion for $H_k g_k$*)

Step 1. $q := g_k$;

Step 2. **for** $i = k - 1, k - 2, \dots, k - m$
 $\alpha_i := \rho_i s_i^T q$;
 $q := q - \alpha_i y_i$;
end (for)

Step 3. $r := H_k^{(0)} q$;

Step 4. **for** $i = k - m, k - m + 1, \dots, k - 1$
 $\beta := \rho_i y_i^T r$;
 $r := r + s_i(\alpha_i - \beta)$
end (for) $\square$

By use of the above algorithm, we obtain $r = H_k g_k$. A choice of $H_k^{(0)}$ is

$$H_k^{(0)} = \frac{s_k^T y_k}{\|y_k\|^2} I. \quad (5.7.7)$$

The limited memory BFGS algorithm can be stated as follows.

Algorithm 5.7.2 (*L-BFGS Method*)

Step 1. Given a starting point $x_0 \in R^n$, an initial symmetric and positive definite matrix $H_0 \in R^{n \times n}$, a nonnegative integer $m \geq 0$, an error tolerance $\epsilon > 0$, $k := 0$.

Step 2. Compute $g_k = \nabla f(x_k)$. If $\|g_k\| \leq \epsilon$, we take $x^* = x_k$, stop; otherwise, compute $d_k = -H_k g_k$ from Algorithm 5.7.1.

Step 3. Find a step size $\alpha_k > 0$ by using Wolfe rule.

Step 4. Set $x_{k+1} = x_k + \alpha_k d_k$.

Step 5. If $k > m$, discard the vector pairs $\{s_{k-m}, y_{k-m}\}$ from storage;

Set $s_k = x_{k+1} - x_k$, $y_k = g_{k+1} - g_k$;

Take $H_k^{(0)} = \frac{s_k^T y_k}{\|y_k\|^2} I$.

Step 6. $k := k + 1$ and go to Step 2. $\square$

The above L-BFGS algorithm is equivalent to the usual BFGS algorithm if the initial matrix H_0 is the same in both algorithms, and if $H_k^{(0)} = H_0$ at each iteration. Normally, for large-scale problems, we take $m \ll n$. In practice, the choice of m is dependent on the dimension of the problem and the storage of employed computer. Usually, we take $3 \leq m \leq 30$.

In the following, we establish the convergence and convergence rate of L-BFGS method.

Lemma 5.7.3 *Let $f(x)$ be a twice continuously differentiable and uniformly convex function, i.e., there exist $0 < m \leq M$ such that*

$$m\|u\|^2 \leq u^T G(x)u \leq M\|u\|^2, \quad \forall x \in L(x_0), \quad u \in R^n, \quad (5.7.8)$$

where $G(x) = \nabla^2 f(x)$ and $L(x_0) = \{x \mid f(x) \leq f(x_0)\}$. Then

$$\frac{\|y\|}{\|s_k\|} \leq M, \quad \frac{\|s_k\|^2}{s_k^T y_k} \leq \frac{1}{m}, \quad \frac{\|y_k\|^2}{s_k^T y_k} \leq M. \quad (5.7.9)$$

Proof. 1) Let $\bar{G} = \int_0^1 G(x_k + \tau s_k) d\tau$. Then

$$y_k = g_{k+1} - g_k = \int_0^1 G(x_k + \tau s_k) s_k d\tau = \bar{G} s_k. \quad (5.7.10)$$

Taking the norm, then we obtain

$$\|y_k\| \leq \|s_k\| \int_0^1 \|G(x_k + \tau s_k)\| d\tau. \quad (5.7.11)$$

From the assumptions, it follows that $L(x_0)$ is a bounded, closed and convex set, then $x_k + \tau s_k \in L(x_0)$. Then $\|G(x_k + \tau s_k)\| \leq M$. Thus we have that $\|y_k\| \leq M\|s_k\|$ which is the first conclusion in (5.7.9).

2) By use of (5.7.10), we have that

$$s_k^T y_k = \int_0^1 s_k^T G(x_k + \tau s_k) s_k d\tau \geq m\|s_k\|^2, \quad (5.7.12)$$

which means

$$\frac{\|s_k\|^2}{s_k^T y_k} \leq \frac{1}{m}.$$

3) Since $y_k = \bar{G}s_k$, then

$$\begin{aligned} \frac{\|y_k\|^2}{s_k^T y_k} &= \frac{y_k^T y_k}{s_k^T y_k} = \frac{s_k^T \bar{G}^{\frac{1}{2}} \bar{G} \bar{G}^{\frac{1}{2}} s_k}{s_k^T \bar{G}^{\frac{1}{2}} \bar{G}^{\frac{1}{2}} s_k} \\ &= \frac{\gamma_k^T \bar{G} \gamma_k}{\gamma_k^T \gamma_k} \leq M, \end{aligned} \quad (5.7.13)$$

where $\gamma_k = \bar{G}^{\frac{1}{2}} s_k$. $\square$

Theorem 5.7.4 *Let $f(x)$ be a twice continuously differentiable and uniformly convex function. Then the iterative sequence $\{x_k\}$ generated from L-BFGS Algorithm 5.7.2 converges to the unique minimizer x^* of $f(x)$.*

Proof. From Lemma 5.7.3, we have

$$\frac{\|s_k\|^2}{s_k^T y_k} \leq M, \quad \frac{\|y_k\|^2}{s_k^T y_k} \leq M. \quad (5.7.14)$$

Then

$$\|V_k\| \leq 1 + M. \quad (5.7.15)$$

Let $\bar{m} = \min\{k, m\}$. Without loss of generality, we assume that $\|H_k^{(0)}\| \leq M$. Then by (5.7.5) and (5.7.14)-(5.7.15) we get

$$\begin{aligned} \|H_k\| &\leq M(1+M)^{2\bar{m}} + \sum_{j=1}^{\bar{m}} M(1+M)^{2(\bar{m}-j)} \\ &\leq M(1+M)^{2\bar{m}}(\bar{m}+1). \end{aligned} \quad (5.7.16)$$

On the other hand, write $B_k^{(0)} = (H_k^{(0)})^{-1}$. From (5.7.4) we have

$$B_k^{(j+1)} = B_k^{(j)} - \frac{B_k^{(j)} s_{k-\bar{m}+j} s_{k-\bar{m}+j}^T (B_k^{(j)})^T}{s_{k-\bar{m}+j}^T B_k^{(j)} s_{k-\bar{m}+j}} + \frac{y_{k-\bar{m}+j} y_{k-\bar{m}+j}^T}{y_{k-\bar{m}+j}^T s_{k-\bar{m}+j}}, \quad j = 0, 1, \bar{m} - 1.$$

Then

$$B_k^{(\bar{m})} = B_k = H_k^{-1}.$$

Since $\text{Tr}(xy^T) = x^T y$ for $x, y \in R^n$ and $\text{Tr}(A+B) = \text{Tr}(A) + \text{Tr}(B)$ for $n \times n$ matrices A and B , then it follows from (5.7.14) that

$$\begin{aligned} \text{Tr}(B_k^{(j+1)}) &= \text{Tr}(B_k^{(j)}) - \frac{\|B_k^{(j)} s_{k-\bar{m}+j}\|^2}{s_{k-\bar{m}+j}^T B_k^{(j)} s_{k-\bar{m}+j}} + \frac{\|y_{k-\bar{m}+j}\|^2}{y_{k-\bar{m}+j}^T s_{k-\bar{m}+j}} \\ &\leq \text{Tr}(B_k^{(j)}) + M. \end{aligned} \quad (5.7.17)$$

Repeatedly applying (5.7.17) $\bar{m}$ times, and using (5.7.7) and (5.7.14), we obtain that

$$\begin{aligned}\mathrm{Tr}(B_k) &= \mathrm{Tr}(B_k^{(\bar{m})}) \leq \mathrm{Tr}(B_k^{(0)}) + \bar{m}M \\ &= \mathrm{Tr}((H_k^{(0)})^{-1}) + \bar{m}M \\ &\leq (n + \bar{m})M.\end{aligned}\tag{5.7.18}$$

Let the eigenvalues of B_k be $0 < \lambda_1 \leq \dots \leq \lambda_n$, then the eigenvalues of H_k are

$$0 < \frac{1}{\lambda_n} \leq \frac{1}{\lambda_{n-1}} \leq \dots \leq \frac{1}{\lambda_1}.$$

By use of the property of the Rayleigh quotient and $\mathrm{Tr}(B_k) = \sum_{j=1}^n \lambda_j$, we obtain

$$\begin{aligned}\cos \theta_k &= \frac{-d_k^T g_k}{\|d_k\| \|g_k\|} = \frac{g_k^T H_k g_k}{\|H_k g_k\| \|g_k\|} \\ &\geq \frac{\|g_k\|^2 / \lambda_n}{\|H_k\| \|g_k\|^2} = \frac{1}{\lambda_n \|H_k\|} \\ &\geq \frac{1}{\mathrm{Tr}(B_k) \|H_k\|}.\end{aligned}\tag{5.7.19}$$

Then, it follows from (5.7.16) and (5.7.18) that there is a $\rho > 0$ such that

$$\cos \theta_k \geq \rho\tag{5.7.20}$$

holds for all k . This implies that there is $\bar{\mu} > 0$ such that

$$\theta_k \leq \frac{\pi}{2} - \bar{\mu}, \quad \forall k.\tag{5.7.21}$$

The assumptions of the theorem and Theorem 1.3.19 indicate that the level $L(x_0)$ is bounded, closed and convex. Then, the continuous function $\nabla f(x)$ exists and is uniformly continuous on $L(x_0)$. Noting that α_k is determined by Wolfe rule, then we obtain, by Theorem 2.5.5 and (5.7.21), that the sequence $\{x_k\}$ converges to the unique minimizer x^* of $f(x)$. $\square$

Next, we establish the convergence rate of L-BFGS method.

Lemma 5.7.5 *Let $f(x)$ be a twice continuously differentiable and uniformly convex function. Then*

$$f(x) - f(x^*) \leq \frac{1}{m} \|g(x)\|^2.\tag{5.7.22}$$

Proof. Since $f(x)$ is a convex function, for any $x \in R^n$ we have

$$f(x) - f(x^*) \leq g(x)^T(x - x^*) \leq \|g(x)\| \|x - x^*\|. \quad (5.7.23)$$

Note that

$$g(x) = g(x) - g(x^*) = \int_0^1 G(x^* + \tau(x - x^*))(x - x^*) d\tau. \quad (5.7.24)$$

Writing $\bar{G} = \int_0^1 G(x^* + \tau(x - x^*)) d\tau$, then we have

$$g(x) = \bar{G}(x - x^*). \quad (5.7.25)$$

By use of (5.7.8), we get

$$\begin{aligned} m\|x - x^*\|^2 &\leq (x - x^*)^T \bar{G}(x - x^*) \leq (x - x^*)^T g(x) \\ &\leq \|x - x^*\| \|g(x)\|, \end{aligned} \quad (5.7.26)$$

that is

$$\|x - x^*\| \leq \|g(x)\|/m. \quad (5.7.27)$$

Substituting (5.7.27) into (5.7.23) gives (5.7.22). $\square$

Lemma 5.7.6 *Let $f(x)$ be a twice continuously differentiable and uniformly convex function. Let $x_{k+1} = x_k + \alpha_k d_k$, where α_k is determined by Wolfe rule. Then*

$$c_1 \|g_k\| \cos \theta_k \leq \|s_k\| \leq c_2 \|g_k\| \cos \theta_k \quad (5.7.28)$$

and

$$f(x_{k+1}) - f(x^*) \leq (1 - \rho m c_1 \cos^2 \theta_k) [f(x_k) - f(x^*)], \quad (5.7.29)$$

where $c_1 = (1 - \sigma)/M$, $c_2 = 2(1 - \rho)/m$, ρ and σ are defined by Wolfe rule, θ_k is an angle between d_k and $-g_k$.

Proof. From (5.7.9) we have that

$$\frac{y_k^T s_k}{\|s_k\|^2} \leq \frac{\|y_k\| \|s_k\|}{\|s_k\|^2} = \frac{\|y_k\|}{\|s_k\|} \leq M. \quad (5.7.30)$$

By using Wolfe rule, we have that

$$y_k^T s_k = g_{k+1}^T s_k - g_k^T s_k \geq -(1 - \sigma) g_k^T s_k. \quad (5.7.31)$$

Then the above expressions give

$$\|s_k\|^2 \geq \frac{y_k^T s_k}{M} \geq -\frac{1-\sigma}{M} g_k^T s_k = \frac{1-\sigma}{M} \|g_k\| \|s_k\| \cos \theta_k,$$

that is

$$\|s_k\| \geq \frac{1-\sigma}{M} \|g_k\| \cos \theta_k. \quad (5.7.32)$$

We obtain the left-hand side inequality of (5.7.28).

By Taylor expression and Wolfe rule, we have that

$$g_k^T s_k + \frac{1}{2} s_k^T G(\xi_k) s_k = f(x_{k+1}) - f(x_k) \leq \rho g_k^T s_k, \quad (5.7.33)$$

where ξ_k lies between x_k and x_{k+1} . Then

$$s_k^T G(\xi_k) s_k \leq -2(1-\rho) g_k^T s_k. \quad (5.7.34)$$

Since $L(x_0)$ is a bounded, closed and convex set, $\xi_k \in L(x_0)$. Then we have that

$$m \|s_k\|^2 \leq s_k^T G(\xi_k) s_k. \quad (5.7.35)$$

The inequalities (5.7.34) and (5.7.35) yield

$$\|s_k\| \leq \frac{2(1-\rho)}{M} \|g_k\| \cos \theta_k,$$

which is the right-hand side of (5.7.28).

Finally, we prove (5.7.29). By using Wolfe rule and (5.7.28), we have that

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq \rho g_k^T s_k = -\rho \|g_k\| \|s_k\| \cos \theta_k \\ &\leq -\rho c_1 \|g_k\|^2 \cos^2 \theta_k. \end{aligned} \quad (5.7.36)$$

From Lemma 5.7.5, we have

$$\|g_k\|^2 \geq m[f(x_k) - f(x^*)]. \quad (5.7.37)$$

So, we can substitute (5.7.37) into (5.7.36) to obtain that

$$f(x_{k+1}) - f(x_k) \leq -\rho m c_1 \cos^2 \theta_k [f(x_k) - f(x^*)], \quad (5.7.38)$$

which gives result (5.7.29) by subtracting $f(x^*)$ from both sides. $\square$

Theorem 5.7.7 *Let $f(x)$ be a twice continuously differentiable and uniformly convex function. Assume that the iterative sequence $\{x_k\}$ generated by L-BFGS Algorithm 5.7.2 converges to the unique minimizer x^* of $f(x)$. Then the rate of convergence is at least R -linear.*

Proof. From (5.7.29) we have

$$f(x_{k+1}) - f(x^*) \leq \delta(f(x_k) - f(x^*)),$$

where $\delta \in (0, 1)$. Also, since $f(x)$ is a uniformly convex function, there are $0 < m_1 \leq M_1$ such that

$$m_1 \|u\|^2 \leq u^T G(x) u \leq M_1 \|u\|^2, \quad \forall x \in L(x_0), u \in R^n. \quad (5.7.39)$$

By using Taylor expression of $f(x_k)$ at x^* and (5.7.39), we obtain that

$$f(x_k) - f(x^*) \geq \frac{m_1}{2} \|x_k - x^*\|^2. \quad (5.7.40)$$

Hence

$$\begin{aligned} \|x_k - x^*\| &\leq \sqrt{\frac{2}{m_1}} (f(x_k) - f(x^*))^{\frac{1}{2}} \\ &\leq \sqrt{\frac{2}{m_1}} \delta^{\frac{1}{2}} (f(x_{k-1}) - f(x^*))^{\frac{1}{2}} \\ &\leq \dots \\ &\leq \sqrt{\frac{2}{m_1}} (\delta^{\frac{1}{2}})^k (f(x_0) - f(x^*))^{\frac{1}{2}}. \end{aligned} \quad (5.7.41)$$

The above inequality shows that the sequence $\{x_k\}$ is R -linearly convergent. $\square$

This theorem indicates that L-BFGS method often converges slowly, which leads to a relatively large number of function evaluations. Also, it is inefficient on highly ill-conditioned optimization problems. Though there are some weaknesses, L-BFGS method is a main choice for large-scale problems in which the true Hessian is not sparse, because, in this case, it may outperform other rival algorithms. For further details of L-BFGS method, please consult Liu and Nocedal [200] and Nash and Nocedal [228].

At the end of this section, we mention a memoryless BFGS formula. For BFGS formula (5.7.1) and (5.7.3), if we set $H_k = I$ at each iteration, we have

$$H_{k+1} = V_k^T V_k + \rho_k s_k s_k^T \quad (5.7.42)$$

$$= \left(I - \frac{s_k y_k^T}{s_k^T y_k} \right) \left(I - \frac{y_k s_k^T}{s_k^T y_k} \right) + \frac{s_k s_k^T}{s_k^T y_k}. \quad (5.7.43)$$

The above formula satisfies quasi-Newton condition and positive definiteness, and is called the memoryless BFGS formula. Obviously, if $m = 1$ and $H_k^{(0)} = I, \forall k$, the limited memory BFGS method is just the memoryless BFGS method.

Exercises

1. Using DFP method minimize the Rosenbrock function in Appendix 1.1 and the Extended Rosenbrock function in Appendix 1.2.
2. Using BFGS method minimize the Extended Rosenbrock function in Appendix 1.2 and the Powell singular function in Appendix 1.4.
3. State the properties of DFP and BFGS formulas and their relations.
4. Prove that if f is strong convex, $y_k^T s_k > 0$ holds.
5. Prove that H_{k+1}^{BFGS} given by (5.1.49) is the unique solution of problem (5.1.79).
6. Prove Theorem 5.2.1.
7. State the properties of Broyden class and Huang class, and their relations.
8. Prove Theorem 5.4.3.
9. Describe the motivation of self-scaling strategy in variable metric methods by observing DFP method.
10. Do programming of L-BFGS algorithm in §5.9 in MATLAB or FORTRAN.